

# ❏ Credit Risk Analysis ❏

## ❏ Overview

- **Background**

In the financial services industry, lending institutions face significant challenges in **managing credit risk** while maintaining profitability. Accurately assessing a borrower's ability to repay is critical to **minimizing defaults** and ensuring sustainable lending practices.

- **Goal of the Project**

Using historical customer, financial, and credit behavior data, this project aims to build a predictive model that estimates the likelihood of a borrower defaulting on a loan. This enables financial institutions to **improve risk assessment, optimize loan approval decisions, and reduce potential losses.**

## ❏ Key Features

### Target Variable

- `loan_status` : *Loan status (e.g., fully paid, default, charged off)*

### Customer Demographics & Profile

- `person_age` : Age of the customer
- `gender` : Gender
- `marital_status` : Marital status
- `education_level` : Education level

### Financial & Employment Information

- `person_income` : Annual income
- `person_emp_length` : Employment length (in years)
- `employment_type` : Type of employment
- `person_home_ownership` : Home ownership status

### Loan Characteristics

- `loan_amnt` : Loan amount
- `loan_term_months` : Loan term (in months)

- `loan_int_rate` : Interest rate
- `loan_grade` : Loan risk grade (A-G)
- `loan_intent` : Loan purpose
- `loan_percent_income` : Loan-to-income percentage
- `loan_to_income_ratio` : Loan-to-income ratio

### Credit History

- `cb_person_cred_hist_length` : Credit history length
- `cb_person_default_on_file` : Historical default indicator

### Debt & Repayment Capacity

- `other_debt` : Other existing debts
- `debt_to_income_ratio` : Debt-to-income ratio

### Credit Behavior

- `open_accounts` : Number of open credit accounts
- `credit_utilization_ratio` : Credit utilization ratio
- `past_delinquencies` : Number of past delinquencies

### Geographic Information

- `country` : Country
- `state` : State/region
- `city` : City
- `city_latitude` : Latitude
- `city_longitude` : Longitude

### □ Files Provided

- `Credit_Risk_Dataset_Onyx_Data_September_25 - Credit Risk Data.csv` : Main dataset used for analysis and modeling

### □ Project Objective

The primary objective of this project is to **perform exploratory data analysis (EDA)** and develop predictive models to classify whether a borrower is likely to default on a loan.

By leveraging these insights, financial institutions can:

- Improve **credit risk assessment**
- Enhance **loan approval strategies**
- Reduce **default rates and financial losses**

# Import Libraries

```
In [1]: # =====  
# Core Libraries  
# =====  
import pandas as pd          # Data handling  
import numpy as np           # Numerical operations  
import seaborn as sns        # Statistical plots  
import matplotlib.pyplot as plt # Visualization  
  
# =====  
# Project Utilities  
# =====  
from utils.logger import LoggerConfig  
from utils.utils import Utils  
from utils.config_loader import load_config  
from config.config import CFG  
from utils.colors_utils import ColorUtils  
from utils.stats_utils import StatUtils  
from utils.validation import DataValidator  
from utils.feature_utils import FeatureUtils  
  
# =====  
# Modeling Utilities  
# =====  
from models.benchmark import *  
from models.oof import *  
  
# =====  
# Hyperparameter Tuning  
# =====  
from models.catboost_tuning import *  
from models.lgbm_tunning import *  
from models.hgbc_tuning import *  
  
# =====  
# Explainability  
# =====  
from utils.explainability import *  
  
# =====  
# Notebook Display  
# =====  
from IPython.display import display, HTML  
  
# =====
```

```

# Data Splitting & Preprocessing
# =====
from sklearn.model_selection import StratifiedShuffleSplit, StratifiedKFold

from sklearn.preprocessing import OneHotEncoder, RobustScaler, StandardScaler
from sklearn.impute import SimpleImputer

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# =====
# Evaluation
# =====
from sklearn.metrics import classification_report

# =====
# Machine Learning Models
# =====
from sklearn.svm import LinearSVC, SVC
from sklearn.neighbors import KNeighborsClassifier

from sklearn.ensemble import (
    RandomForestClassifier,
    GradientBoostingClassifier,
    ExtraTreesClassifier,
    AdaBoostClassifier,
    BaggingClassifier,
    HistGradientBoostingClassifier
)

from xgboost import XGBClassifier
from lightgbm import LGBMClassifier
from catboost import CatBoostClassifier

from sklearn.neural_network import MLPClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression, RidgeClassifier

# =====
# Optimization
# =====
import optuna
optuna.logging.set_verbosity(optuna.logging.WARNING)

# =====
# Environment Settings
# =====
import warnings
warnings.filterwarnings("ignore")

pd.set_option("display.max_columns", 500)
pd.set_option("display.max_colwidth", None)

```

## Load Data

```
In [2]: # Initialize notebook logger
logger = LoggerConfig.setup_logger()

# Load datasets safely
df = Utils.read_csv_safe(path=CFG.DATA_PATH, logger=logger,
                          encodings=CFG.DEFAULT_ENCODING).pipe(Utils.clean_col)

2026-04-18 19:01:09 | INFO      | Logging configured: level=DEBUG, format=colored
2026-04-18 19:01:10 | INFO      | Loaded file with encoding: utf-8 | Shape: (32581, 29)
```

## Data Preview and Info

```
In [3]: # Data Preview
display(df.head())
```

|   | client_ID  | person_age | person_income | person_home_ownership | person_en |
|---|------------|------------|---------------|-----------------------|-----------|
| 0 | CUST_00001 | 22         | 59000         | RENT                  |           |
| 1 | CUST_00002 | 21         | 9600          | OWN                   |           |
| 2 | CUST_00003 | 25         | 9600          | MORTGAGE              |           |
| 3 | CUST_00004 | 23         | 65500         | RENT                  |           |
| 4 | CUST_00005 | 24         | 54400         | RENT                  |           |

```
In [4]: # Display information about the DataFrames
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32581 entries, 0 to 32580
Data columns (total 29 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   client_ID                             32581 non-null  object
 1   person_age                           32581 non-null  int64
 2   person_income                        32581 non-null  int64
 3   person_home_ownership                32581 non-null  object
 4   person_emp_length                    31686 non-null  float64
 5   loan_intent                           32581 non-null  object
 6   loan_grade                           32581 non-null  object
 7   loan_amnt                            32581 non-null  int64
 8   loan_int_rate                        29465 non-null  float64
 9   loan_status                          32581 non-null  int64
10  loan_percent_income                  32581 non-null  float64
11  cb_person_default_on_file            32581 non-null  object
12  cb_person_cred_hist_length           32581 non-null  int64
13  gender                               32581 non-null  object
14  marital_status                       32581 non-null  object
15  education_level                      32581 non-null  object
16  country                              32581 non-null  object
17  state                                32581 non-null  object
18  city                                 32581 non-null  object
19  city_latitude                        32581 non-null  float64
20  city_longitude                       32581 non-null  float64
21  employment_type                      32581 non-null  object
22  loan_term_months                     32581 non-null  int64
23  loan_to_income_ratio                 32581 non-null  float64
24  other_debt                           32581 non-null  float64
25  debt_to_income_ratio                 32581 non-null  float64
26  open_accounts                        32581 non-null  int64
27  credit_utilization_ratio             32581 non-null  float64
28  past_delinquencies                   32581 non-null  int64
dtypes: float64(9), int64(8), object(12)
memory usage: 7.2+ MB

```

## □ Insights from Initial Data Exploration

### Dataset Size and Structure

- The dataset contains **32,581 records** with **29 features**.
- The data is moderately sized and suitable for both **EDA** and **machine learning modeling**.

### Feature Overview

- The dataset includes comprehensive information for credit risk assessment:
  - **Numerical features:** `person_age`, `person_income`, `loan_amnt`, `loan_int_rate`, `loan_percent_income`,

- **debt\_to\_income\_ratio**, **credit\_utilization\_ratio**, **open\_accounts**, **past\_delinquencies**.
- **Categorical features:** **person\_home\_ownership**, **loan\_intent**, **loan\_grade**, **gender**, **marital\_status**, **education\_level**, **employment\_type**, **cb\_person\_default\_on\_file**, **location-related features**.
- The target variable is **loan\_status**, represented as a **binary outcome (0/1)**, indicating loan repayment status.

### Data Quality and Completeness

- Most features are complete, but some missing values are observed:
  - **person\_emp\_length** has missing values (~2.7%).
  - **loan\_int\_rate** has notable missing values (~9.6%).
- Data types are consistent:
  - **Numerical:** int64 and float64
  - **Categorical:** object (string)

```
In [5]: # Numerical Features
num_features = ["person_age", "person_income", "person_emp_length", "loan_amnt",
               "loan_to_income_ratio", "cb_person_cred_hist_length", "other_loan_amnt",
               "credit_utilization_ratio"]

# Categorical Features
cat_features = ["person_home_ownership", "loan_intent", "loan_grade", "cb_person_default_on_file",
               "education_level", "employment_type", "country", "state", "city"]

# Data Describe
cm = sns.light_palette("blue", as_cmap=True)
display(df[num_features].describe().T.style.background_gradient(cmap=cm))
```

|                            | count        | mean         | std          | n         |
|----------------------------|--------------|--------------|--------------|-----------|
| person_age                 | 32581.000000 | 27.734600    | 6.348078     | 20.0000   |
| person_income              | 32581.000000 | 66074.848470 | 61983.119168 | 4000.0000 |
| person_emp_length          | 31686.000000 | 4.789686     | 4.142630     | 0.0000    |
| loan_amnt                  | 32581.000000 | 9589.371106  | 6322.086646  | 500.0000  |
| loan_int_rate              | 29465.000000 | 11.011695    | 3.240459     | 5.4200    |
| loan_percent_income        | 32581.000000 | 0.170203     | 0.106782     | 0.0000    |
| loan_to_income_ratio       | 32581.000000 | 0.170553     | 0.107049     | 0.0000    |
| cb_person_cred_hist_length | 32581.000000 | 5.804211     | 4.055001     | 2.0000    |
| other_debt                 | 32581.000000 | 11567.959773 | 13060.930898 | 225.2073  |
| debt_to_income_ratio       | 32581.000000 | 0.345208     | 0.129406     | 0.0645    |
| open_accounts              | 32581.000000 | 8.042018     | 4.328087     | 0.0000    |
| credit_utilization_ratio   | 32581.000000 | 0.499884     | 0.259530     | 0.0500    |

## □ Descriptive Statistics

- **person\_age**: Avg ~27.7; median ~26; range (~20 → 144) → mostly young population with extreme outliers (likely data quality issues).
- **person\_income**: Avg ~66K; median ~55K; very wide range (~4K → 6M) → highly skewed distribution with strong outliers.
- **person\_emp\_length**: Avg ~4.8 years; median ~4; range (~0 → 123) → reasonable spread but contains unrealistic high values.
- **loan\_amnt**: Avg ~9.6K; median ~8K; range (~500 → 35K) → moderately distributed loan sizes.
- **loan\_int\_rate**: Avg ~11%; range (~5.4 → 23.2) → typical lending rates with moderate variability.
- **loan\_percent\_income**: Avg ~17%; range (~0 → 83%) → indicates some borrowers take disproportionately high loans relative to income.
- **loan\_to\_income\_ratio**: Similar distribution to **loan\_percent\_income** → potential redundancy between features.
- **cb\_person\_cred\_hist\_length**: Avg ~5.8 years; range (~2 → 30) → relatively short credit history for most borrowers.
- **other\_debt**: Avg ~11.5K; extremely wide range (~225 → 1.18M) → highly skewed with significant outliers.



- `debt_to_income_ratio`: Avg ~0.35; range (~0.06 → 1.05) → high values indicate financial stress and elevated default risk.
- `open_accounts`: Avg ~8; range (~0 → 15) → moderate credit activity across customers.
- `credit_utilization_ratio`: Avg ~0.50; range (~0.05 → 0.95) → many users utilize a large portion of their credit limit.

```
In [6]: df = Utils.reduce_mem_usage(cols=num_features, df=df)
df = Utils.convert_cat(features=cat_features, df=df)

# Display information about the DataFrames
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32581 entries, 0 to 32580
Data columns (total 29 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   client_ID                             32581 non-null  object
 1   person_age                           32581 non-null  int16
 2   person_income                         32581 non-null  int32
 3   person_home_ownership                 32581 non-null  category
 4   person_emp_length                     31686 non-null  float32
 5   loan_intent                           32581 non-null  category
 6   loan_grade                           32581 non-null  category
 7   loan_amnt                            32581 non-null  int32
 8   loan_int_rate                         29465 non-null  float32
 9   loan_status                           32581 non-null  int64
10   loan_percent_income                  32581 non-null  float32
11   cb_person_default_on_file            32581 non-null  category
12   cb_person_cred_hist_length           32581 non-null  int8
13   gender                               32581 non-null  category
14   marital_status                       32581 non-null  category
15   education_level                      32581 non-null  category
16   country                              32581 non-null  category
17   state                                32581 non-null  category
18   city                                 32581 non-null  category
19   city_latitude                        32581 non-null  float64
20   city_longitude                       32581 non-null  float64
21   employment_type                      32581 non-null  category
22   loan_term_months                     32581 non-null  category
23   loan_to_income_ratio                 32581 non-null  float32
24   other_debt                           32581 non-null  float32
25   debt_to_income_ratio                 32581 non-null  float32
26   open_accounts                        32581 non-null  int8
27   credit_utilization_ratio             32581 non-null  float32
28   past_delinquencies                   32581 non-null  category
dtypes: category(13), float32(7), float64(2), int16(1), int32(2), int64(1),
int8(2), object(1)
memory usage: 2.6+ MB

```

```

In [7]: # Data Describe
display(df[cat_features].describe().T.style.background_gradient(cmap="Blues")

```

|                                  | count | unique | top         | freq  |
|----------------------------------|-------|--------|-------------|-------|
| <b>person_home_ownership</b>     | 32581 | 4      | RENT        | 16446 |
| <b>loan_intent</b>               | 32581 | 6      | EDUCATION   | 6453  |
| <b>loan_grade</b>                | 32581 | 7      | A           | 10777 |
| <b>cb_person_default_on_file</b> | 32581 | 2      | N           | 26836 |
| <b>gender</b>                    | 32581 | 2      | Male        | 16371 |
| <b>marital_status</b>            | 32581 | 4      | Single      | 16368 |
| <b>education_level</b>           | 32581 | 4      | High School | 13185 |
| <b>employment_type</b>           | 32581 | 4      | Full-time   | 19473 |
| <b>country</b>                   | 32581 | 3      | UK          | 10944 |
| <b>state</b>                     | 32581 | 9      | BC          | 3679  |
| <b>city</b>                      | 32581 | 18     | Victoria    | 1852  |
| <b>loan_term_months</b>          | 32581 | 4      | 36          | 12944 |
| <b>past_delinquencies</b>        | 32581 | 7      | 0           | 19702 |

## □ Categorical Features Overview

- **person\_home\_ownership**: 4 categories; most customers are **RENT** (~50%) → suggests many borrowers do not own property, potentially increasing risk exposure.
- **loan\_intent**: 6 categories; **EDUCATION** is the most common → indicates strong demand for education-related loans.
- **loan\_grade**: 7 levels (A-G); majority in **Grade A** → dataset is skewed toward lower-risk borrowers.
- **cb\_person\_default\_on\_file**: Binary; majority **No (~82%)** → relatively low historical default rate.
- **gender**: Balanced distribution (Male slightly higher) → unlikely to strongly bias model.
- **marital\_status**: **Single** is the dominant group → may correlate with different financial behaviors.
- **education\_level**: Most borrowers have **High School** education → mid-level education dominates the dataset.
- **employment\_type**: Majority are **Full-time** → indicates relatively stable income sources.

- `country`: 3 categories; **UK** appears most frequently → dataset concentrated in specific regions.
- `state`: 9 categories; moderate cardinality → useful for regional analysis.
- `city`: 18 categories; relatively higher cardinality → may require encoding strategies beyond simple one-hot.
- `loan_term_months`: 4 categories; **36 months** most common → standard loan duration dominates.
- `past_delinquencies`: 7 categories; majority have **0 past delinquencies** → indicates generally good repayment behavior.

## Data Quality Checks

### Missing Value

```
In [8]: logger.info("Start checking NULL values")
logger.info("-" * 80)
logger.info(f"Checking missing values for {CFG.DATASET_NAME}...")
DataValidator.checkNULL(df=df, logger=logger)
logger.info("-" * 80)
logger.info("Completed checking for NULL values across datasets.")
```

```

2026-04-18 19:01:10 | INFO      | Start checking NULL values
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Checking missing values for Credit Risk Dat
aset...
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Checking missing values for Credit Risk Dat
aset...
2026-04-18 19:01:10 | WARNING   |
+-----+-----+-----+
| Feature          | Missing_Count | Missing_% |
+-----+-----+-----+
| loan_int_rate     | 3116          | 9.56       |
| person_emp_length | 895           | 2.75       |
+-----+-----+-----+
2026-04-18 19:01:10 | WARNING   | Total missing values: 4,011 out of 32,581 r
ows.
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Completed checking for NULL values across d
atasets.

```

## Checking duplicate Value

```

In [9]: logger.info("Start checking duplicate values")
logger.info("-" * 80)
logger.info(f"Data duplicate Analysis...")
DataValidator.checkDuplicate(df=df, logger=logger)
logger.info("-" * 80)
logger.info("Duplicate value inspection successfully completed across dataset")

```

```

2026-04-18 19:01:10 | INFO      | Start checking duplicate values
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Data duplicate Analysis...
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Data duplicate Analysis...
2026-04-18 19:01:10 | INFO      | No duplicate rows found out of 32,581 total
rows.
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Duplicate value inspection successfully com
pleted across datasets.

```

## Checking Outlier Value

```

In [10]: logger.info("Start checking outlier dataset")
logger.info("-" * 80)
outlier_data = DataValidator.checkOutlier(df=df, dataset_name="Bank Marketin
list_feature=num_features, logger=

```

```
logger.info("-" * 80)
logger.info("Outlier checking successfully completed across datasets.")
```

```
2026-04-18 19:01:10 | INFO      | Start checking outlier dataset
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Checking outliers for Bank Marketing...
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Checking outliers for Bank Marketing...
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] person_age: 1,494 outliers
detected (lower=12.500, upper=40.500)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] person_income: 1,484 outli
ers detected (lower=-22550.000, upper=140250.000)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] person_emp_length: 853 out
liers detected (lower=-5.500, upper=14.500)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] loan_amnt: 1,689 outliers
detected (lower=-5800.000, upper=23000.000)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] loan_int_rate: 6 outliers
detected (lower=-0.455, upper=21.825)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] loan_percent_income: 651 o
utliers detected (lower=-0.120, upper=0.440)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] loan_to_income_ratio: 733
outliers detected (lower=-0.120, upper=0.438)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] cb_person_cred_hist_lengt
h: 1,142 outliers detected (lower=-4.500, upper=15.500)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] other_debt: 1,559 outliers
detected (lower=-8376.475, upper=28326.574)
2026-04-18 19:01:10 | WARNING   | [Bank Marketing] debt_to_income_ratio: 463
outliers detected (lower=-0.007, upper=0.681)
2026-04-18 19:01:10 | DEBUG     | [Bank Marketing] open_accounts: No outliers
detected
2026-04-18 19:01:10 | DEBUG     | [Bank Marketing] credit_utilization_ratio:
No outliers detected
2026-04-18 19:01:10 | INFO      | -----
-----
2026-04-18 19:01:10 | INFO      | Outlier checking successfully completed acr
oss datasets.
```

## □ Data Quality Insights: Missing Values, Duplicates, and Outliers

### Missing Values Analysis

- Missing values were identified in **2 features**:

- `loan_int_rate`: 3,116 missing (~9.56%) → significant and requires careful imputation.
- `person_emp_length`: 895 missing (~2.75%) → relatively minor but still needs handling.
- Total missing values: **4,011** across **32,581 records**.
- Overall, the dataset is largely complete, with missingness concentrated in specific financial attributes.

### Duplicate Records Analysis

- No duplicate rows were detected across the dataset.
- This ensures data integrity and prevents bias from repeated observations.

### Outlier Analysis

- Outliers are present across multiple numerical features, indicating high variability in borrower profiles.
  - Strong outliers detected in `person_age`, `person_income`, `loan_amnt`, and `other_debt`.
  - Financial ratio features such as `loan_percent_income`, `loan_to_income_ratio`, and `debt_to_income_ratio` also show notable outliers.
  - `person_emp_length` and `cb_person_cred_hist_length` contain unrealistic extreme values, suggesting potential data quality issues.
- Some features like `open_accounts` and `credit_utilization_ratio` show stable distributions with **no outliers detected**.
- Outliers in financial datasets often reflect real-world extreme cases (e.g., high debt or income), so **removal should be handled carefully**.

## Exploratory Data Analysis (EDA)

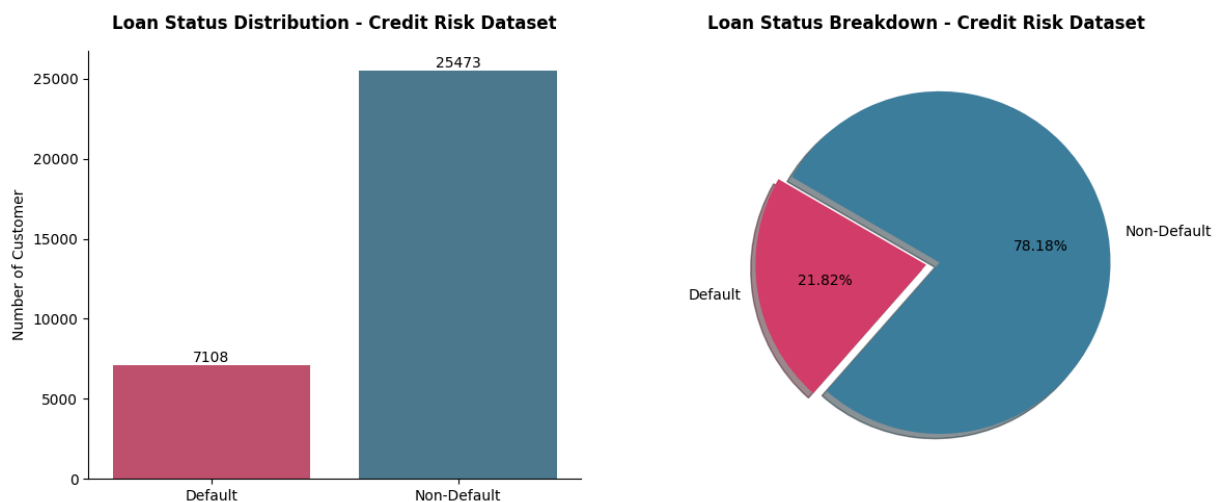
# Loan Status Distribution

```
In [11]: mapping = {0: "Non-Default", 1: "Default"}
df["loan_status"] = df["loan_status"].map(mapping)

order = ["Default", "Non-Default"]
loan_status_distribution = df[CFG.TARGET_FEATURE].value_counts().reindex(order)
fig, ax = plt.subplots(nrows=1, ncols=2, sharey=False, figsize=(12, 5))
n_color = ColorUtils.color(n_colors=df[CFG.TARGET_FEATURE].nunique())
# ax[0]
sns.countplot(data=df, x = CFG.TARGET_FEATURE, palette=n_color,
               ax=ax[0], order=order)
ax[0].set_title(f"Loan Status Distribution - {CFG.DATASET_NAME}", weight="bold")
for container in ax[0].containers:
    ax[0].bar_label(container, fmt="%d", label_type="edge", fontsize=10)
ax[0].set_ylabel("Number of Customer")
ax[0].set_xlabel("")
sns.despine(ax=ax[0], top=True, right=True, left=False, bottom=False)

# ax[1]
ax[1].pie(
    loan_status_distribution,
    labels=order,
    colors = n_color,
    autopct = "%1.2f%%",
    startangle = 150,
    explode = (0, 0.08),
    shadow= True
)
ax[1].set_title(f"Loan Status Breakdown - {CFG.DATASET_NAME}", weight="bold")

plt.tight_layout()
plt.subplots_adjust(hspace=0.3, wspace=0.2)
plt.show()
```





## □ Insight

| Dataset     | Non-Default (0) | Default (1) | Default Rate |
|-------------|-----------------|-------------|--------------|
| Credit Risk | 25,473          | 7,108       | 21.82%       |

- **Majority class:** `Non-Default` (~78.18%) → most customers successfully repay their loans.
- **Minority class:** `Default` (~21.82%) → represents borrowers with repayment risk.
- The dataset shows a **moderate class imbalance**, with roughly a **1:4 ratio** between default and non-default cases.
- The default rate (~22%) is **significantly higher** than typical banking datasets → providing a stronger signal for model learning.
- This imbalance suggests the need for:
  - Evaluation metrics beyond accuracy (e.g., **ROC-AUC**, **F1-score**)
  - Potential use of **class weighting** or **resampling techniques**

## Numerical Feature Distributions

```
In [12]: def plot_numerical_features(df=df, numerical_features = num_features):
        """
        Plot distribution of numerical features using Histogram and Boxplot.
        """

        # Create subplot grid
        fig, ax = plt.subplots(len(numerical_features), 2, figsize=(12, len(numerical_features)))
        # Ensure ax is always 2D (important if only 1 feature)
        ax = np.atleast_2d(ax)

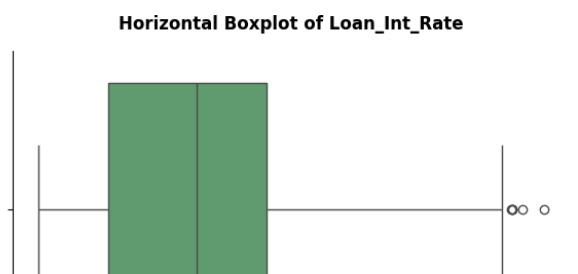
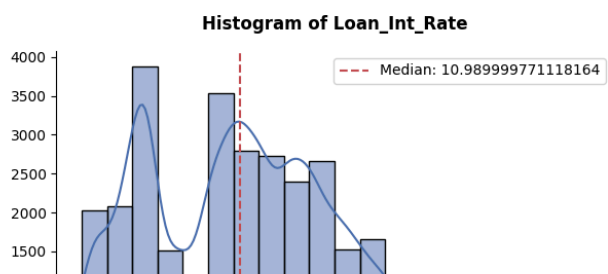
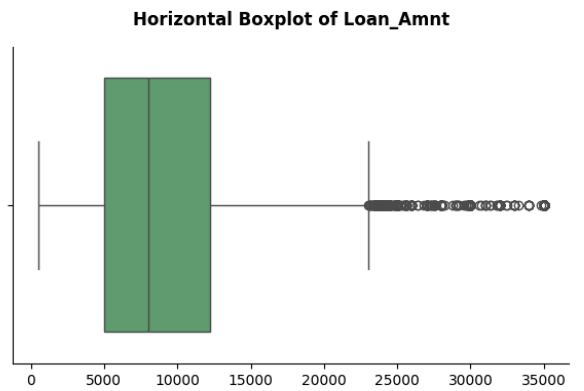
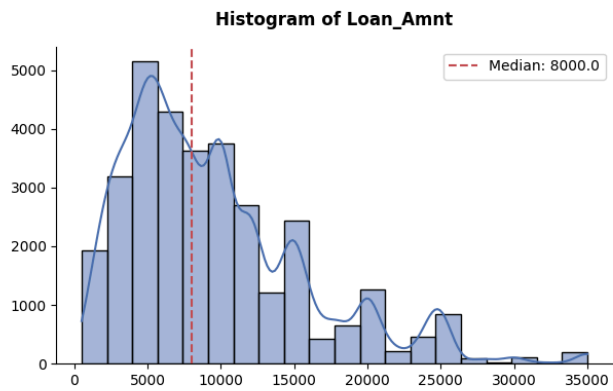
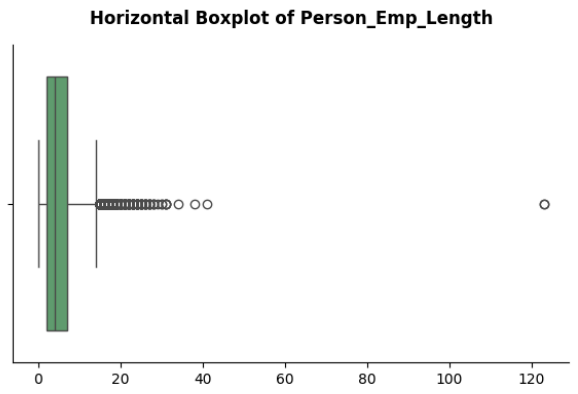
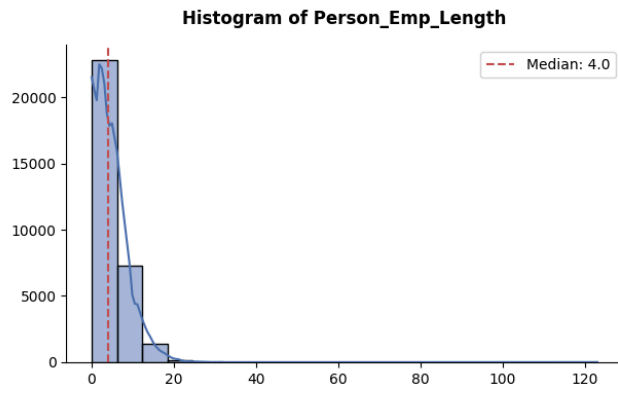
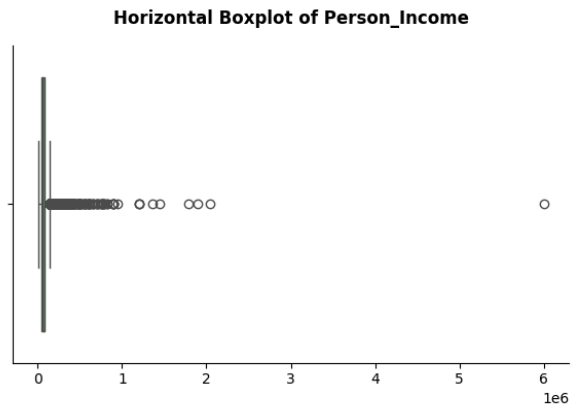
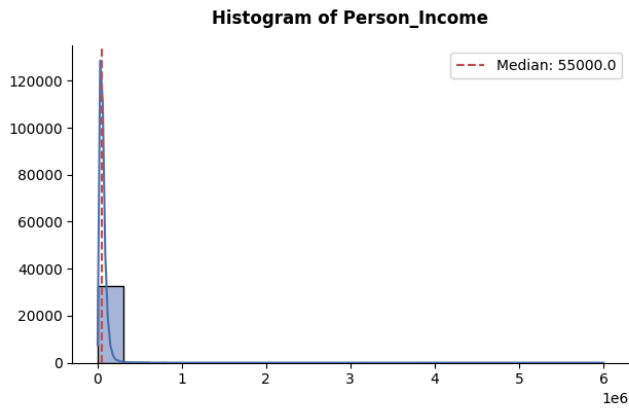
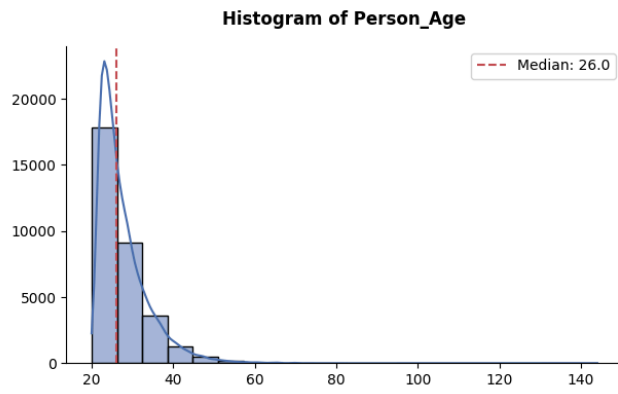
        for i, feature in enumerate(numerical_features):
            # Histogram + KDE
            sns.histplot(data=df[feature], color="#4C72B0", bins = 20, kde=True,
                        ax=ax[i, 0].set_title(f"Histogram of {feature.title()}", weight="bold",
                        ax[i, 0].set_xlabel(""),
                        ax[i, 0].set_ylabel(""))

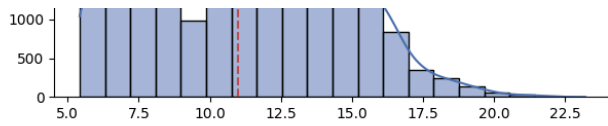
            # Draw vertical line for median
            ax[i, 0].axvline(df[feature].median(), color="#C44E52", linestyle="--")
            ax[i, 0].legend(loc="upper right")
            sns.despine(ax=ax[i, 0], top=True, right=True, left=False, bottom=False)
```

```
# Horizontal Boxplot
sns.boxplot(data=df[feature], color="#55A868", orient="h", ax=ax[i,
ax[i, 1].set_title(f"Horizontal Boxplot of {feature.title()}", weight
ax[i, 1].set_xlabel("")
sns.despine(ax=ax[i, 1], top=True, right=True, left=False, bottom=False)

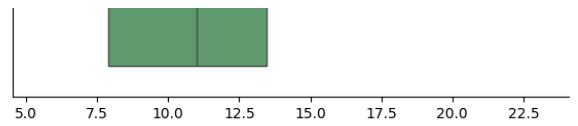
# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()

plot_numerical_features()
```

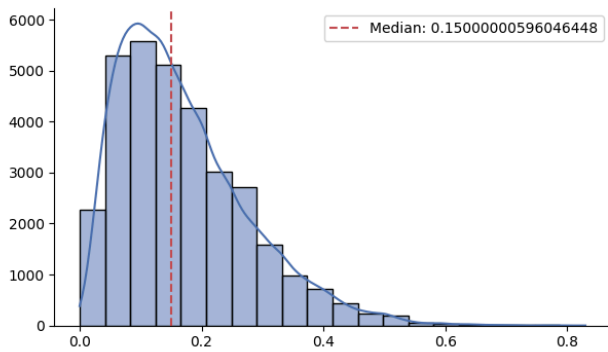




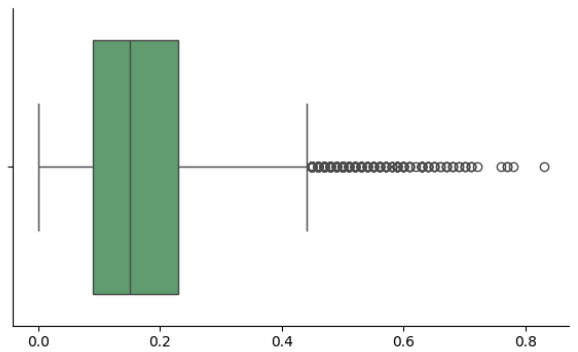
Histogram of Loan\_Percent\_Income



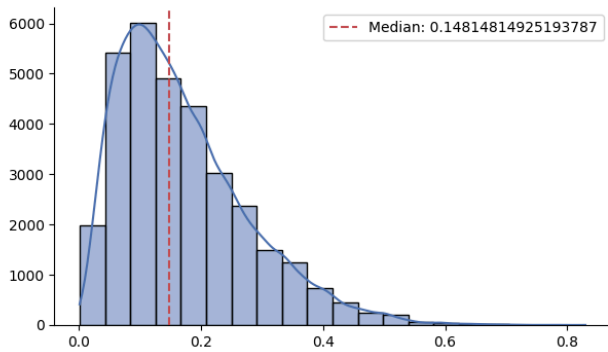
Horizontal Boxplot of Loan\_Percent\_Income



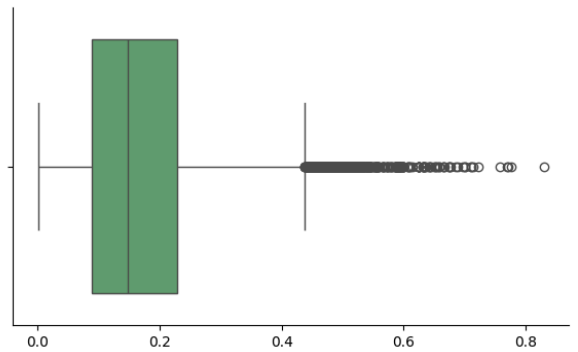
Histogram of Loan\_To\_Income\_Ratio



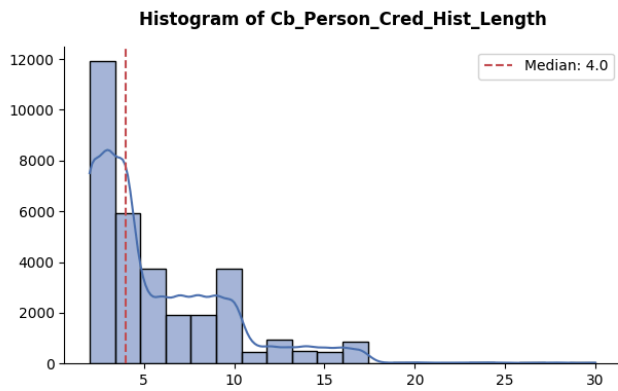
Horizontal Boxplot of Loan\_To\_Income\_Ratio



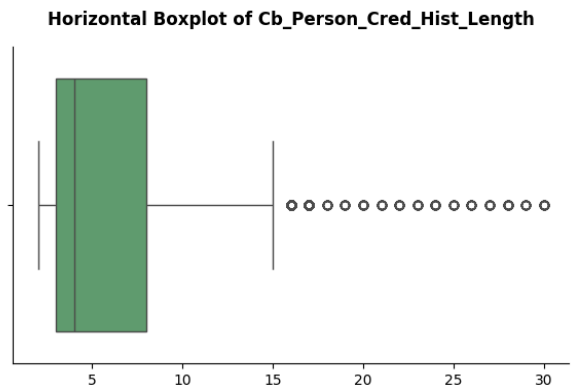
Histogram of Cb\_Person\_Cred\_Hist\_Length



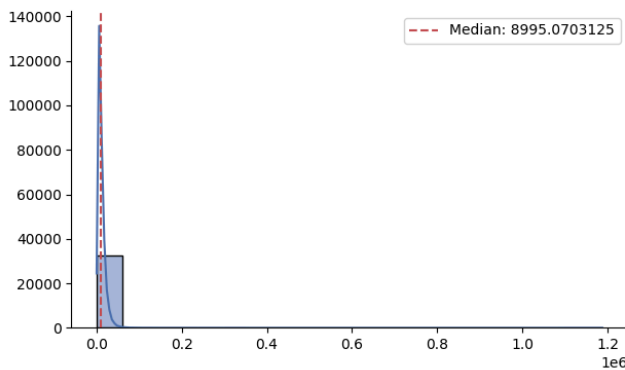
Horizontal Boxplot of Cb\_Person\_Cred\_Hist\_Length



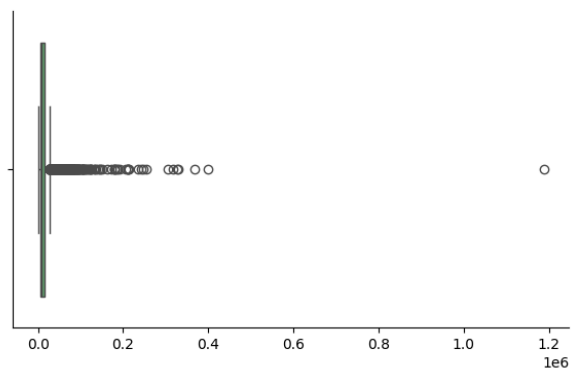
Histogram of Other\_Debt



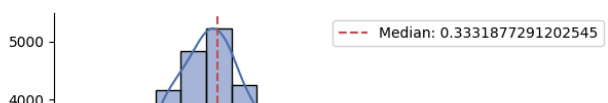
Horizontal Boxplot of Other\_Debt

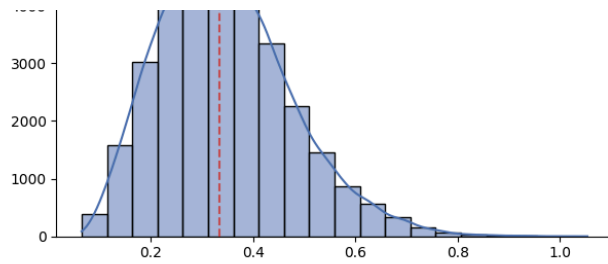


Histogram of Debt\_To\_Income\_Ratio

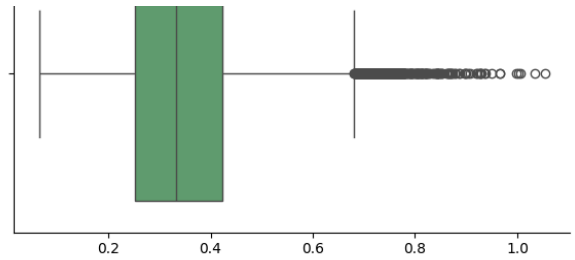


Horizontal Boxplot of Debt\_To\_Income\_Ratio

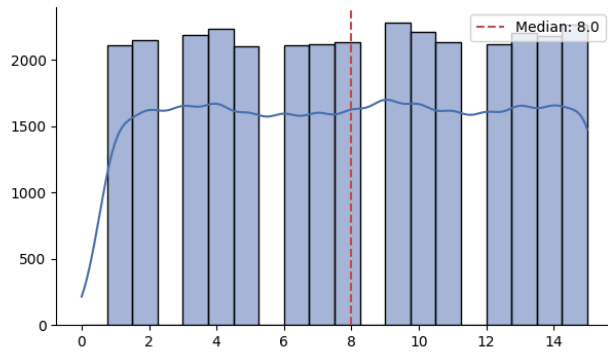




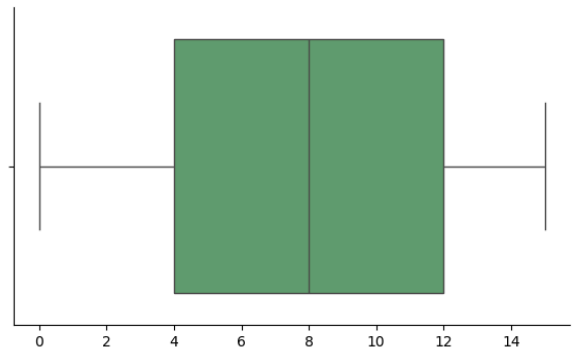
**Histogram of Open\_Accounts**



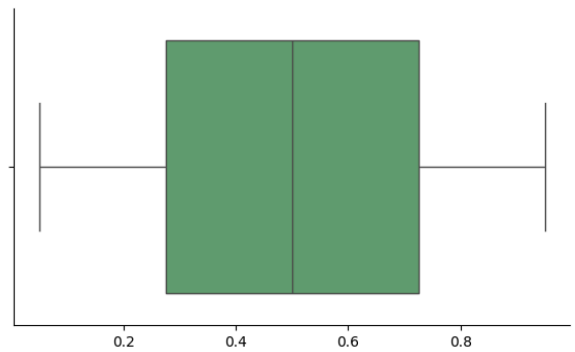
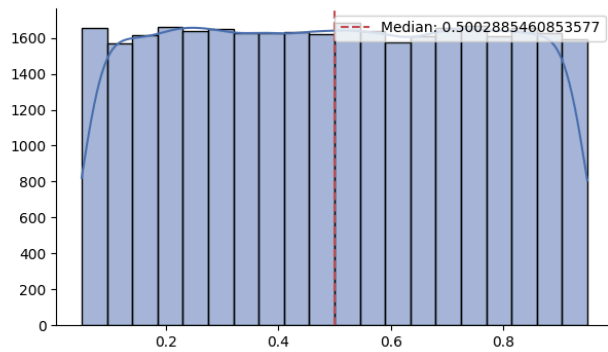
**Horizontal Boxplot of Open\_Accounts**



**Histogram of Credit\_Utilization\_Ratio**



**Horizontal Boxplot of Credit\_Utilization\_Ratio**



```
In [13]: skew_feature = DataValidator.check_skewness(data=df, dataset_name= CFG.DATAS
                                                    numerical_features=num_features,
                                                    logger.info("-" * 80)
                                                    logger.info("Checking skewness successfully completed the dataset.")
```

```

2026-04-18 19:01:15 | INFO | Skewness analysis for Credit Risk Dataset
2026-04-18 19:01:15 | INFO | Total numerical features checked: 12
2026-04-18 19:01:15 | DEBUG | -----
2026-04-18 19:01:15 | DEBUG | Feature | Skewness
| Remark
2026-04-18 19:01:15 | DEBUG | -----
2026-04-18 19:01:15 | DEBUG | person_income | +32.8653
| Highly skewed
2026-04-18 19:01:15 | DEBUG | other_debt | +28.4619
| Highly skewed
2026-04-18 19:01:15 | DEBUG | person_emp_length | +2.6145
| Highly skewed
2026-04-18 19:01:15 | DEBUG | person_age | +2.5814
| Highly skewed
2026-04-18 19:01:15 | DEBUG | cb_person_cred_hist_length | +1.6618
| Highly skewed
2026-04-18 19:01:15 | DEBUG | loan_amnt | +1.1925
| Highly skewed
2026-04-18 19:01:15 | DEBUG | loan_to_income_ratio | +1.0651
| Highly skewed
2026-04-18 19:01:15 | DEBUG | loan_percent_income | +1.0647
| Highly skewed
2026-04-18 19:01:15 | DEBUG | debt_to_income_ratio | +0.6195
| Moderately skewed
2026-04-18 19:01:15 | DEBUG | loan_int_rate | +0.2086
| Approximately symmetric
2026-04-18 19:01:15 | DEBUG | open_accounts | -0.0071
| Approximately symmetric
2026-04-18 19:01:15 | DEBUG | credit_utilization_ratio | -0.0001
| Approximately symmetric
2026-04-18 19:01:15 | DEBUG | -----
2026-04-18 19:01:15 | INFO | 9 features are skewed (|skew| > 0.5).
2026-04-18 19:01:15 | INFO | Recommendation: Consider log / Box-Cox / Ye
o-Johnson transformation.
2026-04-18 19:01:15 | INFO | -----
2026-04-18 19:01:15 | INFO | Checking skewness successfully completed th
e dataset.

```

## 📊 Insight

- Most numerical features in the dataset are **right-skewed**, indicating that values are concentrated at lower ranges with **long tails and extreme outliers**.
- **9 out of 12 features** show significant skewness (`|skew| > 0.5`), highlighting strong non-normal distributions.

- Features such as `person_income` and `other_debt` exhibit **extreme skewness** (skew > 25), suggesting the presence of very large outliers and heavy-tailed distributions.
- Moderately skewed features include `debt_to_income_ratio`, indicating some imbalance but less extreme compared to financial amount variables.
- Features like `loan_int_rate`, `open_accounts`, and `credit_utilization_ratio` are **approximately symmetric**, making them more stable for modeling.
- The strong skewness in financial variables reflects real-world behavior, where a small number of customers have **exceptionally high income or debt levels**.
- **Data transformation techniques** such as **log transformation**, **Box-Cox**, or **Yeo-Johnson** are recommended to reduce skewness and improve model performance.

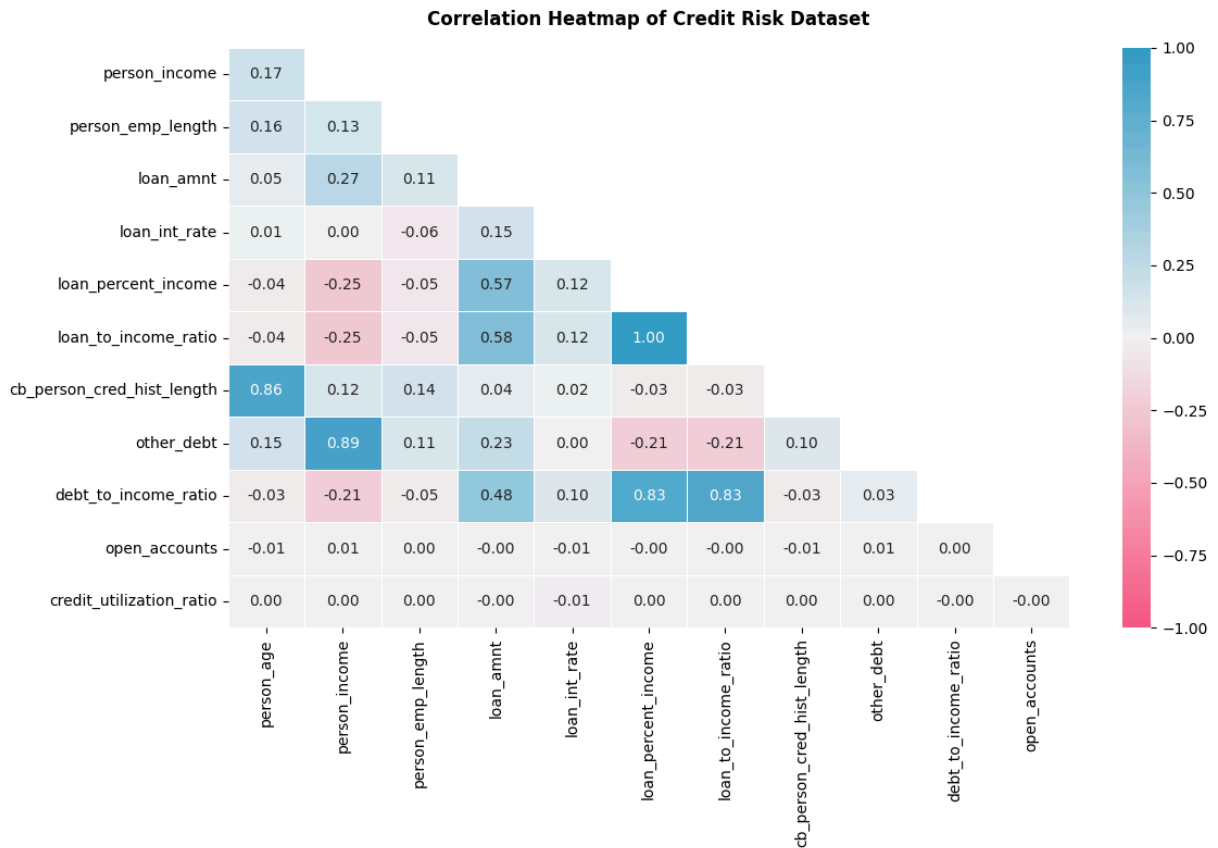
## Correlation Analysis of Numerical Features

```
In [14]: corr_matrix = df[num_features].corr(numeric_only=True)
# one_like can build a matrix of boolean(True, False) with the same shape as
ones_corr = np.ones_like(corr_matrix, dtype=bool)
mask = np.triu(ones_corr)
adjusted_mask = mask[1:, :-1]
adjusted_cereal_corr = corr_matrix.iloc[1:, :-1]

fig, ax = plt.subplots(figsize = (12, 8))
# That method uses HUSL colors, so you need hue, saturation, and lightness.
# I used hsluv.org to select the colors of this chart.
cmap = sns.diverging_palette(0, 230, 90, 60, as_cmap=True)

sns.heatmap(data=adjusted_cereal_corr, mask=adjusted_mask,
            annot=True, fmt=".2f", cmap=cmap,
            vmin=-1, vmax=1, linecolor="white", linewidths=0.5)

title = f"Correlation Heatmap of {CFG.DATASET_NAME}"
ax.set_title(title, loc="center", fontsize=12, weight="bold", pad=15)
plt.tight_layout()
plt.show()
```



## Correlation Insights

- Most numerical variables exhibit **weak correlations**, indicating that customer financial behaviors are largely independent.
- A **very strong correlation** is observed between `loan_percent_income` and `loan_to_income_ratio` (~1.00), confirming that these features are **redundant** and carry nearly identical information.
- `debt_to_income_ratio` shows a **strong correlation** (~0.83) with both `loan_percent_income` and `loan_to_income_ratio`, indicating overlapping financial risk signals.
- `person_income` and `other_debt` are also **highly correlated** (~0.89), suggesting that higher-income individuals tend to carry higher absolute debt levels.
- A strong relationship between `person_age` and `cb_person_cred_hist_length` (~0.86) reflects expected real-world behavior, as older individuals typically have longer credit histories.



- Moderate correlations exist between `loan_amnt` and financial ratios ( $\sim 0.48$ – $0.58$ ), indicating that larger loans are associated with higher financial burden.
- Features such as `open_accounts` and `credit_utilization_ratio` show **near-zero correlation** with other variables, suggesting they provide **unique, independent signals** for modeling.
- Overall, while most features are independent, a few clusters of correlated variables suggest potential **multicollinearity**, which should be addressed through feature selection or transformation.

## Categorical Feature Distributions

```
In [15]: def plot_categorical_distribution_across_datasets(df: pd.DataFrame, feature:
        """
        Display distribution summary of a categorical feature with styled table.

        Parameters
        -----
        df : pd.DataFrame
            Input dataset.
        feature : str
            Categorical column to analyze.
        """

        # --- Summarize categorical feature ---
        summary = (
            df[feature]
            .value_counts()
            .rename_axis("Category")
            .reset_index(name="Data_Count")
        )

        # Calculate percentage distribution
        summary["Data_%"] = round(summary["Data_Count"] / summary["Data_Count"].fi

        # Ensure Category column is string
        summary["Category"] = summary["Category"].astype(str)

        # Fill NA if any
        summary[["Data_Count", "Data_%"]] = summary[["Data_Count", "Data_%"]].fi

        # ---- Styling with color gradient ----
        styled_table = (
            summary.style
            .background_gradient(cmap="Blues", subset=["Data_Count"])
            .background_gradient(cmap="Blues", subset=["Data_%"])
```

```

        .format({
            "Data_Count": "{:,}",
            "Data_%": "{:.2f}%"
        })
        .set_caption(f"Categorical Feature Distribution: {feature.title()}")
        .set_table_styles([
            {"selector": "caption", "props": [("font-size", "14px"), ("font-
            {"selector": "th", "props": [("text-align", "center")]}},
            {"selector": "td", "props": [("text-align", "center")]}
        ])
    )

    display(styled_table)
    print("*" * 80)

# ----- Run for all categorical features -----
for feature in cat_features:
    plot_categorical_distribution_across_datasets(df=df, feature=feature)

```

### Categorical Feature Distribution:

#### Person\_Home\_Ownership

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | RENT     | 16,446     | 50.48% |
| 1 | MORTGAGE | 13,444     | 41.26% |
| 2 | OWN      | 2,584      | 7.93%  |
| 3 | OTHER    | 107        | 0.33%  |

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### Categorical Feature Distribution:

#### Loan\_Intent

|   | Category          | Data_Count | Data_% |
|---|-------------------|------------|--------|
| 0 | EDUCATION         | 6,453      | 19.81% |
| 1 | MEDICAL           | 6,071      | 18.63% |
| 2 | VENTURE           | 5,719      | 17.55% |
| 3 | PERSONAL          | 5,521      | 16.95% |
| 4 | DEBTCONSOLIDATION | 5,212      | 16.00% |
| 5 | HOMEIMPROVEMENT   | 3,605      | 11.06% |

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### Categorical Feature Distribution:

#### Loan\_Grade

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | A        | 10,777     | 33.08% |
| 1 | B        | 10,451     | 32.08% |
| 2 | C        | 6,458      | 19.82% |
| 3 | D        | 3,626      | 11.13% |
| 4 | E        | 964        | 2.96%  |
| 5 | F        | 241        | 0.74%  |
| 6 | G        | 64         | 0.20%  |

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### Categorical Feature Distribution:

#### Cb\_Person\_Default\_On\_File

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | N        | 26,836     | 82.37% |
| 1 | Y        | 5,745      | 17.63% |

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### Categorical Feature Distribution:

#### Gender

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | Male     | 16,371     | 50.25% |
| 1 | Female   | 16,210     | 49.75% |

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### Categorical Feature Distribution:

#### Marital\_Status

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | Single   | 16,368     | 50.24% |
| 1 | Married  | 11,393     | 34.97% |
| 2 | Divorced | 3,173      | 9.74%  |
| 3 | Widowed  | 1,647      | 5.06%  |

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### Categorical Feature Distribution:

#### Education\_Level

|   | Category    | Data_Count | Data_% |
|---|-------------|------------|--------|
| 0 | High School | 13,185     | 40.47% |
| 1 | Bachelor    | 11,390     | 34.96% |
| 2 | Master      | 6,508      | 19.97% |
| 3 | PhD         | 1,498      | 4.60%  |

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### Categorical Feature Distribution:

#### Employment\_Type

|   | Category      | Data_Count | Data_% |
|---|---------------|------------|--------|
| 0 | Full-time     | 19,473     | 59.77% |
| 1 | Part-time     | 6,510      | 19.98% |
| 2 | Self-employed | 4,926      | 15.12% |
| 3 | Unemployed    | 1,672      | 5.13%  |

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### Categorical Feature Distribution:

#### Country

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | UK       | 10,944     | 33.59% |
| 1 | USA      | 10,852     | 33.31% |
| 2 | Canada   | 10,785     | 33.10% |

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**Categorical Feature Distribution:**

| State |            |            |        |
|-------|------------|------------|--------|
|       | Category   | Data_Count | Data_% |
| 0     | BC         | 3,679      | 11.29% |
| 1     | California | 3,679      | 11.29% |
| 2     | England    | 3,654      | 11.22% |
| 3     | Scotland   | 3,648      | 11.20% |
| 4     | Wales      | 3,642      | 11.18% |
| 5     | Texas      | 3,608      | 11.07% |
| 6     | New York   | 3,565      | 10.94% |
| 7     | Ontario    | 3,553      | 10.91% |
| 8     | Quebec     | 3,553      | 10.91% |

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**Categorical Feature Distribution: City**

|    | Category      | Data_Count | Data_% |
|----|---------------|------------|--------|
| 0  | Victoria      | 1,852      | 5.68%  |
| 1  | London        | 1,851      | 5.68%  |
| 2  | San Francisco | 1,841      | 5.65%  |
| 3  | Glasgow       | 1,841      | 5.65%  |
| 4  | Los Angeles   | 1,838      | 5.64%  |
| 5  | Cardiff       | 1,831      | 5.62%  |
| 6  | Vancouver     | 1,827      | 5.61%  |
| 7  | Houston       | 1,811      | 5.56%  |
| 8  | Swansea       | 1,811      | 5.56%  |
| 9  | Edinburgh     | 1,807      | 5.55%  |
| 10 | Manchester    | 1,803      | 5.53%  |
| 11 | Ottawa        | 1,802      | 5.53%  |
| 12 | Montreal      | 1,799      | 5.52%  |
| 13 | Dallas        | 1,797      | 5.52%  |
| 14 | Buffalo       | 1,796      | 5.51%  |
| 15 | New York City | 1,769      | 5.43%  |
| 16 | Quebec City   | 1,754      | 5.38%  |
| 17 | Toronto       | 1,751      | 5.37%  |

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### Categorical Feature Distribution:

#### Loan\_Term\_Months

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | 36       | 12,944     | 39.73% |
| 1 | 60       | 9,923      | 30.46% |
| 2 | 24       | 6,374      | 19.56% |
| 3 | 12       | 3,340      | 10.25% |

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### Categorical Feature Distribution:

#### Past\_Delinquencies

|   | Category | Data_Count | Data_% |
|---|----------|------------|--------|
| 0 | 0        | 19,702     | 60.47% |
| 1 | 1        | 9,829      | 30.17% |
| 2 | 2        | 2,586      | 7.94%  |
| 3 | 3        | 405        | 1.24%  |
| 4 | 4        | 54         | 0.17%  |
| 5 | 5        | 4          | 0.01%  |
| 6 | 6        | 1          | 0.00%  |

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## □ Categorical Feature Distributions

- The dataset is dominated by **non-homeowners**, with most customers either **RENT (~50%)** or **MORTGAGE (~41%)**, indicating moderate financial leverage.
- Loan purposes are well distributed, with **education, medical, and venture** loans being the most common (~17-20% each), suggesting diverse borrowing needs.
- The majority of loans fall into **high-quality grades (A & B ~65%)**, indicating that most borrowers are considered low-risk by lending standards.

- Most customers have **no prior default history (~82%)**, reinforcing a generally reliable borrower base.
- The dataset is **gender-balanced (~50/50)**, minimizing gender bias in modeling.
- **Single individuals (~50%)** dominate the dataset, followed by married customers (~35%), suggesting varying financial responsibilities.
- Most borrowers have **mid-level education**, with **High School (~40%)** and **Bachelor (~35%)** being the most common.
- **Full-time employment (~60%)** is the dominant employment type, indicating stable income for the majority of customers.
- Geographically, the dataset is **evenly distributed across countries (UK, USA, Canada)**, providing balanced regional representation.
- State and city distributions are relatively **uniform**, suggesting no strong geographic concentration or bias.
- **Loan terms** are concentrated around **36 months (~40%)** and **60 months (~30%)**, reflecting standard lending durations.
- Most customers have **no past delinquencies (~60%)**, while a smaller portion (~30%) has only **1 prior delinquency**, indicating generally good repayment behavior.

## Bivariate Analysis

```
In [16]: for feature in num_features:
display(HTML(f"<h2 style='text-align:center; font-size:22px; color:blue;
StatUtils.analyze_numeric_feature_by_target(feature=feature, df=df, targ
```

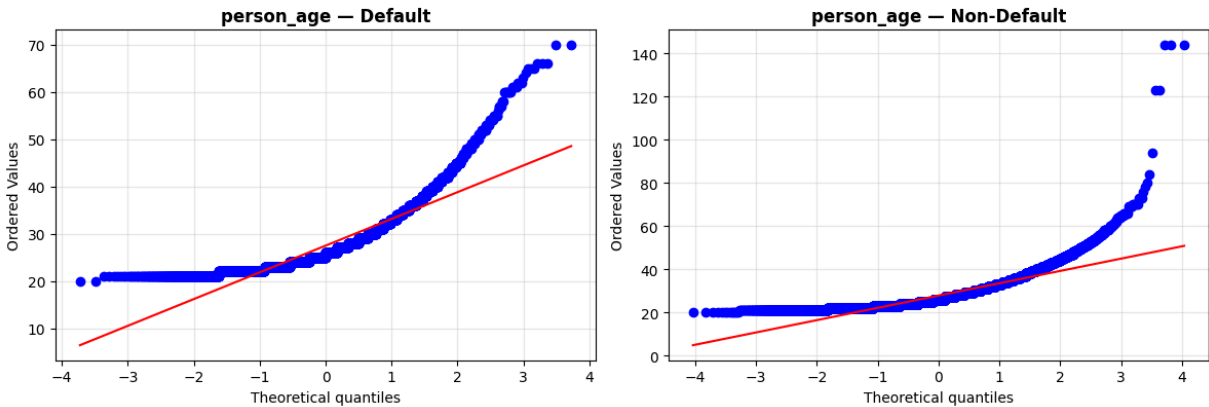
### Distribution of Person\_Age by Loan Status

- **Overall Mean:** 27.73
- **Overall Median:** 26.0
- **Overall Std:** 6.35

|   | loan_status | Count | Mean      | Median    | Std      |
|---|-------------|-------|-----------|-----------|----------|
| 0 | Non-Default | 25473 | 27.807129 | 26.000000 | 6.369669 |
| 1 | Default     | 7108  | 27.474676 | 26.000000 | 6.263642 |

```
2026-04-18 19:01:15 | INFO      | Checking normality of feature 'person_age'
by groups of 'loan_status'...
2026-04-18 19:01:15 | INFO      | Evaluating distribution characteristics (sk
ewness & kurtosis)...
2026-04-18 19:01:15 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:15 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:15 | DEBUG     | [loan_status=Default] skew=1.9864, kurtosis
=5.5547 → Highly skewed, Heavy/light tails
2026-04-18 19:01:15 | DEBUG     | [loan_status=Non-Default] skew=2.7400, kurt
osis=21.9279 → Highly skewed, Heavy/light tails
2026-04-18 19:01:15 | INFO      | Generating Q-Q plots for visual assessmen
t...
```

Q-Q Plots of person\_age by loan\_status

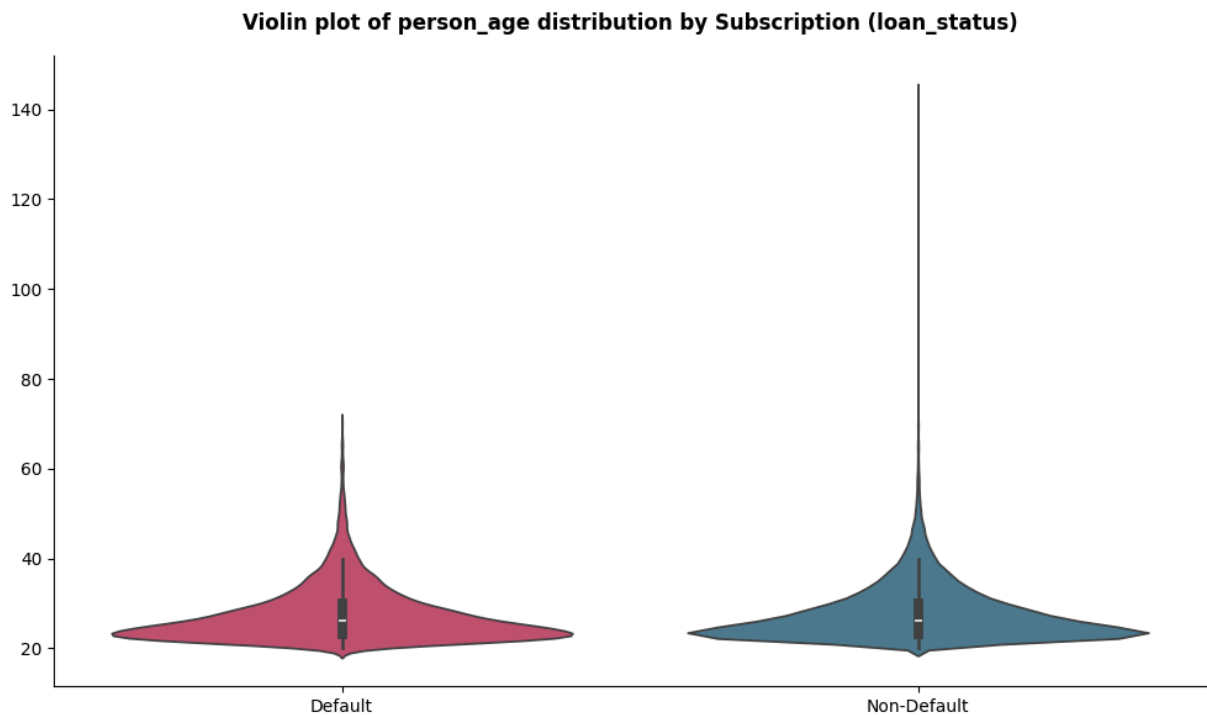


Skewness & Kurtosis of person\_age by loan\_status

|   | Feature    | Group       | Skewness | Kurtosis | Remark                           |
|---|------------|-------------|----------|----------|----------------------------------|
| 0 | person_age | Default     | 1.9864   | 5.5547   | Highly skewed, Heavy/light tails |
| 1 | person_age | Non-Default | 2.7400   | 21.9279  | Highly skewed, Heavy/light tails |

```
2026-04-18 19:01:16 | WARNING   | Normality assumption violated in at least o
ne group.
2026-04-18 19:01:16 | INFO      | Recommendation: Use non-parametric tests (K
ruskal-Wallis / Mann-Whitney U).
2026-04-18 19:01:16 | INFO      | Running Mann-Whitney U Test: 'person_age' b
y 'loan_status'...
2026-04-18 19:01:16 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:16 | INFO      | U statistic: 86323515.0
2026-04-18 19:01:16 | INFO      | p-value    : 0.000000
2026-04-18 19:01:16 | INFO      | Result: p-value ≤ 0.05 → Reject H0
2026-04-18 19:01:16 | INFO      | Conclusion: Significant difference between
the two groups.
2026-04-18 19:01:16 | INFO      | Effect size interpretation: Negligible
```





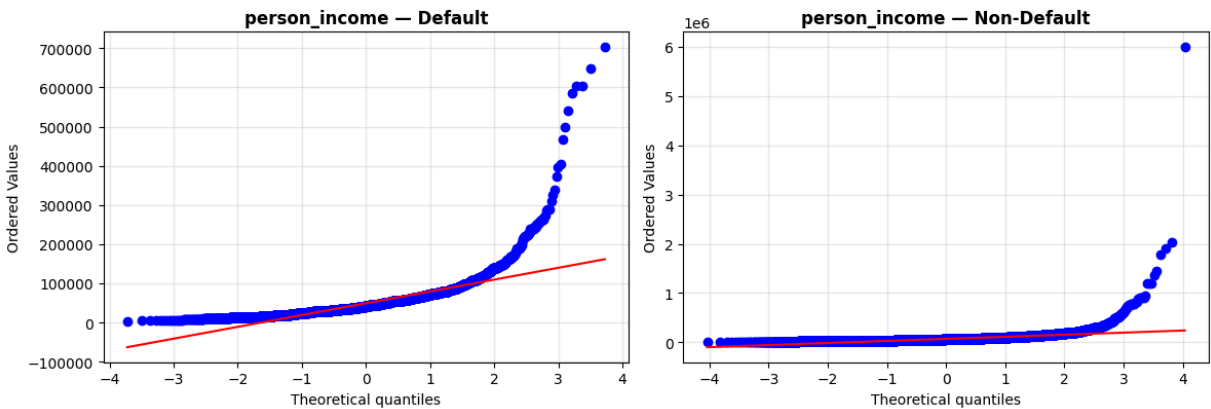
## Distribution of Person\_Income by Loan Status

- **Overall Mean:** 66074.85
- **Overall Median:** 55000.0
- **Overall Std:** 61983.12

|   | loan_status | Count | Mean         | Median       | Std          |
|---|-------------|-------|--------------|--------------|--------------|
| 0 | Non-Default | 25473 | 70804.361559 | 55000.000000 | 66568.802606 |
| 1 | Default     | 7108  | 49125.652203 | 49800.000000 | 60737.6527   |

```
2026-04-18 19:01:16 | INFO      | Checking normality of feature 'person_income' by groups of 'loan_status'...
2026-04-18 19:01:16 | INFO      | Evaluating distribution characteristics (skewness & kurtosis)...
2026-04-18 19:01:16 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:16 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:16 | DEBUG     | [loan_status=Default] skew=5.4699, kurtosis=62.0899 → Highly skewed, Heavy/light tails
2026-04-18 19:01:16 | DEBUG     | [loan_status=Non-Default] skew=33.5291, kurtosis=2577.7449 → Highly skewed, Heavy/light tails
2026-04-18 19:01:16 | INFO      | Generating Q-Q plots for visual assessment...
```

Q-Q Plots of person\_income by loan\_status



Skewness & Kurtosis of person\_income by loan\_status

| Feature | Group                    | Skewness | Kurtosis  | Remark                           |
|---------|--------------------------|----------|-----------|----------------------------------|
| 0       | person_incomeDefault     | 5.4699   | 62.0899   | Highly skewed, Heavy/light tails |
| 1       | person_incomeNon-Default | 33.5291  | 2577.7449 | Highly skewed, Heavy/light tails |

2026-04-18 19:01:16 | WARNING | Normality assumption violated in at least one group.

2026-04-18 19:01:16 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-04-18 19:01:16 | INFO | Running Mann-Whitney U Test: 'person\_income' by 'loan\_status'...

2026-04-18 19:01:16 | DEBUG | Number of groups detected: 2

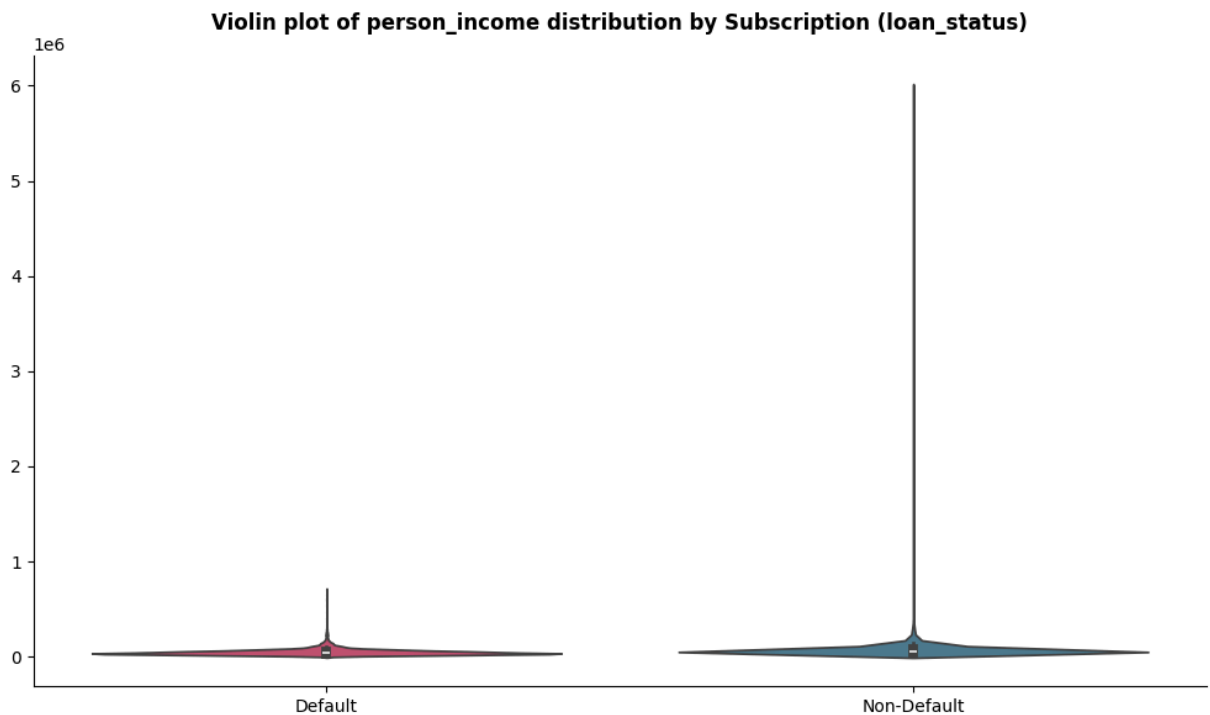
2026-04-18 19:01:16 | INFO | U statistic: 56093141.5

2026-04-18 19:01:16 | INFO | p-value : 0.000000

2026-04-18 19:01:16 | INFO | Result: p-value ≤ 0.05 → Reject H0

2026-04-18 19:01:16 | INFO | Conclusion: Significant difference between the two groups.

2026-04-18 19:01:16 | INFO | Effect size interpretation: Small



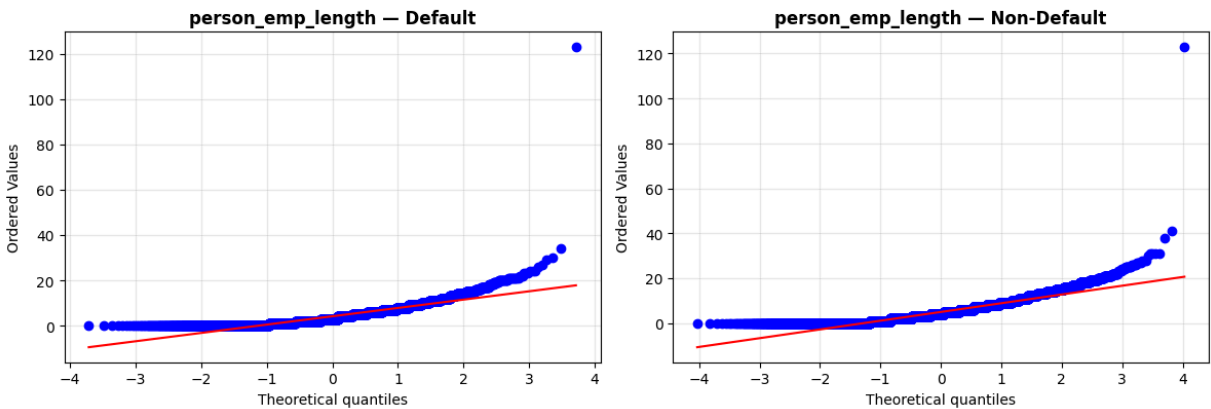
## Distribution of Person\_Emp\_Length by Loan Status

- **Overall Mean:** 4.79
- **Overall Median:** 4.0
- **Overall Std:** 4.14

|   | loan_status | Count | Mean     | Median   | Std      |
|---|-------------|-------|----------|----------|----------|
| 0 | Non-Default | 24860 | 4.968745 | 4.000000 | 4.129022 |
| 1 | Default     | 6826  | 4.137562 | 3.000000 | 4.126956 |

```
2026-04-18 19:01:17 | INFO      | Checking normality of feature 'person_emp_l
length' by groups of 'loan_status'...
2026-04-18 19:01:17 | INFO      | Evaluating distribution characteristics (sk
ewness & kurtosis)...
2026-04-18 19:01:17 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:17 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:17 | DEBUG     | [loan_status=Default] skew=4.7058, kurtosis
=102.7128 → Highly skewed, Heavy/light tails
2026-04-18 19:01:17 | DEBUG     | [loan_status=Non-Default] skew=2.0762, kurt
osis=28.7722 → Highly skewed, Heavy/light tails
2026-04-18 19:01:17 | INFO      | Generating Q-Q plots for visual assessmen
t...
```

Q-Q Plots of person\_emp\_length by loan\_status



Skewness & Kurtosis of person\_emp\_length by loan\_status

| Feature | Group                           | Skewness | Kurtosis | Remark                           |
|---------|---------------------------------|----------|----------|----------------------------------|
| 0       | person_emp_length - Default     | 4.7058   | 102.7128 | Highly skewed, Heavy/light tails |
| 1       | person_emp_length - Non-Default | 2.0762   | 28.7722  | Highly skewed, Heavy/light tails |

2026-04-18 19:01:17 | WARNING | Normality assumption violated in at least one group.

2026-04-18 19:01:17 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-04-18 19:01:17 | INFO | Running Mann-Whitney U Test: 'person\_emp\_length' by 'loan\_status'...

2026-04-18 19:01:17 | DEBUG | Number of groups detected: 2

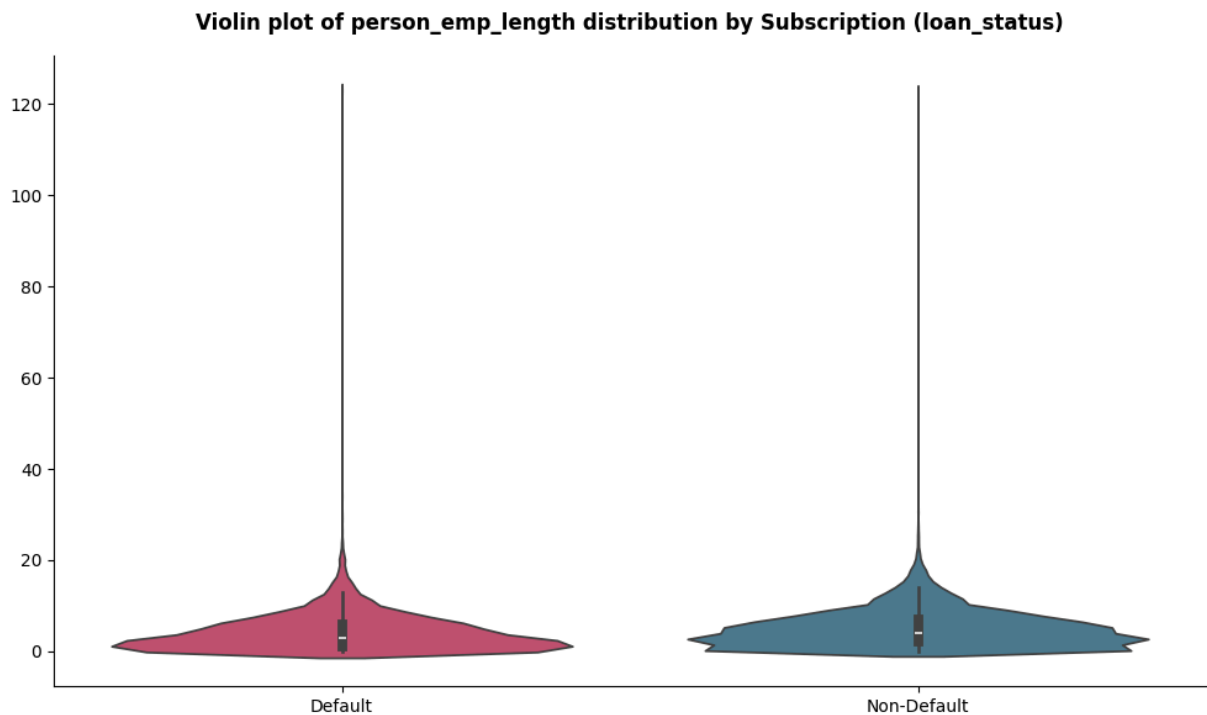
2026-04-18 19:01:17 | INFO | U statistic: 73273849.0

2026-04-18 19:01:17 | INFO | p-value : 0.000000

2026-04-18 19:01:17 | INFO | Result: p-value ≤ 0.05 → Reject H0

2026-04-18 19:01:17 | INFO | Conclusion: Significant difference between the two groups.

2026-04-18 19:01:17 | INFO | Effect size interpretation: Negligible



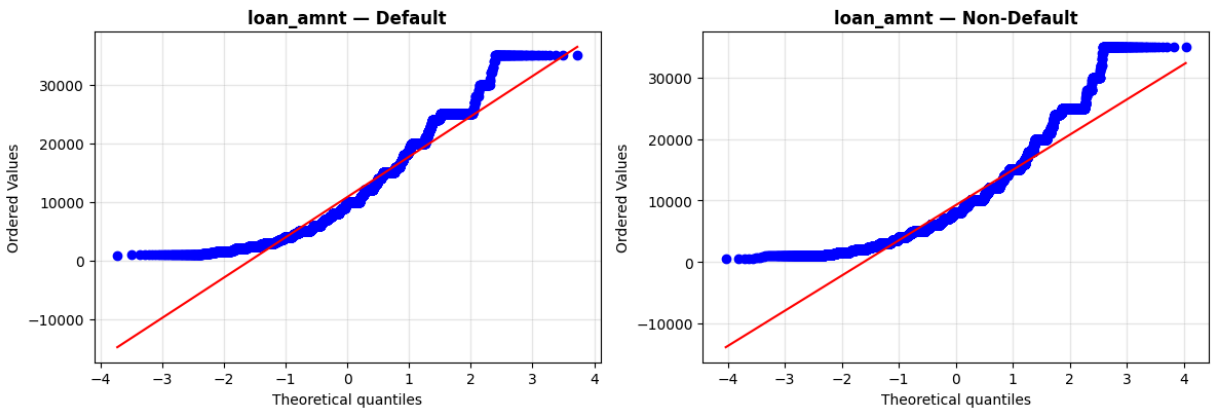
## Distribution of Loan\_Amnt by Loan Status

- **Overall Mean:** 9589.37
- **Overall Median:** 8000.0
- **Overall Std:** 6322.09

|   | loan_status | Count | Mean         | Median      | Std        |
|---|-------------|-------|--------------|-------------|------------|
| 0 | Default     | 7108  | 10850.502956 | 8000.000000 | 143.345344 |
| 1 | Non-Default | 25473 | 9237.464178  | 8000.000000 | 26.375457  |

```
2026-04-18 19:01:17 | INFO      | Checking normality of feature 'loan_amnt' by groups of 'loan_status'...
2026-04-18 19:01:17 | INFO      | Evaluating distribution characteristics (skewness & kurtosis)...
2026-04-18 19:01:17 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:17 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:17 | DEBUG     | [loan_status=Default] skew=0.8989, kurtosis=0.3265 → Moderately skewed, Normal tails
2026-04-18 19:01:17 | DEBUG     | [loan_status=Non-Default] skew=1.2699, kurtosis=1.8433 → Highly skewed, Heavy/light tails
2026-04-18 19:01:17 | INFO      | Generating Q-Q plots for visual assessment...
```

Q-Q Plots of loan\_amnt by loan\_status



Skewness & Kurtosis of loan\_amnt by loan\_status

| Feature     | Group       | Skewness | Kurtosis | Remark                           |
|-------------|-------------|----------|----------|----------------------------------|
| 0 loan_amnt | Default     | 0.8989   | 0.3265   | Moderately skewed, Normal tails  |
| 1 loan_amnt | Non-Default | 1.2699   | 1.8433   | Highly skewed, Heavy/light tails |

2026-04-18 19:01:18 | WARNING | Normality assumption violated in at least one group.

2026-04-18 19:01:18 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-04-18 19:01:18 | INFO | Running Mann-Whitney U Test: 'loan\_amnt' by 'loan\_status'...

2026-04-18 19:01:18 | DEBUG | Number of groups detected: 2

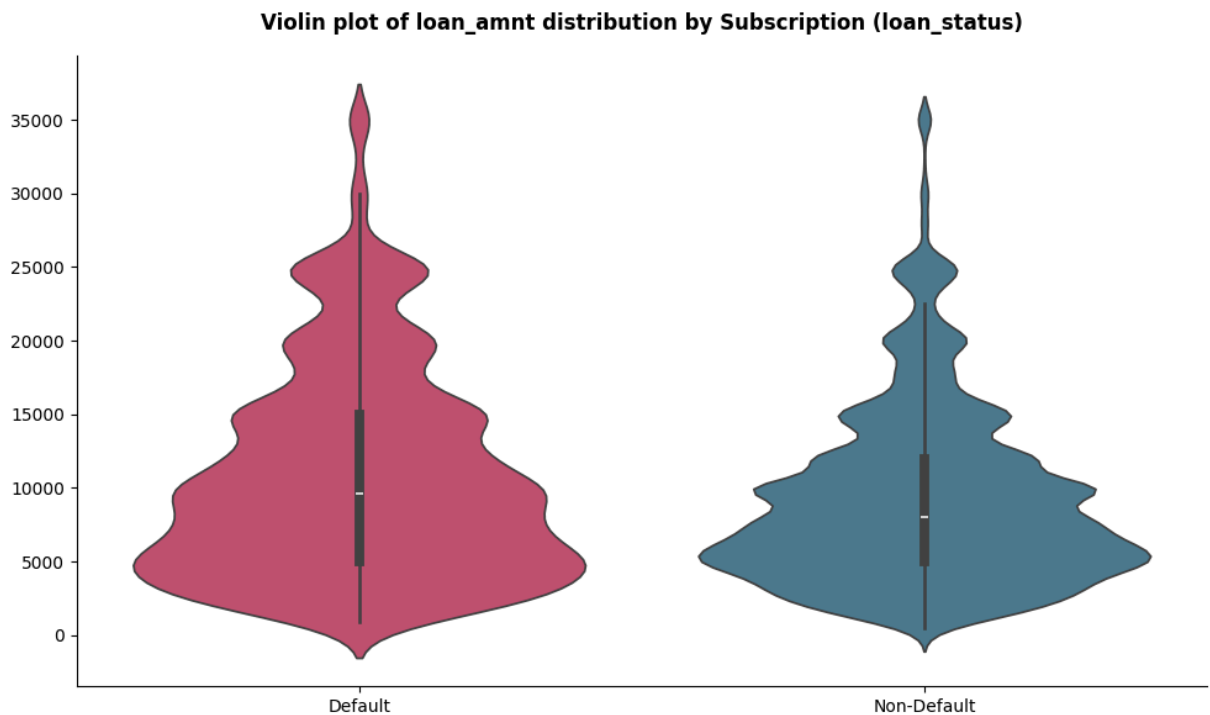
2026-04-18 19:01:18 | INFO | U statistic: 101157877.5

2026-04-18 19:01:18 | INFO | p-value : 0.000000

2026-04-18 19:01:18 | INFO | Result: p-value ≤ 0.05 → Reject H0

2026-04-18 19:01:18 | INFO | Conclusion: Significant difference between the two groups.

2026-04-18 19:01:18 | INFO | Effect size interpretation: Negligible



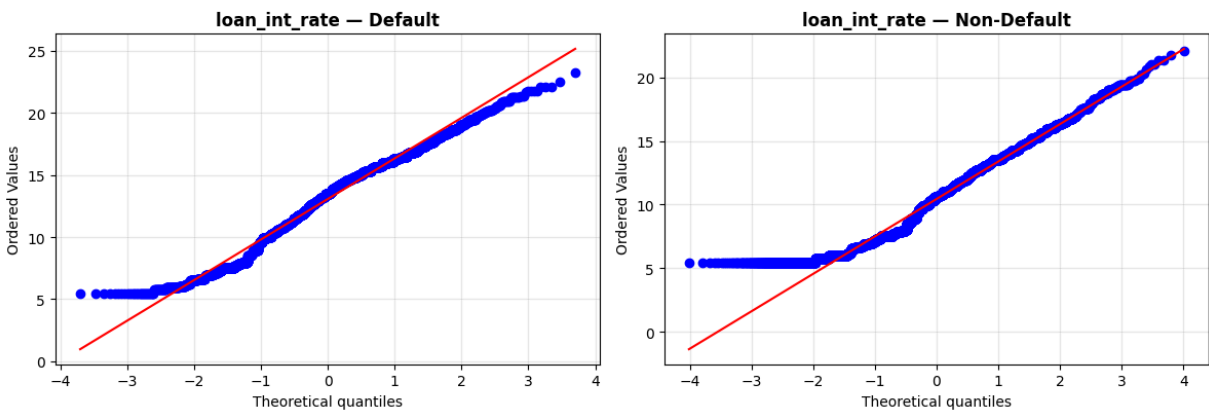
## Distribution of Loan\_Int\_Rate by Loan Status

- **Overall Mean:** 11.01
- **Overall Median:** 10.989999771118164
- **Overall Std:** 3.24

|   | loan_status | Count | Mean      | Median    | Std      |
|---|-------------|-------|-----------|-----------|----------|
| 0 | Default     | 6464  | 13.060207 | 13.490000 | 3.293167 |
| 1 | Non-Default | 23001 | 10.435999 | 10.590000 | 2.982190 |

```
2026-04-18 19:01:18 | INFO      | Checking normality of feature 'loan_int_rate' by groups of 'loan_status'...
2026-04-18 19:01:18 | INFO      | Evaluating distribution characteristics (skewness & kurtosis)...
2026-04-18 19:01:18 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:18 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:18 | DEBUG     | [loan_status=Default] skew=-0.2630, kurtosis=-0.5206 → Approximately symmetric, Normal tails
2026-04-18 19:01:18 | DEBUG     | [loan_status=Non-Default] skew=0.2314, kurtosis=-0.6834 → Approximately symmetric, Normal tails
2026-04-18 19:01:18 | INFO      | Generating Q-Q plots for visual assessment...
```

Q-Q Plots of loan\_int\_rate by loan\_status



Skewness & Kurtosis of loan\_int\_rate by loan\_status

|   | Feature       | Group       | Skewness | Kurtosis | Remark                                |
|---|---------------|-------------|----------|----------|---------------------------------------|
| 0 | loan_int_rate | Default     | -0.2630  | -0.5206  | Approximately symmetric, Normal tails |
| 1 | loan_int_rate | Non-Default | 0.2314   | -0.6834  | Approximately symmetric, Normal tails |

2026-04-18 19:01:18 | INFO | All groups approximately follow a normal distribution.

2026-04-18 19:01:18 | INFO | Check homogeneity of variances across groups using Levene's test (median-centered)

2026-04-18 19:01:18 | DEBUG | Number of groups detected: 2

Homogeneity of Variance — loan\_int\_rate by loan\_status

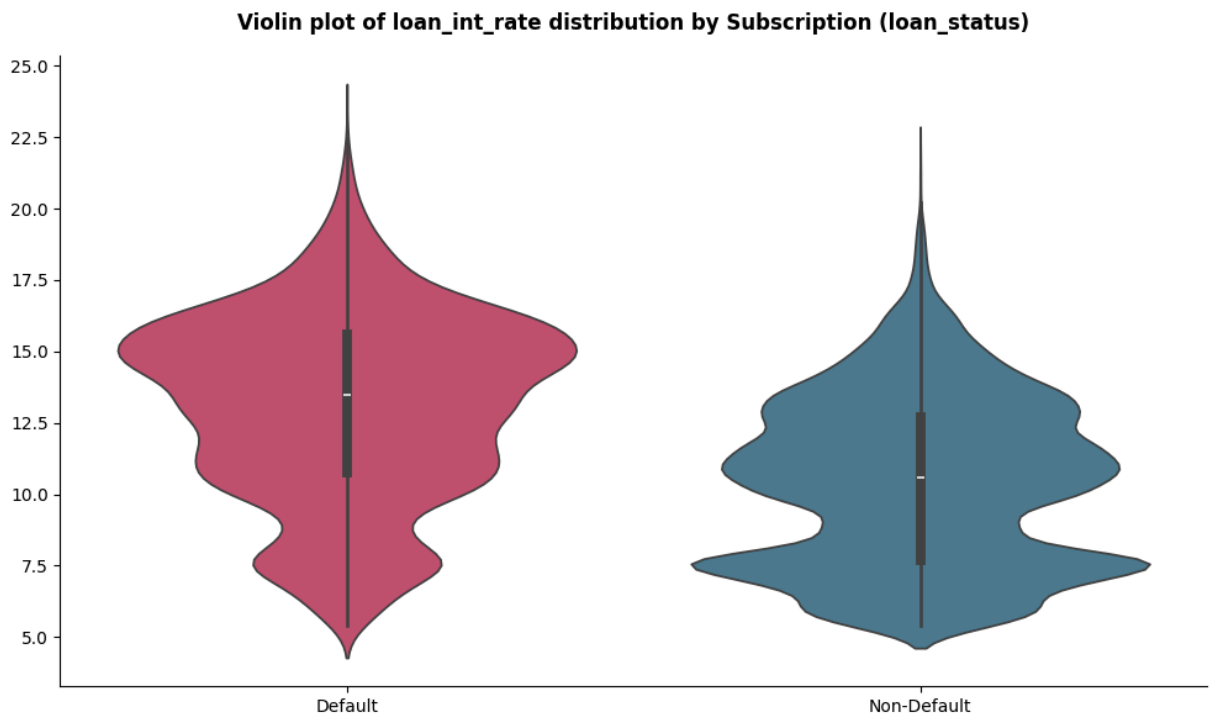
|   | Metric                 | Value    |
|---|------------------------|----------|
| 0 | Levene's Statistic     | 66.9627  |
| 1 | p-value                | 0.000000 |
| 2 | Max/Min Variance Ratio | 1.22     |



```

2026-04-18 19:01:18 | INFO      | Interpretation:
2026-04-18 19:01:18 | INFO      | - Status: Statistically different variance
s, but practical difference is small.
2026-04-18 19:01:18 | INFO      | - Recommendation: ANOVA or T-Test are accep
table, but Welch's ANOVA or Welch's T-Test is recommended.
2026-04-18 19:01:18 | INFO      | Running Independent Two-Sample T-Test: 'loa
n_int_rate' ~ 'loan_status'
2026-04-18 19:01:18 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:18 | DEBUG     | Test Type: Welch's T-Test (unequal variance
s)
2026-04-18 19:01:18 | INFO      | Comparing groups: 'Default' vs 'Non-Defaul
t'
2026-04-18 19:01:18 | INFO      | p-value: 0.000000
2026-04-18 19:01:18 | INFO      | Cohen's d: 0.835
2026-04-18 19:01:18 | DEBUG     | t-statistic: 57.757
2026-04-18 19:01:18 | DEBUG     | Group means → Default: 13.060, Non-Default:
10.436
2026-04-18 19:01:18 | INFO      | Result: p-value ≤ 0.05 → Reject H0
2026-04-18 19:01:18 | INFO      | Conclusion: Significant difference between
the two groups.
2026-04-18 19:01:18 | INFO      | Effect size interpretation: large effect (|
d| = 0.835)

```



## Distribution of Loan\_Percent\_Income by Loan Status

- **Overall Mean:** 0.17
- **Overall Median:** 0.15000000596046448
- **Overall Std:** 0.11

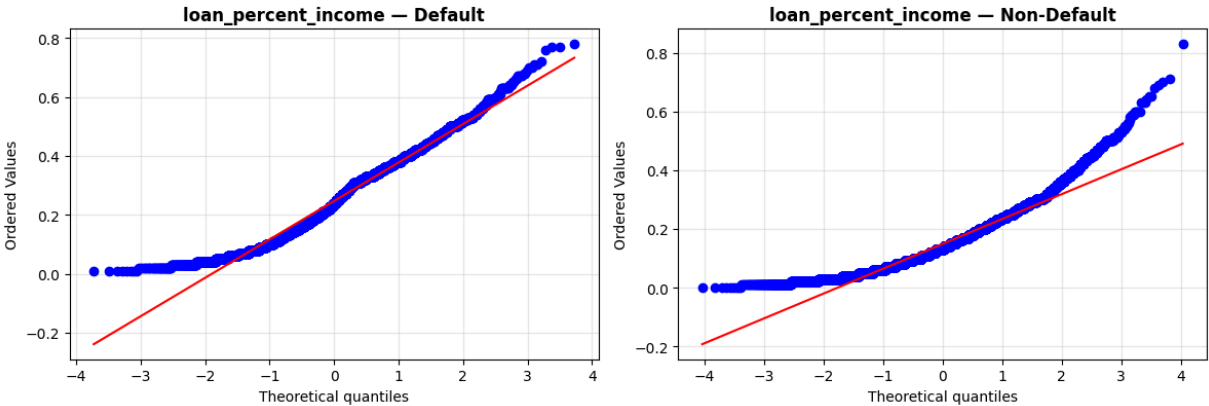
|   | loan_status | Count | Mean     | Median   | Std      |
|---|-------------|-------|----------|----------|----------|
| 0 | Default     | 7108  | 0.246889 | 0.240000 | 0.132148 |
| 1 | Non-Default | 25473 | 0.148805 | 0.130000 | 0.087252 |

```

2026-04-18 19:01:19 | INFO      | Checking normality of feature 'loan_percent
_loan_income' by groups of 'loan_status'...
2026-04-18 19:01:19 | INFO      | Evaluating distribution characteristics (sk
ewness & kurtosis)...
2026-04-18 19:01:19 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:19 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:19 | DEBUG     | [loan_status=Default] skew=0.3550, kurtosis
=-0.4501 → Approximately symmetric, Normal tails
2026-04-18 19:01:19 | DEBUG     | [loan_status=Non-Default] skew=1.0254, kurt
osis=1.6735 → Highly skewed, Heavy/light tails
2026-04-18 19:01:19 | INFO      | Generating Q-Q plots for visual assessmen
t...

```

Q-Q Plots of loan\_percent\_income by loan\_status

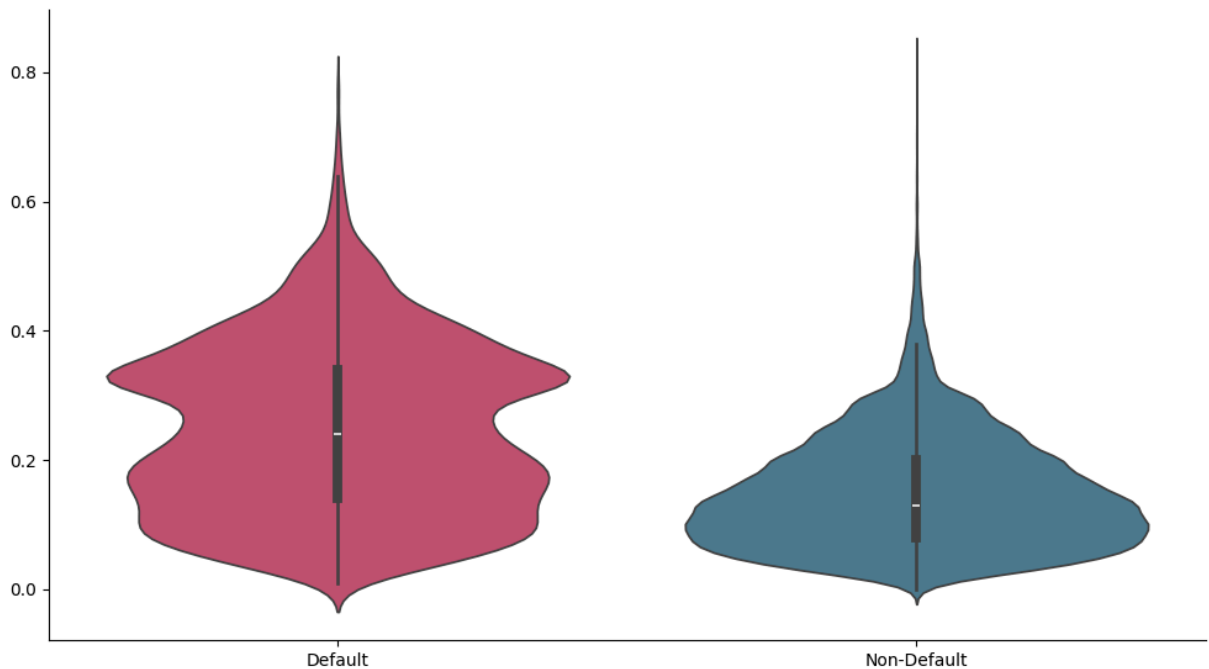


Skewness & Kurtosis of loan\_percent\_income by loan\_status

|   | Feature             | Group       | Skewness | Kurtosis | Remark                                |
|---|---------------------|-------------|----------|----------|---------------------------------------|
| 0 | loan_percent_income | Default     | 0.3550   | -0.4501  | Approximately symmetric, Normal tails |
| 1 | loan_percent_income | Non-Default | 1.0254   | 1.6735   | Highly skewed, Heavy/light tails      |

```
2026-04-18 19:01:19 | WARNING | Normality assumption violated in at least one group.
2026-04-18 19:01:19 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).
2026-04-18 19:01:19 | INFO | Running Mann-Whitney U Test: 'loan_percent_income' by 'loan_status'...
2026-04-18 19:01:19 | DEBUG | Number of groups detected: 2
2026-04-18 19:01:19 | INFO | U statistic: 130515398.0
2026-04-18 19:01:19 | INFO | p-value : 0.000000
2026-04-18 19:01:19 | INFO | Result: p-value ≤ 0.05 → Reject H0
2026-04-18 19:01:19 | INFO | Conclusion: Significant difference between the two groups.
2026-04-18 19:01:19 | INFO | Effect size interpretation: Medium
```

Violin plot of loan\_percent\_income distribution by Subscription (loan\_status)



## Distribution of Loan To Income\_Ratio by Loan Status

- **Overall Mean:** 0.17
- **Overall Median:** 0.14814814925193787
- **Overall Std:** 0.11

|   | loan_status | Count | Mean     | Median   | Std      |
|---|-------------|-------|----------|----------|----------|
| 0 | Default     | 7108  | 0.248750 | 0.239234 | 0.132125 |
| 1 | Non-Default | 25473 | 0.148733 | 0.133333 | 0.087202 |

2026-04-18 19:01:19 | INFO | Checking normality of feature 'loan\_to\_income\_ratio' by groups of 'loan\_status'...

2026-04-18 19:01:19 | INFO | Evaluating distribution characteristics (skewness & kurtosis)...

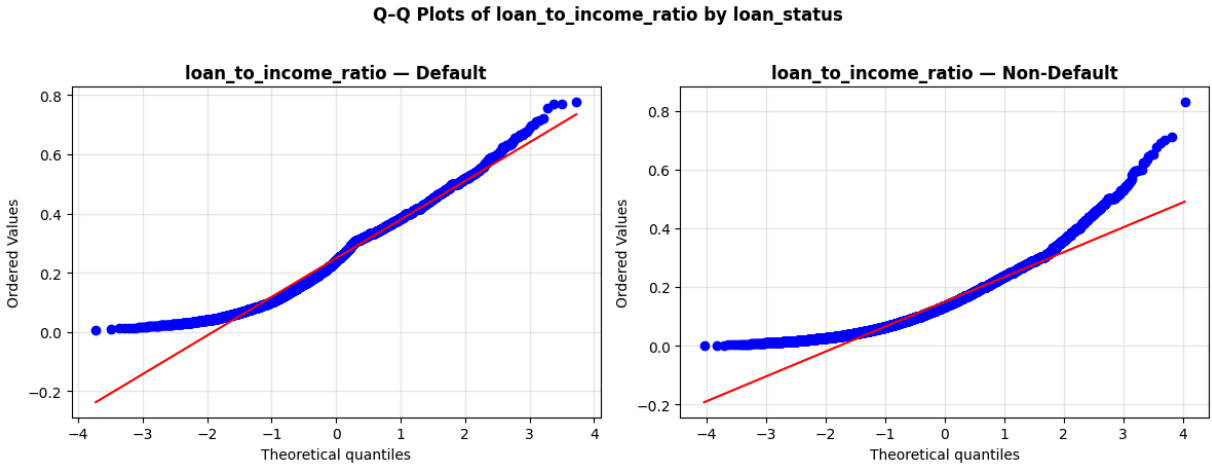
2026-04-18 19:01:19 | DEBUG | Number of groups detected: 2

2026-04-18 19:01:19 | DEBUG | Thresholds → |skew|≤0.5, |kurtosis|≤1.5

2026-04-18 19:01:19 | DEBUG | [loan\_status=Default] skew=0.3436, kurtosis=-0.4629 → Approximately symmetric, Normal tails

2026-04-18 19:01:19 | DEBUG | [loan\_status=Non-Default] skew=1.0286, kurtosis=1.6811 → Highly skewed, Heavy/light tails

2026-04-18 19:01:19 | INFO | Generating Q-Q plots for visual assessment...



Skewness & Kurtosis of loan\_to\_income\_ratio by loan\_status

| Feature                | Group       | Skewness | Kurtosis | Remark                                |
|------------------------|-------------|----------|----------|---------------------------------------|
| 0 loan_to_income_ratio | Default     | 0.3436   | -0.4629  | Approximately symmetric, Normal tails |
|                        | Non-Default | 1.0286   | 1.6811   | Highly skewed, Heavy/light tails      |

2026-04-18 19:01:19 | WARNING | Normality assumption violated in at least one group.

2026-04-18 19:01:19 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-04-18 19:01:19 | INFO | Running Mann-Whitney U Test: 'loan\_to\_income\_ratio' by 'loan\_status'...

2026-04-18 19:01:19 | DEBUG | Number of groups detected: 2

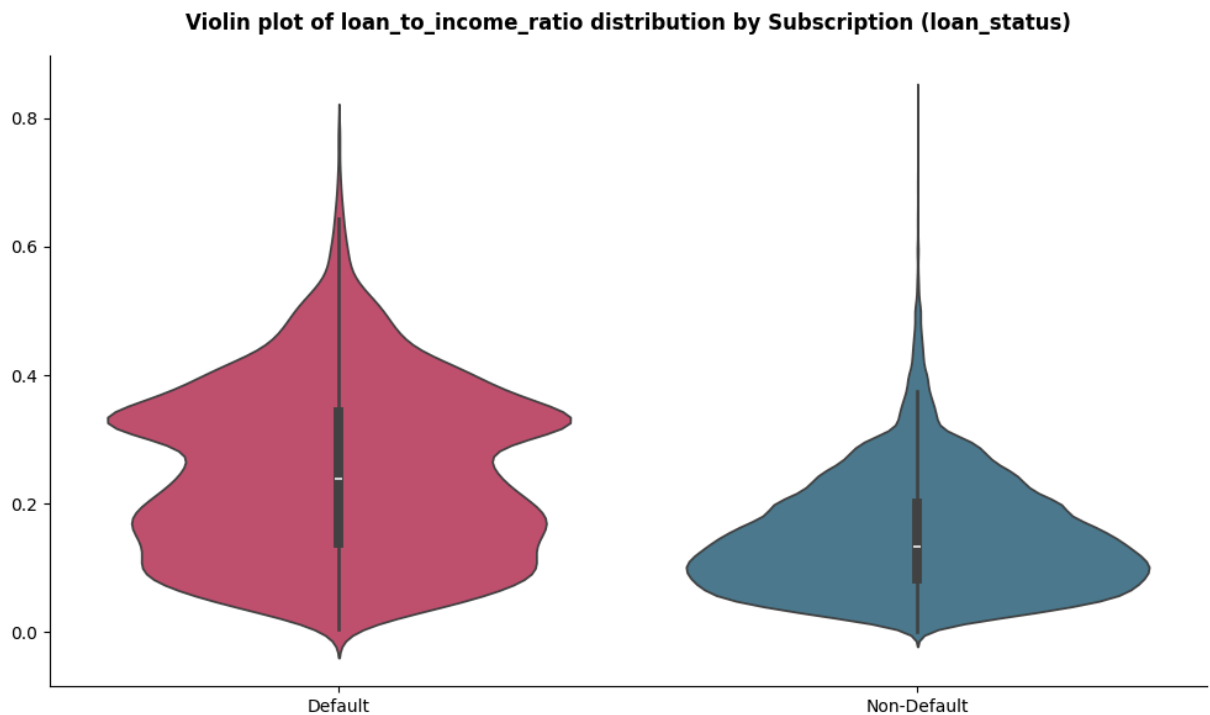
2026-04-18 19:01:19 | INFO | U statistic: 131319126.0

2026-04-18 19:01:19 | INFO | p-value : 0.000000

2026-04-18 19:01:19 | INFO | Result: p-value ≤ 0.05 → Reject H0

2026-04-18 19:01:19 | INFO | Conclusion: Significant difference between the two groups.

2026-04-18 19:01:19 | INFO | Effect size interpretation: Medium



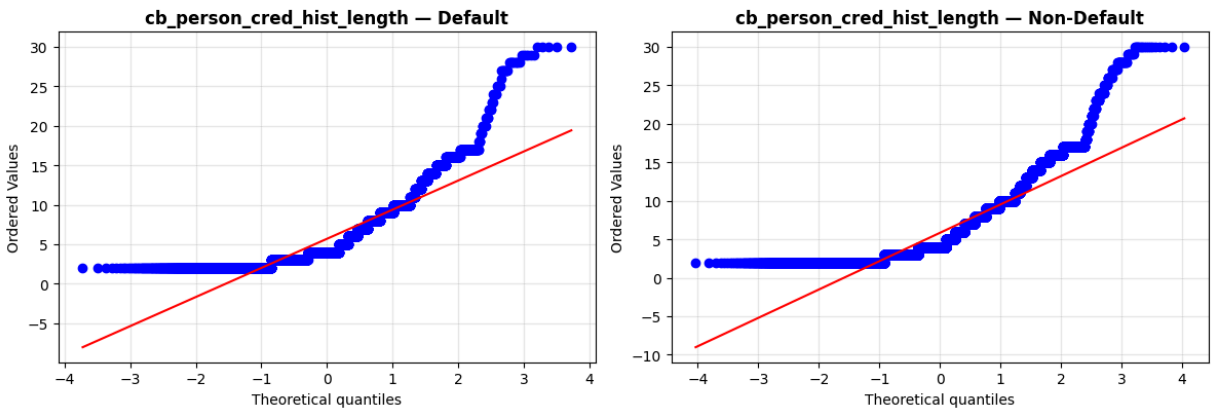
## Distribution of Cb\_Person\_Cred\_Hist\_Length by Loan Status

- **Overall Mean:** 5.80
- **Overall Median:** 4.0
- **Overall Std:** 4.06

|   | loan_status | Count | Mean     | Median   | Std      |
|---|-------------|-------|----------|----------|----------|
| 0 | Non-Default | 25473 | 5.837475 | 4.000000 | 4.040900 |
| 1 | Default     | 7108  | 5.685003 | 4.000000 | 4.103208 |

```
2026-04-18 19:01:20 | INFO      | Checking normality of feature 'cb_person_cred_hist_length' by groups of 'loan_status'...
2026-04-18 19:01:20 | INFO      | Evaluating distribution characteristics (skewness & kurtosis)...
2026-04-18 19:01:20 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:20 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:20 | DEBUG     | [loan_status=Default] skew=1.8187, kurtosis=4.5456 → Highly skewed, Heavy/light tails
2026-04-18 19:01:20 | DEBUG     | [loan_status=Non-Default] skew=1.6173, kurtosis=3.4796 → Highly skewed, Heavy/light tails
2026-04-18 19:01:20 | INFO      | Generating Q-Q plots for visual assessment...
```

Q-Q Plots of cb\_person\_cred\_hist\_length by loan\_status



Skewness & Kurtosis of cb\_person\_cred\_hist\_length by loan\_status

| Feature | Group                      | Skewness                   | Kurtosis | Remark                           |
|---------|----------------------------|----------------------------|----------|----------------------------------|
| 0       | cb_person_cred_hist_length | 1.8187                     | 4.5456   | Highly skewed, Heavy/light tails |
|         | 1                          | cb_person_cred_hist_length | 1.6173   | 3.4796                           |

2026-04-18 19:01:20 | WARNING | Normality assumption violated in at least one group.

2026-04-18 19:01:20 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-04-18 19:01:20 | INFO | Running Mann-Whitney U Test: 'cb\_person\_cred\_hist\_length' by 'loan\_status'...

2026-04-18 19:01:20 | DEBUG | Number of groups detected: 2

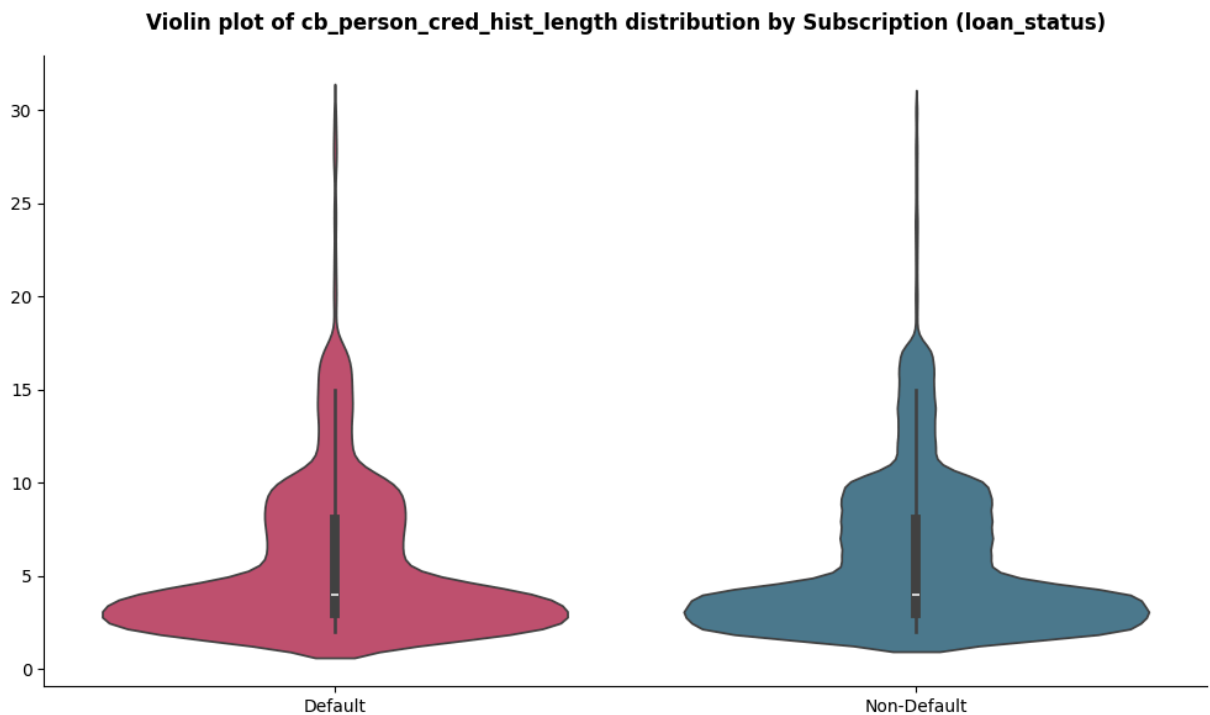
2026-04-18 19:01:20 | INFO | U statistic: 87550955.5

2026-04-18 19:01:20 | INFO | p-value : 0.000018

2026-04-18 19:01:20 | INFO | Result: p-value ≤ 0.05 → Reject H0

2026-04-18 19:01:20 | INFO | Conclusion: Significant difference between the two groups.

2026-04-18 19:01:20 | INFO | Effect size interpretation: Negligible



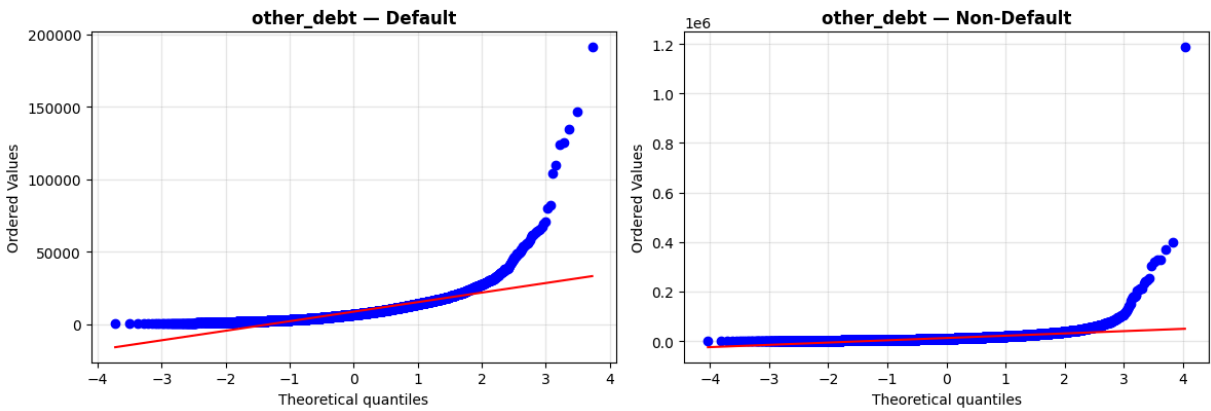
## Distribution of Other\_Debt by Loan Status

- **Overall Mean:** 11567.96
- **Overall Median:** 8995.0703125
- **Overall Std:** 13060.93

|   | loan_status | Count | Mean         | Median      | Std         |
|---|-------------|-------|--------------|-------------|-------------|
| 0 | Non-Default | 25473 | 12384.334967 | 127.9853512 | 1022.312500 |
| 1 | Default     | 7108  | 8642.306646  | 192.3457081 | 145.106934  |

```
2026-04-18 19:01:20 | INFO      | Checking normality of feature 'other_debt'
by groups of 'loan_status'...
2026-04-18 19:01:20 | INFO      | Evaluating distribution characteristics (sk
ewness & kurtosis)...
2026-04-18 19:01:20 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:20 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:20 | DEBUG     | [loan_status=Default] skew=5.9520, kurtosis
=79.7972 → Highly skewed, Heavy/light tails
2026-04-18 19:01:20 | DEBUG     | [loan_status=Non-Default] skew=28.9781, kur
tosis=2042.5092 → Highly skewed, Heavy/light tails
2026-04-18 19:01:20 | INFO      | Generating Q-Q plots for visual assessmen
t...
```

Q-Q Plots of other\_debt by loan\_status



Skewness & Kurtosis of other\_debt by loan\_status

|   | Feature    | Group       | Skewness | Kurtosis  | Remark                           |
|---|------------|-------------|----------|-----------|----------------------------------|
| 0 | other_debt | Default     | 5.9520   | 79.7972   | Highly skewed, Heavy/light tails |
| 1 | other_debt | Non-Default | 28.9781  | 2042.5092 | Highly skewed, Heavy/light tails |

2026-04-18 19:01:21 | WARNING | Normality assumption violated in at least one group.

2026-04-18 19:01:21 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-04-18 19:01:21 | INFO | Running Mann-Whitney U Test: 'other\_debt' by 'loan\_status'...

2026-04-18 19:01:21 | DEBUG | Number of groups detected: 2

2026-04-18 19:01:21 | INFO | U statistic: 64740966.0

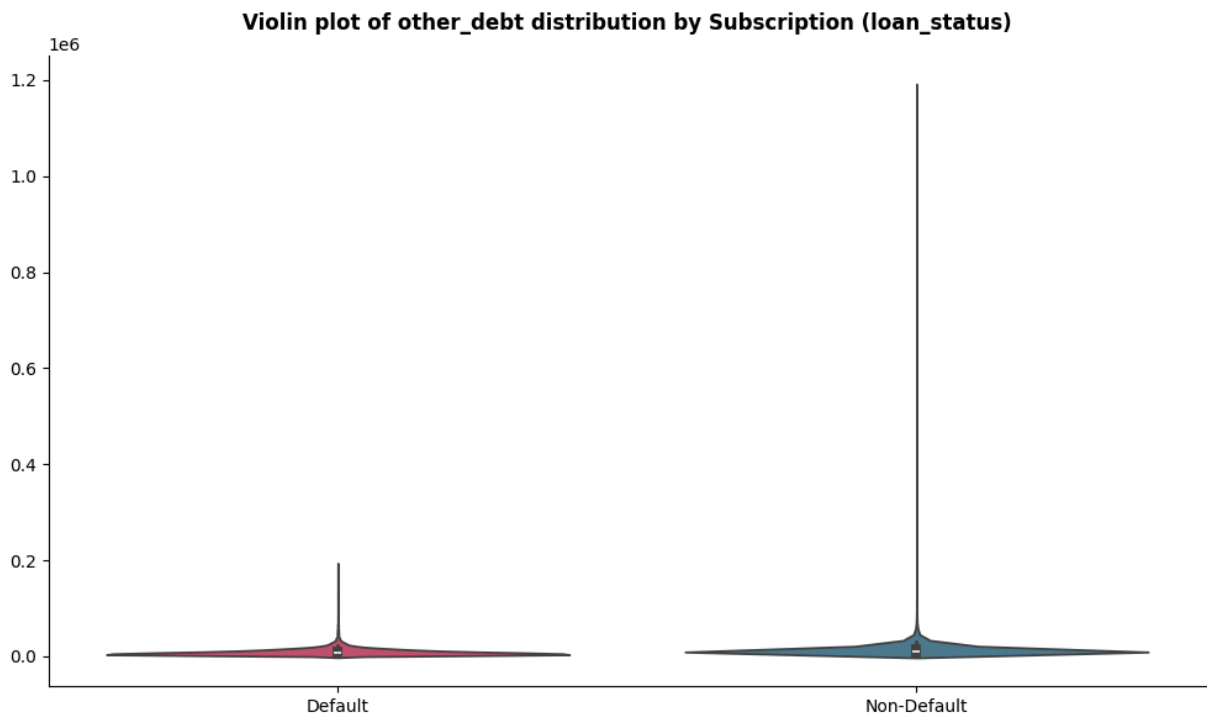
2026-04-18 19:01:21 | INFO | p-value : 0.000000

2026-04-18 19:01:21 | INFO | Result: p-value ≤ 0.05 → Reject H0

2026-04-18 19:01:21 | INFO | Conclusion: Significant difference between the two groups.

2026-04-18 19:01:21 | INFO | Effect size interpretation: Small



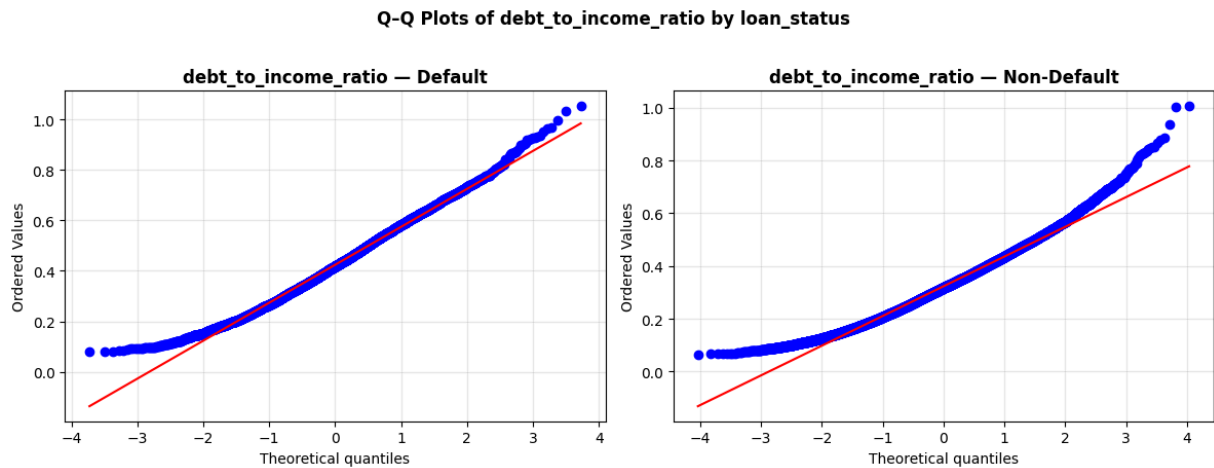


## Distribution of Debt To Income\_Ratio by Loan Status

- **Overall Mean:** 0.35
- **Overall Median:** 0.3331877291202545
- **Overall Std:** 0.13

|   | loan_status | Count | Mean     | Median   | Std      |
|---|-------------|-------|----------|----------|----------|
| 0 | Default     | 7108  | 0.423967 | 0.417944 | 0.150899 |
| 1 | Non-Default | 25473 | 0.323231 | 0.317628 | 0.113366 |

```
2026-04-18 19:01:21 | INFO      | Checking normality of feature 'debt_to_inco
me_ratio' by groups of 'loan_status'...
2026-04-18 19:01:21 | INFO      | Evaluating distribution characteristics (sk
ewness & kurtosis)...
2026-04-18 19:01:21 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:21 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:21 | DEBUG     | [loan_status=Default] skew=0.2597, kurtosis
=-0.2494 → Approximately symmetric, Normal tails
2026-04-18 19:01:21 | DEBUG     | [loan_status=Non-Default] skew=0.4725, kurt
osis=0.3764 → Approximately symmetric, Normal tails
2026-04-18 19:01:21 | INFO      | Generating Q-Q plots for visual assessmen
t...
```



### Skewness & Kurtosis of debt\_to\_income\_ratio by loan\_status

| Feature                | Group       | Skewness | Kurtosis | Remark                                |
|------------------------|-------------|----------|----------|---------------------------------------|
| 0 debt_to_income_ratio | Default     | 0.2597   | -0.2494  | Approximately symmetric, Normal tails |
| 1 debt_to_income_ratio | Non-Default | 0.4725   | 0.3764   | Approximately symmetric, Normal tails |

2026-04-18 19:01:21 | INFO | All groups approximately follow a normal distribution.

2026-04-18 19:01:21 | INFO | Check homogeneity of variances across groups using Levene's test (median-centered)

2026-04-18 19:01:21 | DEBUG | Number of groups detected: 2

### Homogeneity of Variance — debt\_to\_income\_ratio by loan\_status

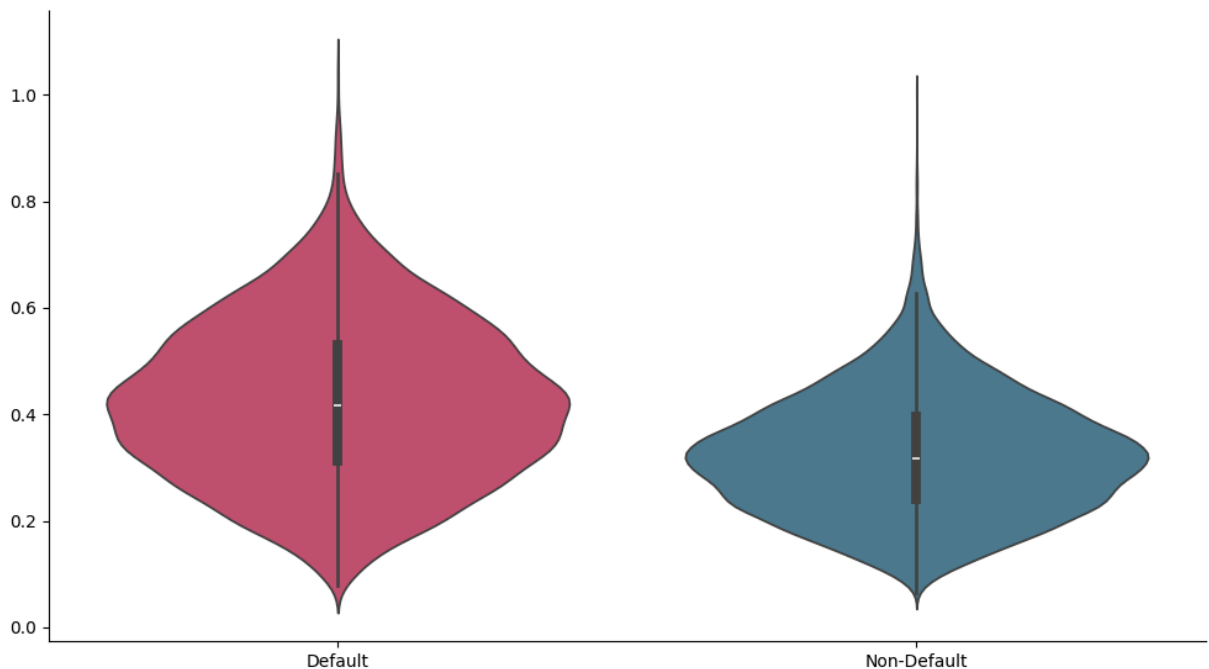
|   | Metric                 | Value     |
|---|------------------------|-----------|
| 0 | Levene's Statistic     | 1078.2762 |
| 1 | p-value                | 0.000000  |
| 2 | Max/Min Variance Ratio | 1.77      |

```

2026-04-18 19:01:21 | INFO      | Interpretation:
2026-04-18 19:01:21 | INFO      | - Status: Statistically different variance
s, but practical difference is small.
2026-04-18 19:01:21 | INFO      | - Recommendation: ANOVA or T-Test are accep
table, but Welch's ANOVA or Welch's T-Test is recommended.
2026-04-18 19:01:21 | INFO      | Running Independent Two-Sample T-Test: 'deb
t_to_income_ratio' ~ 'loan_status'
2026-04-18 19:01:21 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:21 | DEBUG     | Test Type: Welch's T-Test (unequal variance
s)
2026-04-18 19:01:21 | INFO      | Comparing groups: 'Default' vs 'Non-Defaul
t'
2026-04-18 19:01:21 | INFO      | p-value: 0.000000
2026-04-18 19:01:21 | INFO      | Cohen's d: 0.755
2026-04-18 19:01:21 | DEBUG     | t-statistic: 52.313
2026-04-18 19:01:21 | DEBUG     | Group means → Default: 0.424, Non-Default:
0.323
2026-04-18 19:01:21 | INFO      | Result:  $p\text{-value} \leq 0.05 \rightarrow \text{Reject } H_0$ 
2026-04-18 19:01:21 | INFO      | Conclusion: Significant difference between
the two groups.
2026-04-18 19:01:21 | INFO      | Effect size interpretation: large effect (|
d| = 0.755)

```

**Violin plot of debt\_to\_income\_ratio distribution by Subscription (loan\_status)**



## Distribution of Open\_Accounts by Loan Status

- **Overall Mean:** 8.04
- **Overall Median:** 8.0
- **Overall Std:** 4.33

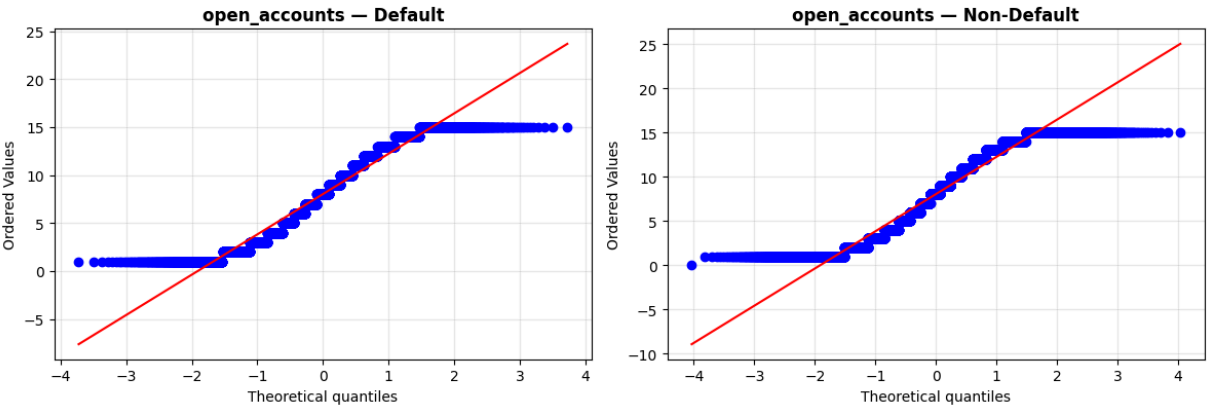
|   | loan_status | Count | Mean     | Median   | Std      |
|---|-------------|-------|----------|----------|----------|
| 0 | Non-Default | 25473 | 8.048286 | 8.000000 | 4.331499 |
| 1 | Default     | 7108  | 8.019555 | 8.000000 | 4.316067 |

```

2026-04-18 19:01:22 | INFO      | Checking normality of feature 'open_accounts' by groups of 'loan_status'...
2026-04-18 19:01:22 | INFO      | Evaluating distribution characteristics (skewness & kurtosis)...
2026-04-18 19:01:22 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:22 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:22 | DEBUG     | [loan_status=Default] skew=0.0058, kurtosis=-1.1989 → Approximately symmetric, Normal tails
2026-04-18 19:01:22 | DEBUG     | [loan_status=Non-Default] skew=-0.0107, kurtosis=-1.2178 → Approximately symmetric, Normal tails
2026-04-18 19:01:22 | INFO      | Generating Q-Q plots for visual assessment...

```

Q-Q Plots of open\_accounts by loan\_status



Skewness & Kurtosis of open\_accounts by loan\_status

|   | Feature       | Group       | Skewness | Kurtosis | Remark                                |
|---|---------------|-------------|----------|----------|---------------------------------------|
| 0 | open_accounts | Default     | 0.0058   | -1.1989  | Approximately symmetric, Normal tails |
| 1 | open_accounts | Non-Default | -0.0107  | -1.2178  | Approximately symmetric, Normal tails |

```

2026-04-18 19:01:22 | INFO      | All groups approximately follow a normal distribution.
2026-04-18 19:01:22 | INFO      | Check homogeneity of variances across groups using Levene's test (median-centered)
2026-04-18 19:01:22 | DEBUG     | Number of groups detected: 2

```

Homogeneity of Variance — open\_accounts by loan\_status

|   | Metric                 | Value    |
|---|------------------------|----------|
| 0 | Levene's Statistic     | 1.6981   |
| 1 | p-value                | 0.192540 |
| 2 | Max/Min Variance Ratio | 1.01     |

2026-04-18 19:01:22 | INFO | Interpretation:

2026-04-18 19:01:22 | INFO | - Status: Variances are statistically and p

2026-04-18 19:01:22 | INFO | ractically homogeneous.

2026-04-18 19:01:22 | INFO | - Recommendation: Use One-Way ANOVA or Inde

2026-04-18 19:01:22 | INFO | pendent Two-Sample T-Test.

2026-04-18 19:01:22 | INFO | Running Independent Two-Sample T-Test: 'ope

2026-04-18 19:01:22 | INFO | n\_accounts' ~ 'loan\_status'

2026-04-18 19:01:22 | DEBUG | Number of groups detected: 2

2026-04-18 19:01:22 | DEBUG | Test Type: Student's T-Test (equal variance

2026-04-18 19:01:22 | DEBUG | s)

2026-04-18 19:01:22 | INFO | Comparing groups: 'Default' vs 'Non-Defaul

2026-04-18 19:01:22 | INFO | t'

2026-04-18 19:01:22 | INFO | p-value: 0.620703

2026-04-18 19:01:22 | INFO | Cohen's d: -0.007

2026-04-18 19:01:22 | DEBUG | t-statistic: -0.495

2026-04-18 19:01:22 | DEBUG | Group means → Default: 8.020, Non-Default:

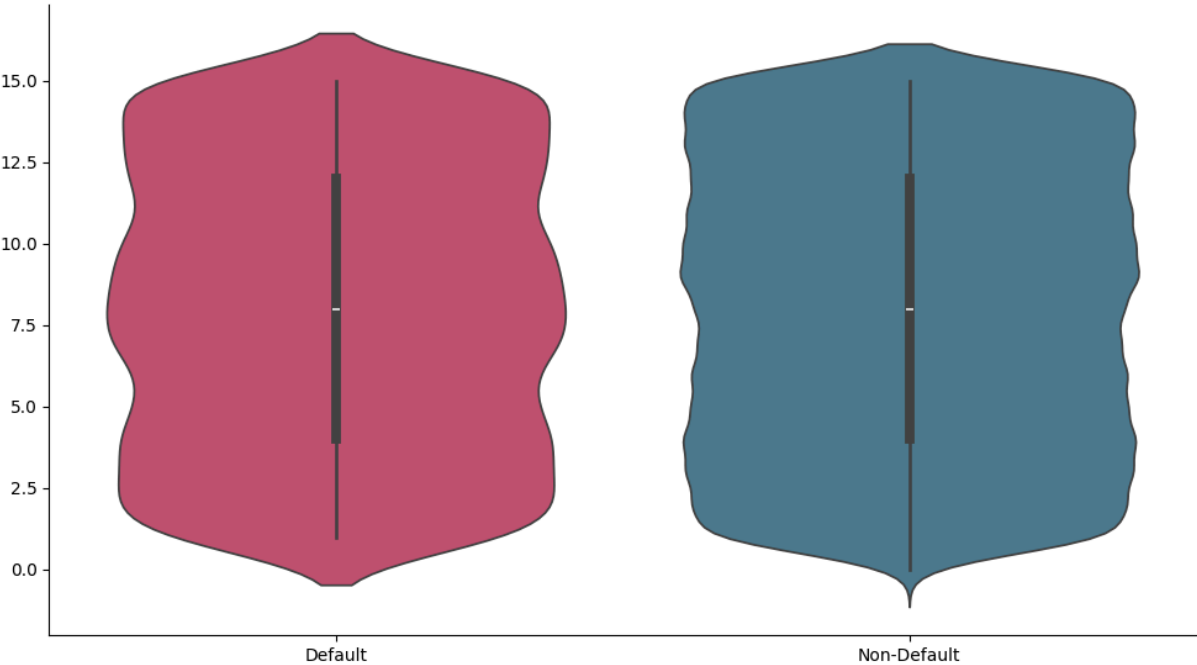
2026-04-18 19:01:22 | DEBUG | 8.048

2026-04-18 19:01:22 | WARNING | Result: p-value > 0.05 → Fail to reject H0

2026-04-18 19:01:22 | WARNING | Conclusion: No statistically significant di

2026-04-18 19:01:22 | WARNING | fference detected.

Violin plot of open\_accounts distribution by Subscription (loan\_status)



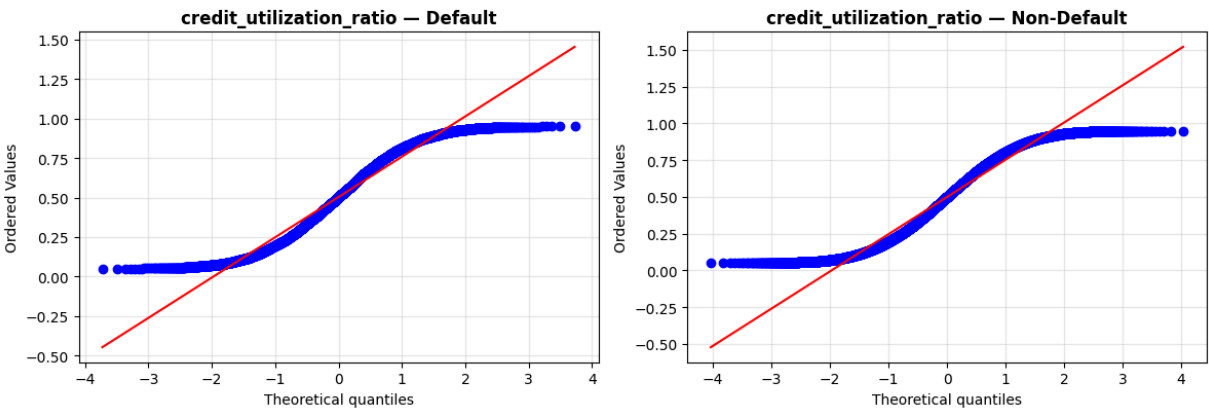
Distribution of Credit Utilization\_Ratio by Loan Status

- **Overall Mean:** 0.50
- **Overall Median:** 0.5002885460853577
- **Overall Std:** 0.26

| loan_status |             | Count | Mean     | Median   | Std      |
|-------------|-------------|-------|----------|----------|----------|
| 0           | Default     | 7108  | 0.503465 | 0.504247 | 0.261262 |
| 1           | Non-Default | 25473 | 0.498885 | 0.499371 | 0.259041 |

```
2026-04-18 19:01:22 | INFO      | Checking normality of feature 'credit_utili
zation_ratio' by groups of 'loan_status'...
2026-04-18 19:01:22 | INFO      | Evaluating distribution characteristics (sk
ewness & kurtosis)...
2026-04-18 19:01:22 | DEBUG     | Number of groups detected: 2
2026-04-18 19:01:22 | DEBUG     | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-04-18 19:01:22 | DEBUG     | [loan_status=Default] skew=-0.0142, kurtosi
s=-1.2148 → Approximately symmetric, Normal tails
2026-04-18 19:01:22 | DEBUG     | [loan_status=Non-Default] skew=0.0037, kurt
osis=-1.1941 → Approximately symmetric, Normal tails
2026-04-18 19:01:22 | INFO      | Generating Q-Q plots for visual assessmen
t...
```

Q-Q Plots of credit\_utilization\_ratio by loan\_status



Skewness & Kurtosis of credit\_utilization\_ratio by loan\_status

| Feature                    | Group       | Skewness | Kurtosis | Remark                                |
|----------------------------|-------------|----------|----------|---------------------------------------|
| 0 credit_utilization_ratio | Default     | -0.0142  | -1.2148  | Approximately symmetric, Normal tails |
| 1 credit_utilization_ratio | Non-Default | 0.0037   | -1.1941  | Approximately symmetric, Normal tails |

```
2026-04-18 19:01:23 | INFO      | All groups approximately follow a normal di
stribution.
2026-04-18 19:01:23 | INFO      | Check homogeneity of variances across group
s using Levene's test (median-centered)
2026-04-18 19:01:23 | DEBUG     | Number of groups detected: 2
```

Homogeneity of Variance — credit\_utilization\_ratio  
by loan\_status

|   | Metric                 | Value    |
|---|------------------------|----------|
| 0 | Levene's Statistic     | 1.5282   |
| 1 | p-value                | 0.216395 |
| 2 | Max/Min Variance Ratio | 1.02     |

2026-04-18 19:01:23 | INFO | Interpretation:

2026-04-18 19:01:23 | INFO | - Status: Variances are statistically and p

2026-04-18 19:01:23 | INFO | ractically homogeneous.

2026-04-18 19:01:23 | INFO | - Recommendation: Use One-Way ANOVA or Inde

2026-04-18 19:01:23 | INFO | pendent Two-Sample T-Test.

2026-04-18 19:01:23 | INFO | Running Independent Two-Sample T-Test: 'cre

2026-04-18 19:01:23 | INFO | dit\_utilization\_ratio' ~ 'loan\_status'

2026-04-18 19:01:23 | DEBUG | Number of groups detected: 2

2026-04-18 19:01:23 | DEBUG | Test Type: Student's T-Test (equal variance

2026-04-18 19:01:23 | DEBUG | s)

2026-04-18 19:01:23 | INFO | Comparing groups: 'Default' vs 'Non-Defaul

2026-04-18 19:01:23 | INFO | t'

2026-04-18 19:01:23 | INFO | p-value: 0.188288

2026-04-18 19:01:23 | INFO | Cohen's d: 0.018

2026-04-18 19:01:23 | DEBUG | t-statistic: 1.316

2026-04-18 19:01:23 | DEBUG | Group means → Default: 0.503, Non-Default:

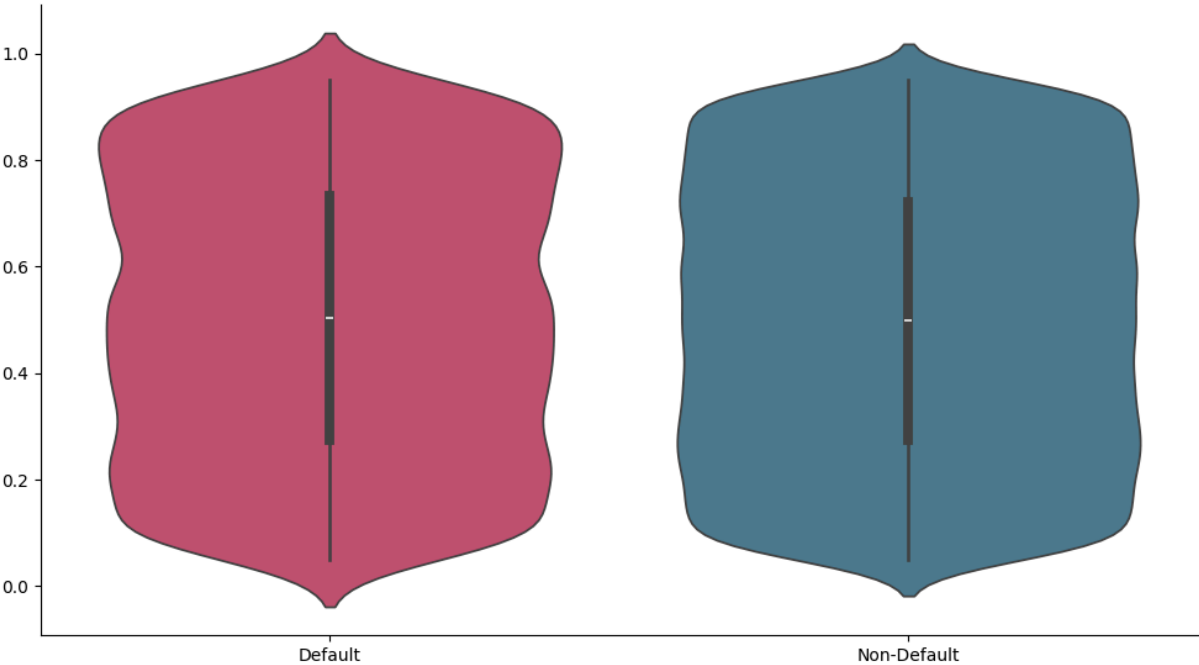
2026-04-18 19:01:23 | DEBUG | 0.499

2026-04-18 19:01:23 | WARNING | Result: p-value > 0.05 → Fail to reject H0

2026-04-18 19:01:23 | WARNING | Conclusion: No statistically significant di

2026-04-18 19:01:23 | WARNING | fference detected.

Violin plot of credit\_utilization\_ratio distribution by Subscription (loan\_status)



## □ Feature-Level Insights

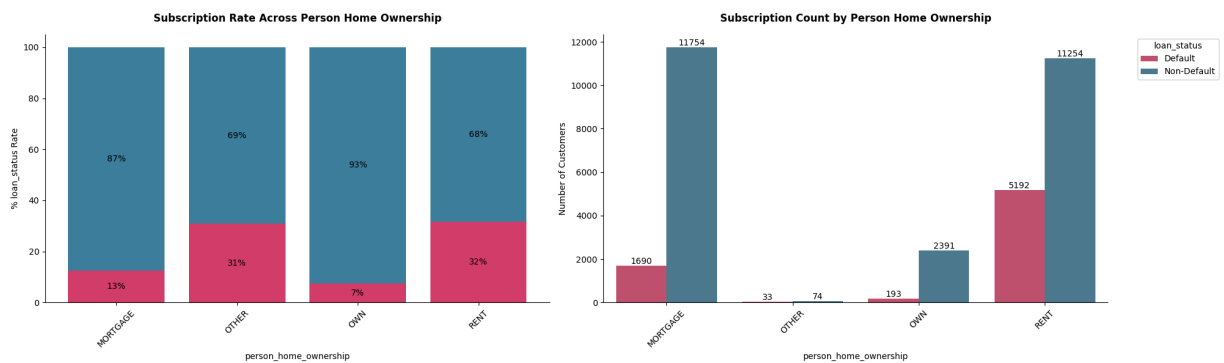
- **Person\_Age:** Age is nearly identical between the two groups (mean ~27), so it is **not a strong discriminator** for default. Although statistically significant ( $p\text{-value} \approx 0$ ), the **effect size is very small** → **negligible practical impact**.
- **Person\_Income:** Clear difference — **Non-default customers have significantly higher income (~70k vs ~49k)** → income is an **important feature**.
- **Person\_Emp\_Length:** Slight difference — **Non-default customers have longer employment (~5 years vs ~4 years)** → weak signal. Despite statistical significance, the **effect size is negligible** and the distribution is highly skewed → low practical impact.
- **Loan\_Amnt:** Loan amount is **higher in the Default group (~10.8k vs ~9.2k)** → larger loans are associated with higher risk.
- **Loan\_Int\_Rate:** Clear difference — **Default group has higher interest rates (~13% vs ~10.4%)** → **very strong feature**. Large effect size ( $d \approx 0.83$ ) → **key driver of default risk**.
- **Loan\_Percent\_Income:** **Higher in Default (~0.25 vs ~0.15)** → borrowers exceed repayment capacity more often → **important feature**. Medium effect size.
- **Loan\_To\_Income\_Ratio:** Very similar to the previous feature → **important but redundant** with `loan_percent_income`. Medium effect size.
- **Cb\_Person\_Cred\_Hist\_Length:** Nearly identical between groups (~5.8 vs ~5.7) → **not a strong discriminator**. Despite statistical significance, **effect size is very small** → **low practical impact**.
- **Other\_Debt:** Counterintuitive result — **Non-default has higher total debt (~12.3k vs ~8.6k)** → absolute debt is **not a reliable risk indicator**. Highly skewed with extreme outliers → **low practical value**; ratios are more meaningful.
- **Debt\_To\_Income\_Ratio:** **Higher in Default (~0.42 vs ~0.32)** → financial burden is a key risk factor. Large effect size ( $d \approx 0.75$ ) → **one of the most important features**.



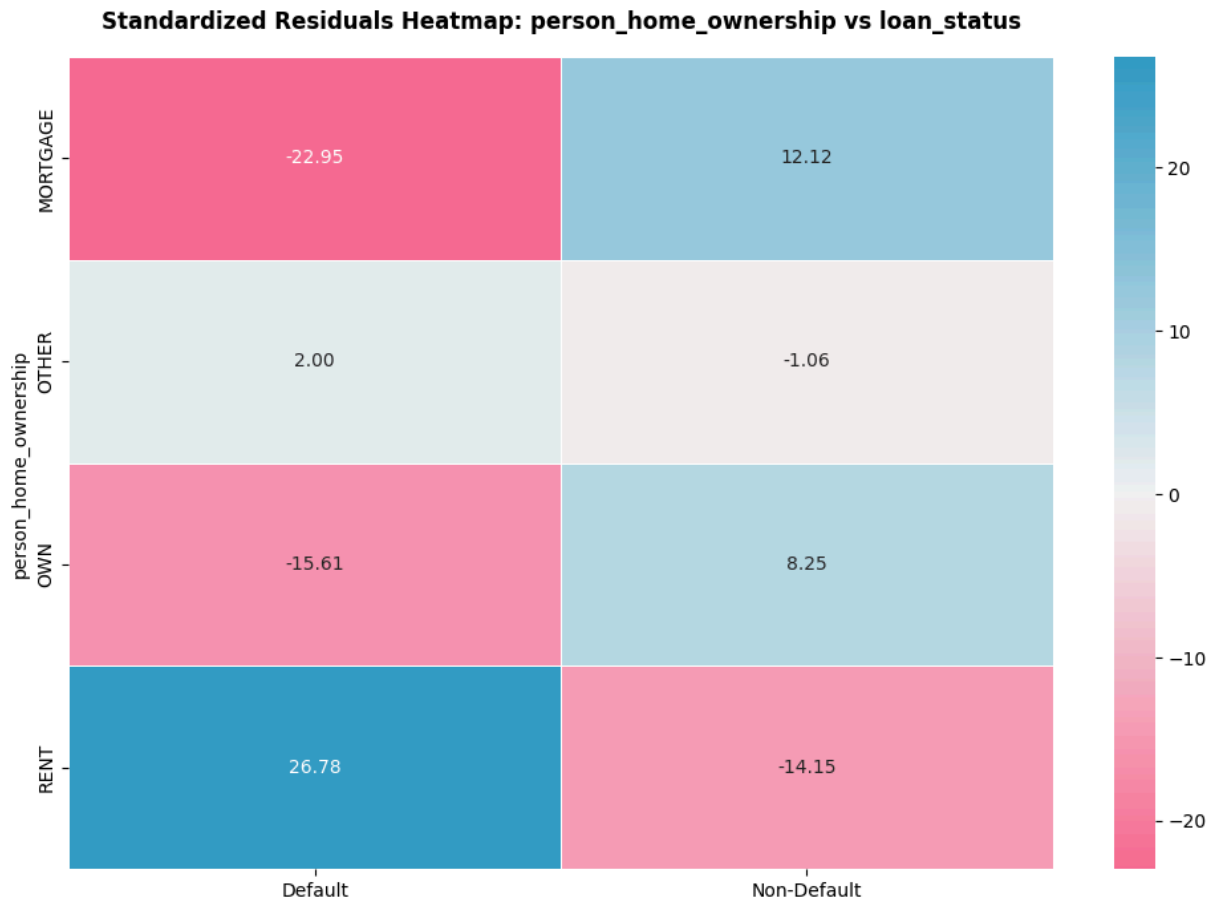
- **Open\_Accounts:** Nearly identical ( $\sim 8$ ) → **no meaningful difference**.  
p-value  $> 0.05$  and effect size  $\approx 0$  → **not predictive**.
- **Credit\_Utilization\_Ratio:** Nearly identical ( $\sim 0.50$ ) → **no meaningful difference**. p-value  $> 0.05$  and effect size  $\approx 0$  → **does not contribute to prediction**.

```
In [17]: for feature in cat_features:
          StatUtils.analyze_categorical_feature_vs_target(cat = feature, target_fe
                                                         logger=logger, df=df)
```

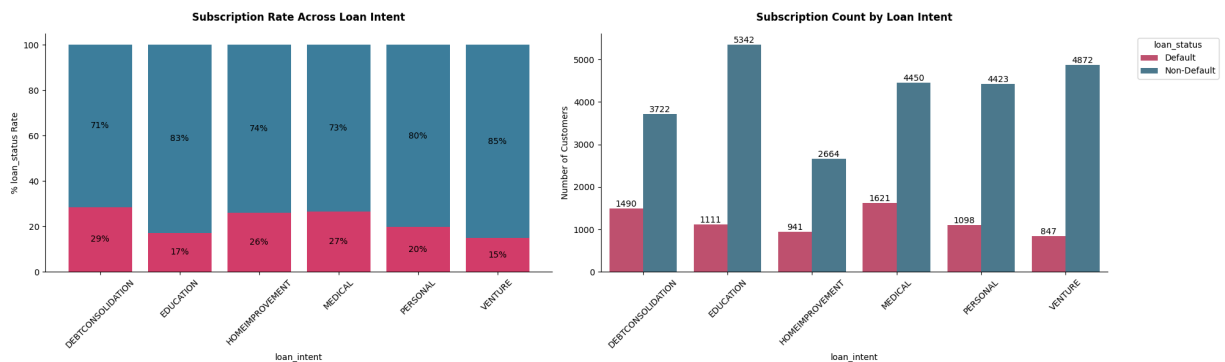
## Distribution of Person Home\_Ownership by Loan\_Status



```
2026-04-18 19:01:23 | INFO      | Chi-Square Test of Independence: 'person_ho
me_ownership' vs. 'loan_status'...
2026-04-18 19:01:23 | INFO      | Chi-squared statistic: 1907.981
2026-04-18 19:01:23 | INFO      | Degrees of freedom: 3
2026-04-18 19:01:23 | INFO      | p-value: 0.000000
2026-04-18 19:01:23 | INFO      | Result: p-value < 0.05 → Reject H₀
2026-04-18 19:01:23 | INFO      | Variables 'person_home_ownership' and 'loan
_status' are statistically associated.
2026-04-18 19:01:23 | DEBUG     | Displaying standardized residuals heatmap.
```

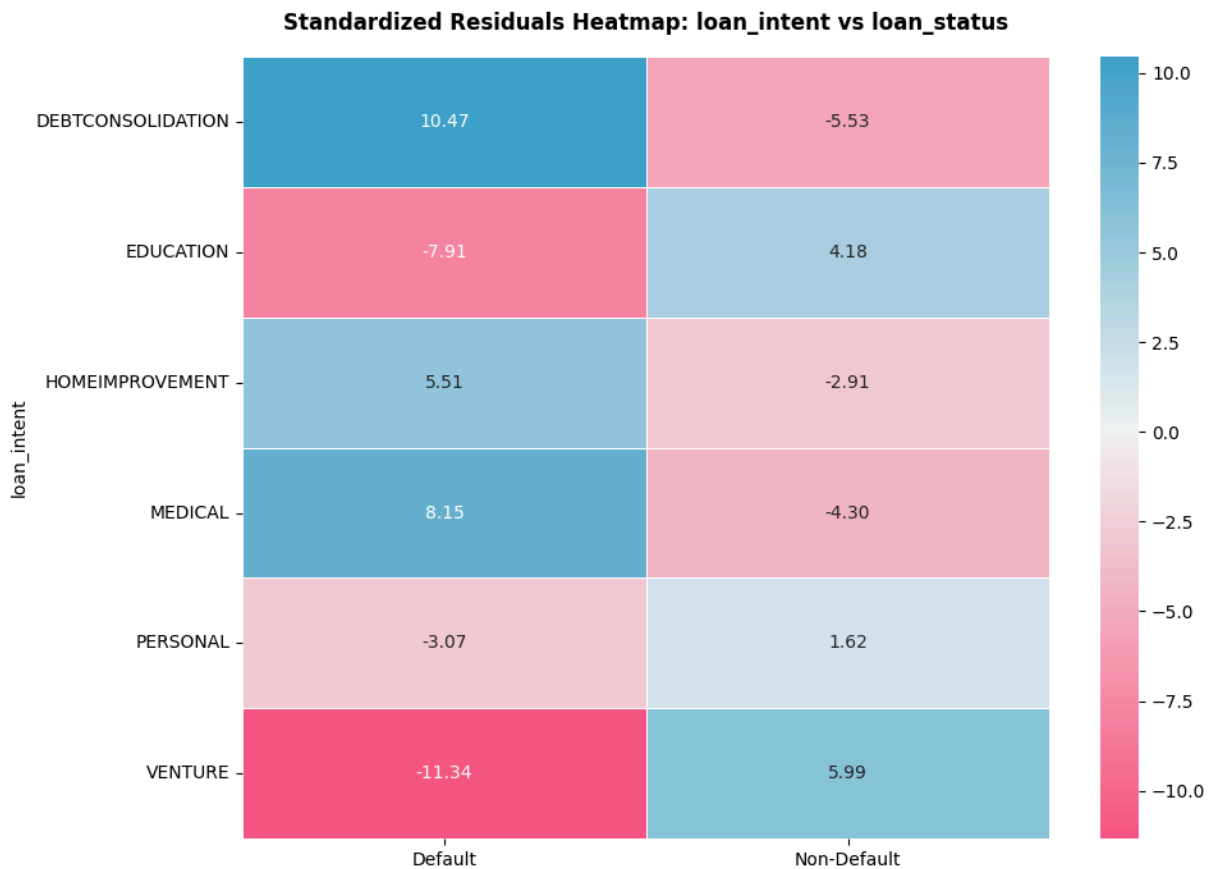


## Distribution of Loan\_Intent by Loan\_Status

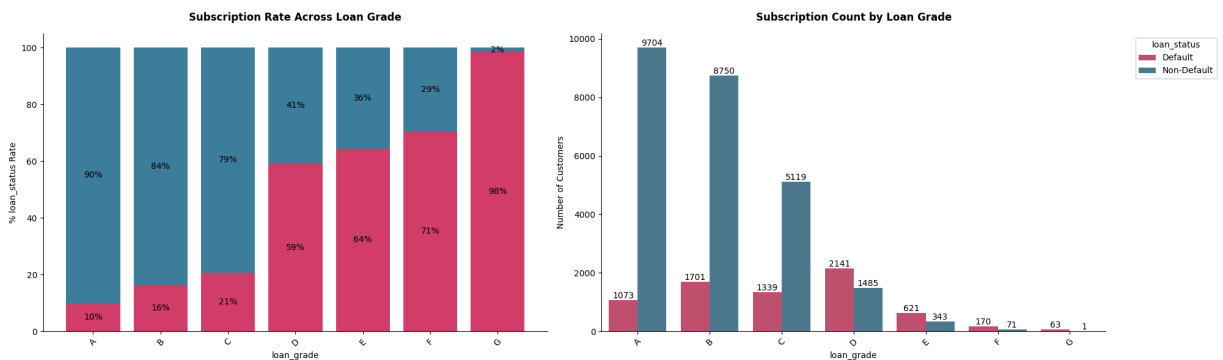


2026-04-18 19:01:24 | INFO  
2026-04-18 19:01:24 | INFO  
2026-04-18 19:01:24 | INFO  
2026-04-18 19:01:24 | INFO  
2026-04-18 19:01:24 | INFO  
2026-04-18 19:01:24 | INFO  
2026-04-18 19:01:24 | DEBUG

| Chi-Square Test of Independence: 'loan\_inte  
| Chi-squared statistic: 520.512  
| Degrees of freedom: 5  
| p-value: 0.000000  
| Result: p-value < 0.05 → Reject H<sub>0</sub>  
| Variables 'loan\_intent' and 'loan\_status' a  
| Displaying standardized residuals heatmap.



## Distribution of Loan\_Grade by Loan\_Status



2026-04-18 19:01:24 | INFO e' vs. 'loan\_status'...

2026-04-18 19:01:24 | INFO

2026-04-18 19:01:24 | INFO

2026-04-18 19:01:24 | INFO

2026-04-18 19:01:24 | INFO

2026-04-18 19:01:24 | INFO e statistically associated.

2026-04-18 19:01:24 | DEBUG

| Chi-Square Test of Independence: 'loan\_grad

| Chi-squared statistic: 5609.184

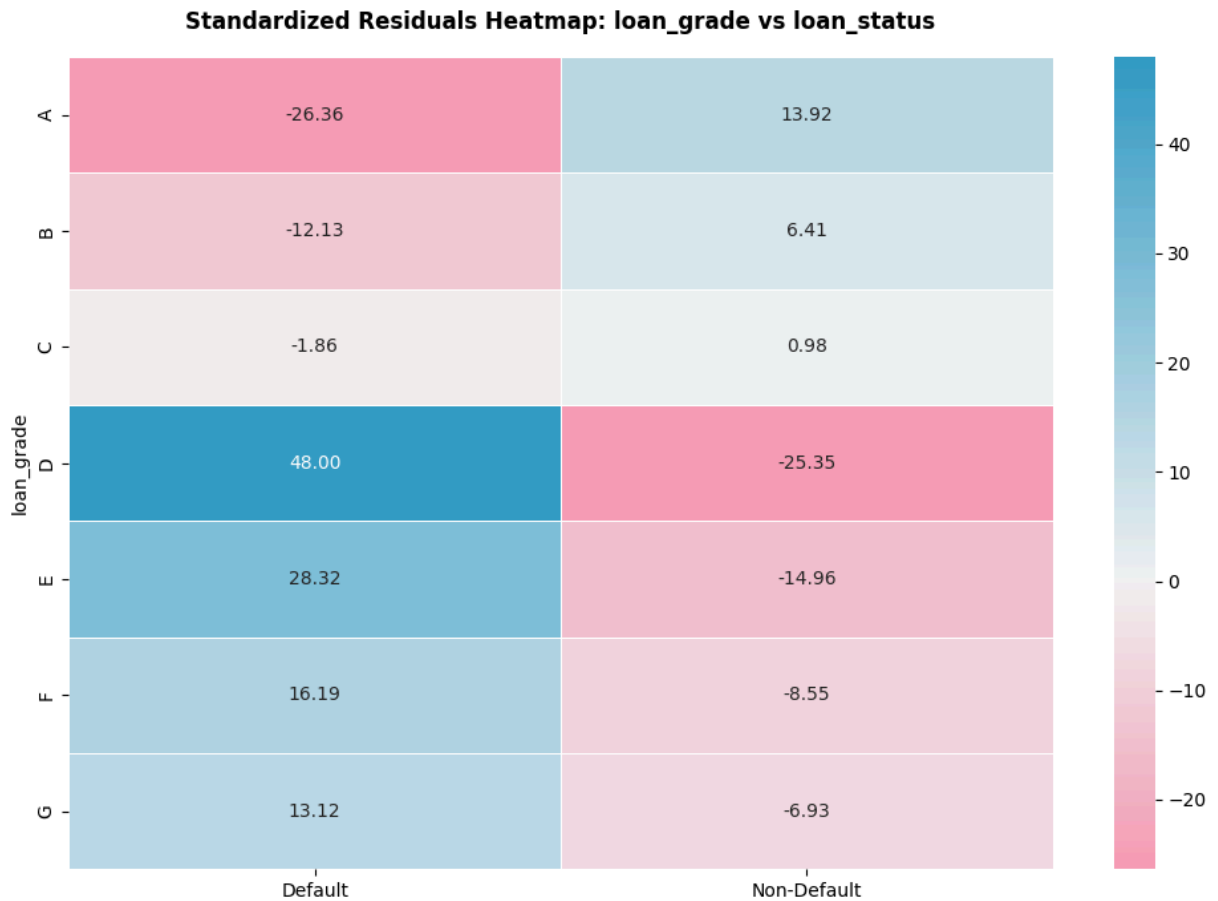
| Degrees of freedom: 6

| p-value: 0.000000

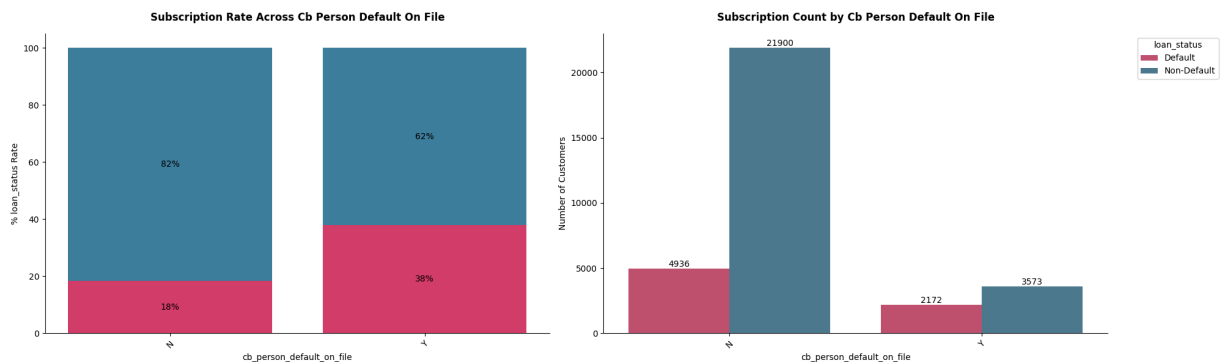
| Result: p-value < 0.05 → Reject H<sub>0</sub>

| Variables 'loan\_grade' and 'loan\_status' ar

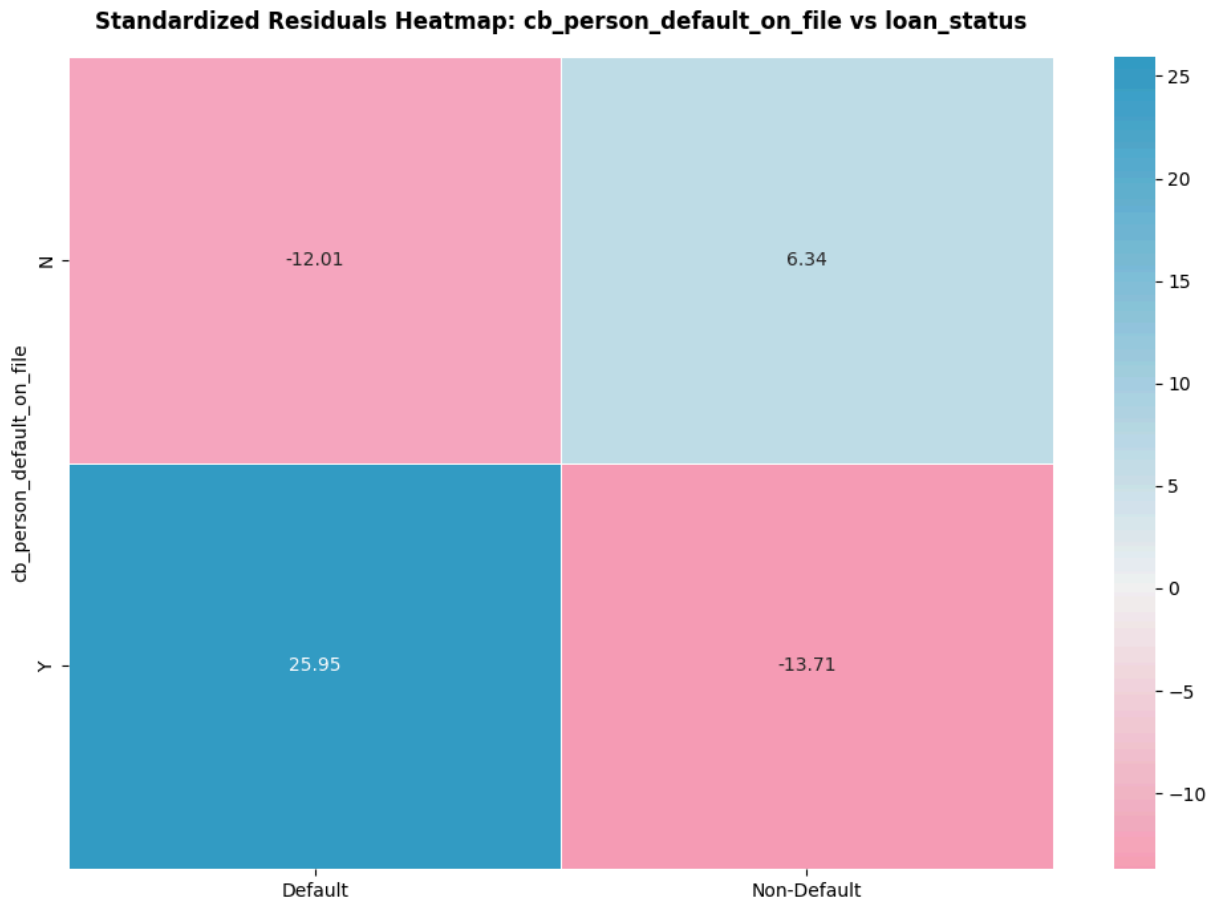
| Displaying standardized residuals heatmap.



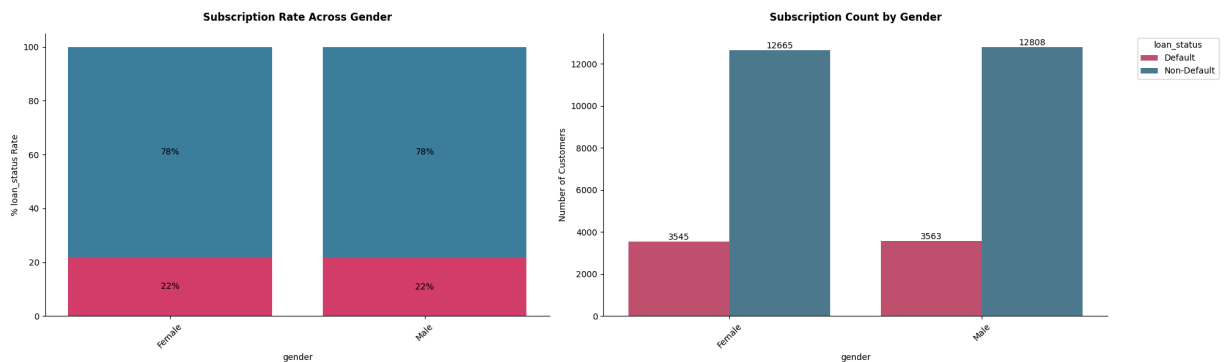
## Distribution of Cb\_Person\_Default\_On\_File by Loan\_Status



```
2026-04-18 19:01:25 | INFO      | Chi-Square Test of Independence: 'cb_person
_default_on_file' vs. 'loan_status'...
2026-04-18 19:01:25 | INFO      | Chi-squared statistic: 1044.440
2026-04-18 19:01:25 | INFO      | Degrees of freedom: 1
2026-04-18 19:01:25 | INFO      | p-value: 0.000000
2026-04-18 19:01:25 | INFO      | Result: p-value < 0.05 → Reject H0
2026-04-18 19:01:25 | INFO      | Variables 'cb_person_default_on_file' and
'loan_status' are statistically associated.
2026-04-18 19:01:25 | DEBUG     | Displaying standardized residuals heatmap.
```

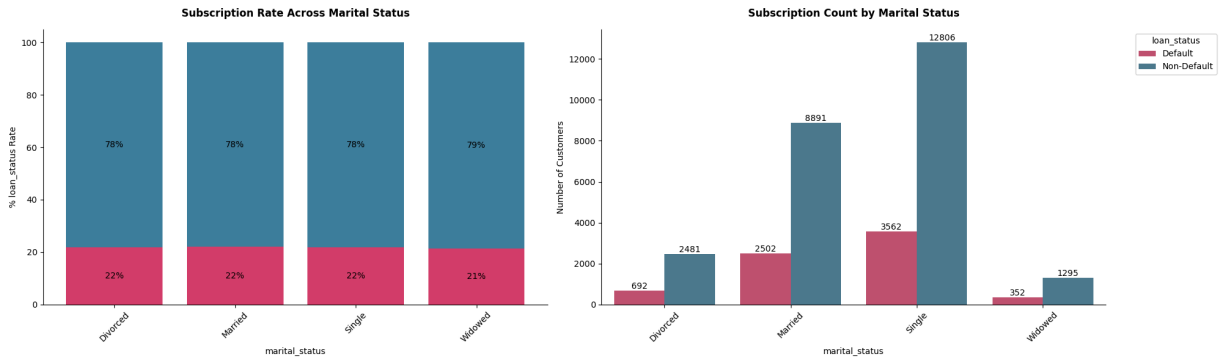


## Distribution of Gender by Loan\_Status



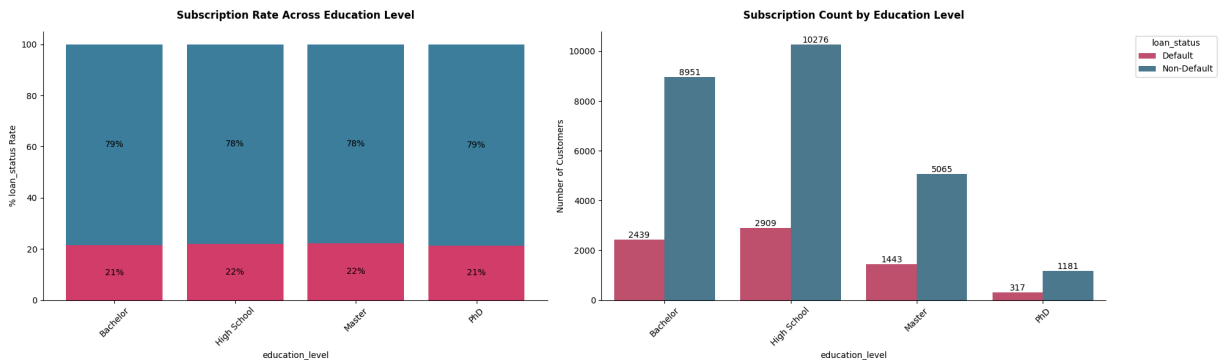
```
2026-04-18 19:01:25 | INFO      | Chi-Square Test of Independence: 'gender' v
s. 'loan_status'...
2026-04-18 19:01:25 | INFO      | Chi-squared statistic: 0.047
2026-04-18 19:01:25 | INFO      | Degrees of freedom: 1
2026-04-18 19:01:25 | INFO      | p-value: 0.828754
2026-04-18 19:01:25 | WARNING   | Result: p-value ≥ 0.05 → Fail to reject H0
2026-04-18 19:01:25 | WARNING   | No statistically significant association de
tected between 'gender' and 'loan_status'.
```

## Distribution of Marital\_Status by Loan\_Status



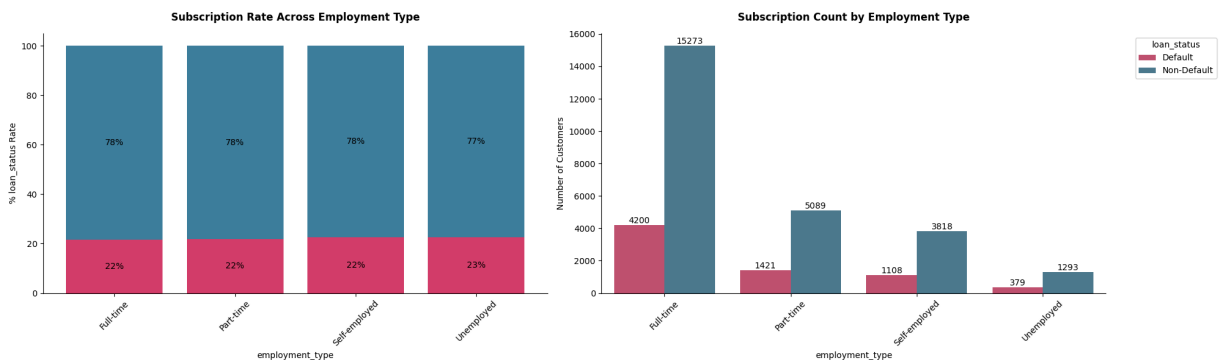
2026-04-18 19:01:26 | INFO | Chi-Square Test of Independence: 'marital\_s  
tatus' vs. 'loan\_status'...  
2026-04-18 19:01:26 | INFO | Chi-squared statistic: 0.358  
2026-04-18 19:01:26 | INFO | Degrees of freedom: 3  
2026-04-18 19:01:26 | INFO | p-value: 0.948688  
2026-04-18 19:01:26 | WARNING | Result:  $p\text{-value} \geq 0.05 \rightarrow$  Fail to reject  $H_0$   
2026-04-18 19:01:26 | WARNING | No statistically significant association de  
tected between 'marital\_status' and 'loan\_status'.

## Distribution of Education\_Level by Loan\_Status



2026-04-18 19:01:26 | INFO | Chi-Square Test of Independence: 'education  
\_level' vs. 'loan\_status'...  
2026-04-18 19:01:26 | INFO | Chi-squared statistic: 2.415  
2026-04-18 19:01:26 | INFO | Degrees of freedom: 3  
2026-04-18 19:01:26 | INFO | p-value: 0.490891  
2026-04-18 19:01:26 | WARNING | Result:  $p\text{-value} \geq 0.05 \rightarrow$  Fail to reject  $H_0$   
2026-04-18 19:01:26 | WARNING | No statistically significant association de  
tected between 'education\_level' and 'loan\_status'.

## Distribution of Employment\_Type by Loan\_Status



2026-04-18 19:01:26 | INFO | Chi-Square Test of Independence: 'employment\_type' vs. 'loan\_status'...

2026-04-18 19:01:26 | INFO | Chi-squared statistic: 2.735

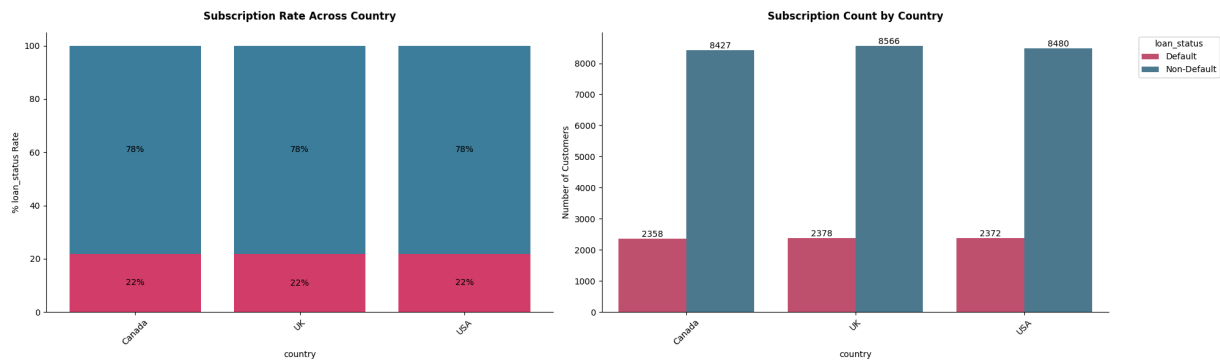
2026-04-18 19:01:26 | INFO | Degrees of freedom: 3

2026-04-18 19:01:26 | INFO | p-value: 0.434350

2026-04-18 19:01:26 | WARNING | Result:  $p\text{-value} \geq 0.05 \rightarrow$  Fail to reject  $H_0$

2026-04-18 19:01:26 | WARNING | No statistically significant association detected between 'employment\_type' and 'loan\_status'.

## Distribution of Country by Loan\_Status



2026-04-18 19:01:27 | INFO | Chi-Square Test of Independence: 'country' vs. 'loan\_status'...

2026-04-18 19:01:27 | INFO | Chi-squared statistic: 0.074

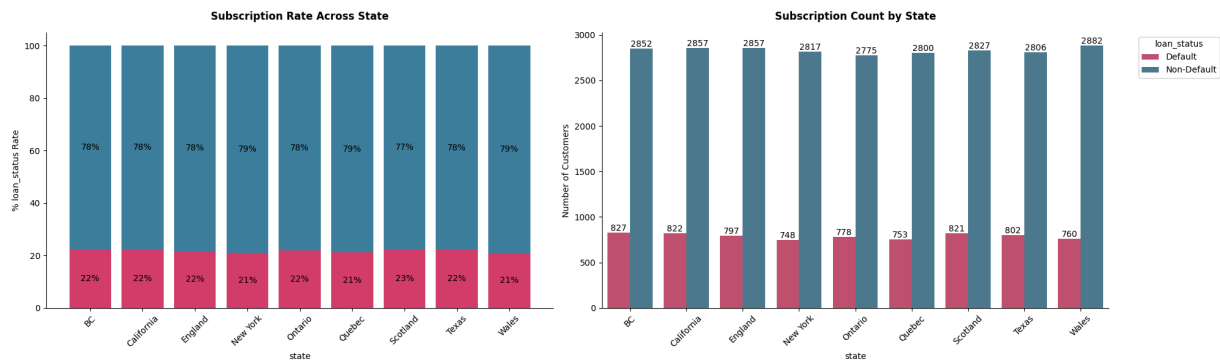
2026-04-18 19:01:27 | INFO | Degrees of freedom: 2

2026-04-18 19:01:27 | INFO | p-value: 0.963558

2026-04-18 19:01:27 | WARNING | Result:  $p\text{-value} \geq 0.05 \rightarrow$  Fail to reject  $H_0$

2026-04-18 19:01:27 | WARNING | No statistically significant association detected between 'country' and 'loan\_status'.

## Distribution of State by Loan\_Status



2026-04-18 19:01:27 | INFO | Chi-Square Test of Independence: 'state' vs. 'loan\_status'...

2026-04-18 19:01:27 | INFO | Chi-squared statistic: 7.120

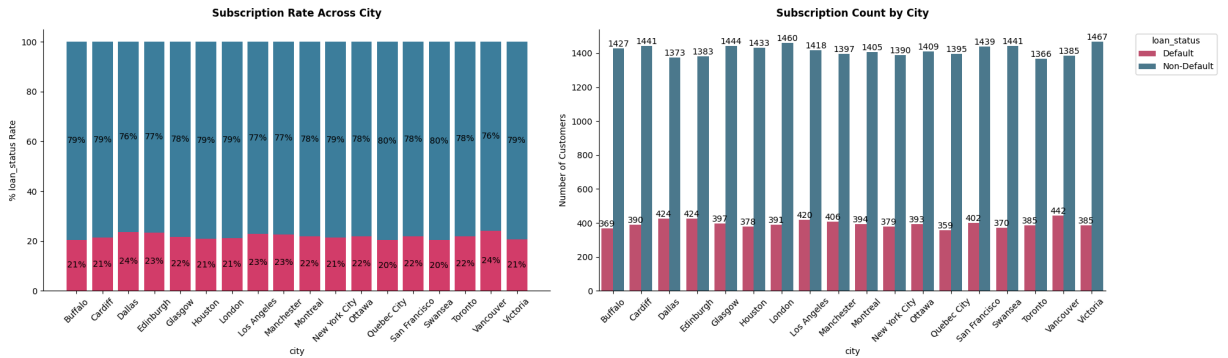
2026-04-18 19:01:27 | INFO | Degrees of freedom: 8

2026-04-18 19:01:27 | INFO | p-value: 0.523784

2026-04-18 19:01:27 | WARNING | Result:  $p\text{-value} \geq 0.05 \rightarrow$  Fail to reject  $H_0$

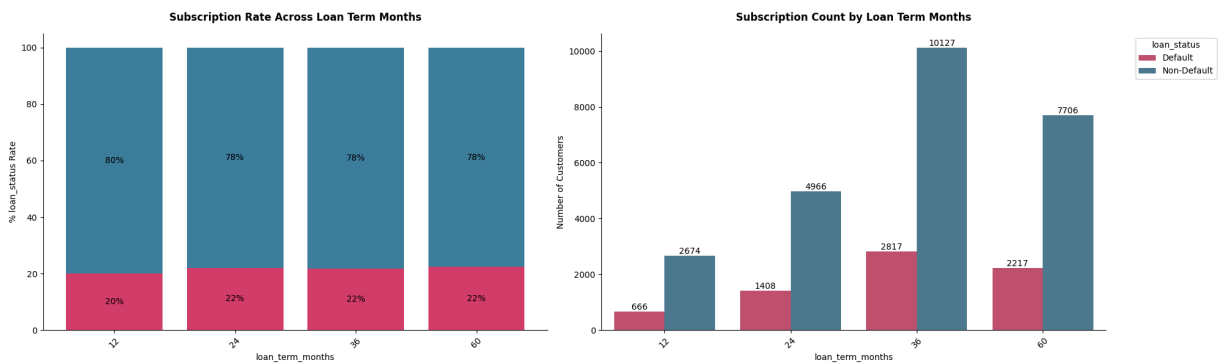
2026-04-18 19:01:27 | WARNING | No statistically significant association detected between 'state' and 'loan\_status'.

## Distribution of City by Loan\_Status



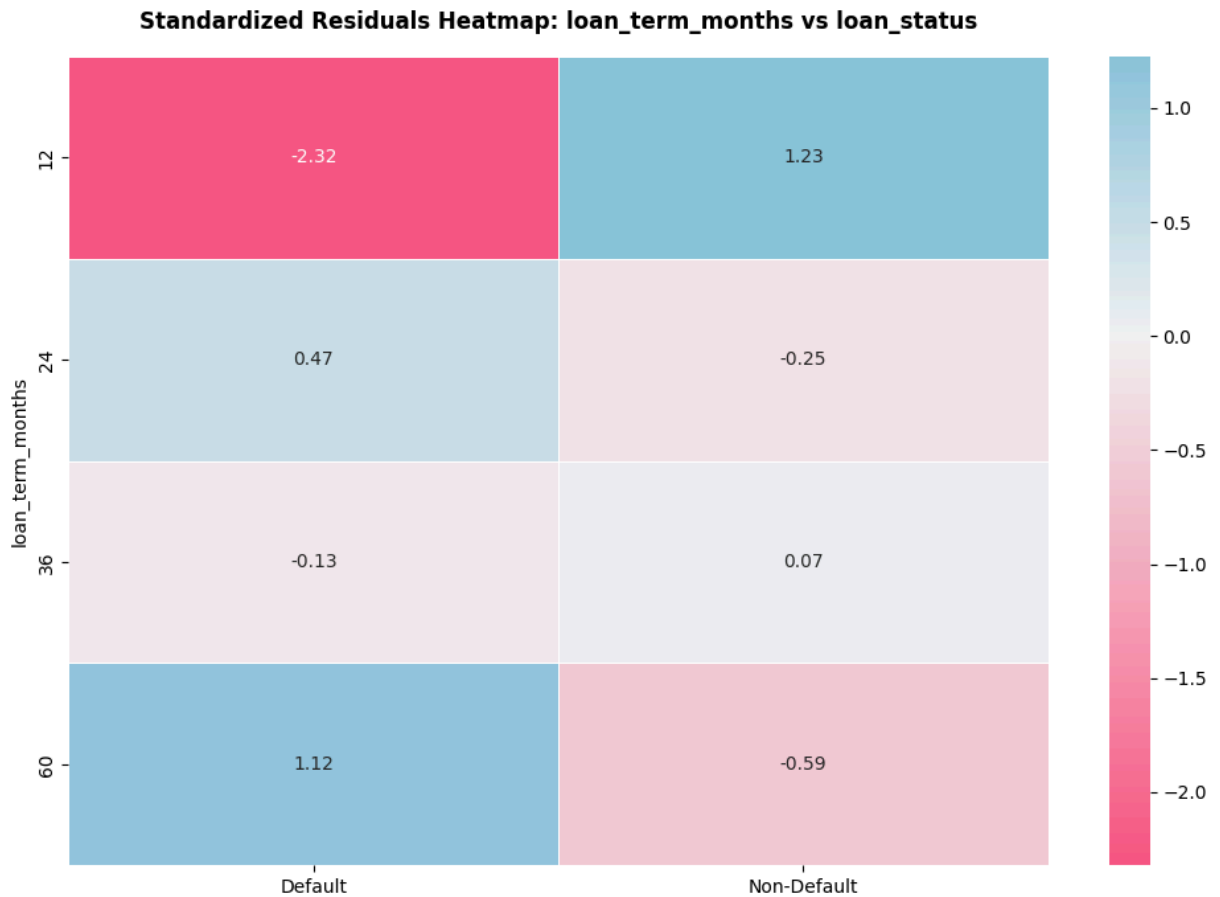
```
2026-04-18 19:01:28 | INFO      | Chi-Square Test of Independence: 'city' vs.
'loan_status'...
2026-04-18 19:01:28 | INFO      | Chi-squared statistic: 22.708
2026-04-18 19:01:28 | INFO      | Degrees of freedom: 17
2026-04-18 19:01:28 | INFO      | p-value: 0.159006
2026-04-18 19:01:28 | WARNING   | Result: p-value ≥ 0.05 → Fail to reject H0
2026-04-18 19:01:28 | WARNING   | No statistically significant association de
tected between 'city' and 'loan_status'.
```

## Distribution of Loan Term Months by Loan Status

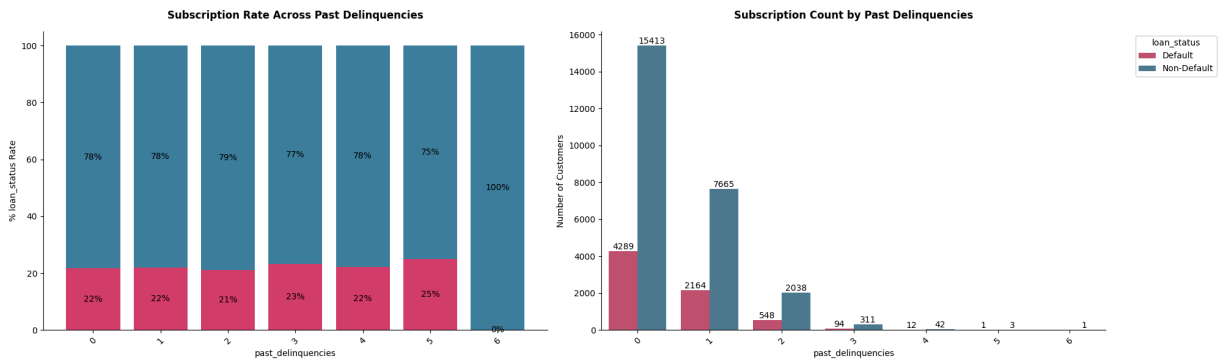


```
2026-04-18 19:01:28 | INFO      | Chi-Square Test of Independence: 'loan_term
_months' vs. 'loan_status'...
2026-04-18 19:01:28 | INFO      | Chi-squared statistic: 8.802
2026-04-18 19:01:28 | INFO      | Degrees of freedom: 3
2026-04-18 19:01:28 | INFO      | p-value: 0.032046
2026-04-18 19:01:28 | INFO      | Result: p-value < 0.05 → Reject H0
2026-04-18 19:01:28 | INFO      | Variables 'loan_term_months' and 'loan_stat
us' are statistically associated.
2026-04-18 19:01:28 | DEBUG     | Displaying standardized residuals heatmap.
```





## Distribution of Past Delinquencies by Loan\_Status



2026-04-18 19:01:29 | INFO | Chi-Square Test of Independence: 'past\_delinquencies' vs. 'loan\_status'...

2026-04-18 19:01:29 | INFO | Chi-squared statistic: 1.618

2026-04-18 19:01:29 | INFO | Chi-squared statistic: 1.618

2026-04-18 19:01:29 | INFO | Degrees of freedom: 6

2026-04-18 19:01:29 | INFO | p-value: 0.951256

2026-04-18 19:01:29 | WARNING | Result: p-value  $\geq 0.05 \rightarrow$  Fail to reject  $H_0$

2026-04-18 19:01:29 | WARNING | No statistically significant association detected between 'past\_delinquencies' and 'loan\_status'.

## 📊 Categorical Feature Insights

- **person\_home\_ownership:** Renters are the **highest-risk group** (~32% default), while homeowners (especially **OWN** ~7%) are the safest → **strong risk indicator**.
- **Loan\_Intent:** Clearly associated with default → **DEBTCONSOLIDATION & MEDICAL** have higher risk, while **EDUCATION & VENTURE** are safer segments.
- **Loan\_Grade:** **Strongest risk driver** → default rate increases sharply from **A** (~10%) → **G** (~98%).
- **Cb\_Person\_Default\_On\_File:** Prior default (**Y**) has much higher risk (~38% vs ~18%) → **strong and intuitive predictor**.
- **Gender:** Default rates are identical (~22%), p-value > 0.05 → **no predictive power**.
- **Marital\_Status:** Nearly identical (~21-22%), p-value > 0.05 → **no meaningful relationship**.
- **Education\_Level:** Very similar (~21-22%), p-value > 0.05 → **no meaningful impact**.
- **Employment\_Type:** Nearly identical (~22-23%), p-value > 0.05 → **no meaningful impact**.
- **Country:** Almost identical (~22%), p-value > 0.05 → **no meaningful impact**.
- **State:** Very similar (~21-23%), p-value > 0.05 → **no meaningful impact**.
- **City:** Consistent (~20-24%), p-value > 0.05 → **no meaningful impact**.
- **Loan\_Term\_Months:** Statistically significant but weak ( $p < 0.05$ ), with similar default rates (~20-22%).
- **Past\_Delinquencies:** No significant relationship (p-value > 0.05), default rates remain consistent (~21-25%).

## Overall Picture

## Overall Picuture

| Feature                    | Key Insight                                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| loan_grade                 | <b>Clear risk segmentation</b> → lower grades (D-G) have much higher default rates → useful for approval and pricing decisions. |
| loan_int_rate              | <b>Higher interest rates indicate higher risk borrowers</b> → strong signal of default likelihood.                              |
| debt_to_income_ratio       | <b>Core risk driver</b> → customers with higher financial burden are much more likely to default.                               |
| loan_percent_income        | <b>Repayment pressure matters</b> → higher loan-to-income ratio increases default risk.                                         |
| person_income              | <b>Higher income = safer borrowers</b> → strong indicator of repayment ability.                                                 |
| person_home_ownership      | <b>Housing stability matters</b> → renters are riskier, while homeowners are significantly safer.                               |
| cb_person_default_on_file  | <b>Past behavior predicts future risk</b> → customers with prior default are much more likely to default again.                 |
| loan_intent                | <b>Loan purpose impacts risk</b> → debt consolidation & medical loans are riskier; education & venture are safer.               |
| loan_amnt                  | <b>Larger loans increase exposure</b> → slightly higher default risk.                                                           |
| loan_to_income_ratio       | <b>Overlaps with loan_percent_income</b> → similar signal, should keep one to avoid redundancy.                                 |
| loan_term_months           | <b>Minimal impact</b> → default rates are similar across different loan durations.                                              |
| person_age                 | <b>Low business impact</b> → similar behavior across age groups.                                                                |
| person_emp_length          | <b>Weak signal of stability</b> → longer employment slightly reduces risk but not significant.                                  |
| cb_person_cred_hist_length | <b>Limited usefulness</b> → credit history length alone does not explain default well.                                          |

| Feature                                                                   | Key Insight                                                                                        |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>other_debt</b>                                                         | <b>Misleading signal</b> → absolute debt is less meaningful than ratios due to skew.               |
| <b>open_accounts</b>                                                      | <b>No meaningful impact</b> → similar behavior across groups.                                      |
| <b>credit_utilization_ratio</b>                                           | <b>No clear signal</b> → does not differentiate default behavior.                                  |
| <b>past_delinquencies</b>                                                 | <b>Weak predictor</b> → less informative than prior default flag.                                  |
| <b>gender / marital_status / education_level / country / state / city</b> | <b>No business impact</b> → default rates are consistent across demographic and geographic groups. |

## Data Pre-processing

### Handle skewness

```
In [18]: logger.info("-" * 80)
skew_feature = DataValidator.check_skewness(data=df, dataset_name=CFG.DATASET_NAME,
                                             numerical_features=num_features,
                                             logger=logger)
logger.info("-" * 80)
logger.info("Checking skewness successfully completed the dataset.")
```

```

2026-04-18 19:01:29 | INFO      | -----
-----
2026-04-18 19:01:29 | INFO      | Skewness analysis for Credit Risk Dataset
2026-04-18 19:01:29 | INFO      | Total numerical features checked: 12
2026-04-18 19:01:29 | INFO      | Skewness analysis for Credit Risk Dataset
2026-04-18 19:01:29 | INFO      | Total numerical features checked: 12
2026-04-18 19:01:29 | DEBUG     | -----
-----
2026-04-18 19:01:29 | DEBUG     | Feature                               | Skewness
| Remark
2026-04-18 19:01:29 | DEBUG     | -----
-----
2026-04-18 19:01:29 | DEBUG     | person_income                        | +32.8653
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | other_debt                           | +28.4619
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | person_emp_length                    | +2.6145
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | person_age                           | +2.5814
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | cb_person_cred_hist_length           | +1.6618
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | loan_amnt                            | +1.1925
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | loan_to_income_ratio                 | +1.0651
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | loan_percent_income                  | +1.0647
| Highly skewed
2026-04-18 19:01:29 | DEBUG     | debt_to_income_ratio                 | +0.6195
| Moderately skewed
2026-04-18 19:01:29 | DEBUG     | loan_int_rate                        | +0.2086
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG     | open_accounts                        | -0.0071
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG     | credit_utilization_ratio             | -0.0001
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG     | -----
-----
2026-04-18 19:01:29 | INFO      | 9 features are skewed (|skew| > 0.5).
2026-04-18 19:01:29 | INFO      | Recommendation: Consider log / Box-Cox / Ye
o-Johnson transformation.
2026-04-18 19:01:29 | INFO      | -----
-----
2026-04-18 19:01:29 | INFO      | Checking skewness successfully completed th
e dataset.

```

```
In [19]: processed_df, transformed_columns, sparse_columns, skewed_columns = FeatureRel
```

```

2026-04-18 19:01:29 | INFO      | Starting skewed feature handling
2026-04-18 19:01:29 | DEBUG     | Total numerical features provided: 9
2026-04-18 19:01:29 | DEBUG     | Total numerical features provided: 9
2026-04-18 19:01:29 | DEBUG     | Excluded columns: []
2026-04-18 19:01:29 | INFO      | Scanning 9 numerical columns
2026-04-18 19:01:29 | INFO      | Detected 9 skewed features
2026-04-18 19:01:29 | DEBUG     | Processing column: person_age
2026-04-18 19:01:29 | INFO      | person_age: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: cb_person_cred_hist_length
2026-04-18 19:01:29 | INFO      | cb_person_cred_hist_length: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: person_income
2026-04-18 19:01:29 | INFO      | person_income: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: other_debt
2026-04-18 19:01:29 | INFO      | other_debt: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: loan_to_income_ratio
2026-04-18 19:01:29 | INFO      | loan_to_income_ratio: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: person_emp_length
2026-04-18 19:01:29 | INFO      | person_emp_length: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: loan_amnt
2026-04-18 19:01:29 | INFO      | loan_amnt: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: loan_percent_income
2026-04-18 19:01:29 | INFO      | loan_percent_income: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | DEBUG     | Processing column: debt_to_income_ratio
2026-04-18 19:01:29 | INFO      | debt_to_income_ratio: applying Yeo-Johnson transformation
2026-04-18 19:01:29 | INFO      | Dropping original columns after transformation: ['person_age', 'cb_person_cred_hist_length', 'person_income', 'other_debt', 'loan_to_income_ratio', 'person_emp_length', 'loan_amnt', 'loan_percent_income', 'debt_to_income_ratio']
2026-04-18 19:01:29 | INFO      | Feature transformation completed. Generated 9 new features

```

```

In [20]: num_features = ["PT_person_age", "PT_person_income", "PT_person_emp_length",
                        "PT_loan_to_income_ratio", "PT_cb_person_cred_hist_length",
                        "credit_utilization_ratio"]
logger.info("-" * 80)
skew_feature = DataValidator.check_skewness(data=processed_df, dataset_name=
                                           numerical_features=num_features,
logger.info("-" * 80)
logger.info("Checking skewness successfully completed the dataset.")

```

```

2026-04-18 19:01:29 | INFO | -----
2026-04-18 19:01:29 | INFO | Skewness analysis for Credit Risk Dataset
2026-04-18 19:01:29 | INFO | Skewness analysis for Credit Risk Dataset
2026-04-18 19:01:29 | INFO | Total numerical features checked: 12
2026-04-18 19:01:29 | DEBUG | -----
2026-04-18 19:01:29 | DEBUG | Feature | Skewness
| Remark
2026-04-18 19:01:29 | DEBUG | -----
2026-04-18 19:01:29 | DEBUG | loan_int_rate | +0.2086
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_person_age | +0.1589
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_cb_person_cred_hist_length | +0.0956
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_loan_to_income_ratio | +0.0906
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_loan_percent_income | +0.0889
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_person_emp_length | -0.0423
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_loan_amnt | -0.0211
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_person_income | -0.0129
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_debt_to_income_ratio | +0.0116
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | open_accounts | -0.0071
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | PT_other_debt | +0.0035
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | credit_utilization_ratio | -0.0001
| Approximately symmetric
2026-04-18 19:01:29 | DEBUG | -----
2026-04-18 19:01:29 | INFO | No highly skewed features detected.
2026-04-18 19:01:29 | INFO | -----
2026-04-18 19:01:29 | INFO | Checking skewness successfully completed th
e dataset.

```

## 📊 Insight

### Observations from the Skewness Analysis:

#### Before Transformation:

- Most numerical features were **heavily right-skewed**, especially `person_income` and `other_debt` (extreme skew > 25).

- A total of **9 out of 12 features** showed significant skewness ( $|\text{skew}| > 0.5$ ), indicating **non-normal distributions and presence of outliers**.
- Only a few features such as `loan_int_rate`, `open_accounts`, and `credit_utilization_ratio` were approximately symmetric.

#### After Transformation (Yeo-Johnson / Power Transform):

- All transformed features (`PT_*`) became **approximately symmetric** ( $\text{skew} \approx 0$ ).
- **No highly skewed features remain**, indicating a successful normalization process.
- Extreme skewness in key financial variables (`person_income`, `other_debt`) was **effectively corrected**.
- Original symmetric features (`loan_int_rate`, `open_accounts`, `credit_utilization_ratio`) remained stable.

#### Conclusion:

- The transformation significantly **improved feature distributions**, reducing the impact of outliers.
- This enhances **model stability, convergence, and performance**, especially for linear and distance-based models.
- The dataset is now better suited for **machine learning modeling** with reduced bias from skewed features.

## Re-check Outlier

```
In [21]: logger.info("Start checking outlier dataset")
logger.info("-" * 80)
outlier_data = DataValidator.checkOutlier(df=processed_df, dataset_name=CFG.
                                          list_feature=num_features, logger=
logger.info("-" * 80)
logger.info("Outlier checking successfully completed across datasets.")
```



```

2026-04-18 19:01:29 | INFO      | Start checking outlier dataset
2026-04-18 19:01:29 | INFO      | -----
2026-04-18 19:01:29 | INFO      | Checking outliers for Credit Risk Data
t...
2026-04-18 19:01:29 | INFO      | -----
2026-04-18 19:01:29 | INFO      | -----
2026-04-18 19:01:29 | INFO      | Checking outliers for Credit Risk Data
t...
2026-04-18 19:01:29 | INFO      | -----
2026-04-18 19:01:29 | DEBUG     | [Credit Risk Dataset] PT_person_age: No out
liers detected
2026-04-18 19:01:29 | WARNING   | [Credit Risk Dataset] PT_person_income: 568
outliers detected (lower=-2.553, upper=2.553)
2026-04-18 19:01:29 | WARNING   | [Credit Risk Dataset] PT_person_emp_length:
44 outliers detected (lower=-2.547, upper=2.697)
2026-04-18 19:01:29 | WARNING   | [Credit Risk Dataset] PT_loan_amnt: 8 outli
ers detected (lower=-2.591, upper=2.553)
2026-04-18 19:01:29 | WARNING   | [Credit Risk Dataset] loan_int_rate: 6 outl
iers detected (lower=-0.455, upper=21.825)
2026-04-18 19:01:29 | DEBUG     | [Credit Risk Dataset] PT_loan_percent_incom
e: No outliers detected
2026-04-18 19:01:29 | DEBUG     | [Credit Risk Dataset] PT_loan_to_income_rat
io: No outliers detected
2026-04-18 19:01:29 | DEBUG     | [Credit Risk Dataset] PT_cb_person_cred_his
t_length: No outliers detected
2026-04-18 19:01:29 | WARNING   | [Credit Risk Dataset] PT_other_debt: 291 ou
tliers detected (lower=-2.666, upper=2.686)
2026-04-18 19:01:29 | WARNING   | [Credit Risk Dataset] PT_debt_to_income_rat
io: 56 outliers detected (lower=-2.775, upper=2.765)
2026-04-18 19:01:29 | DEBUG     | [Credit Risk Dataset] open_accounts: No out
liers detected
2026-04-18 19:01:29 | DEBUG     | [Credit Risk Dataset] credit_utilization_ra
tio: No outliers detected
2026-04-18 19:01:29 | INFO      | -----
2026-04-18 19:01:29 | INFO      | Outlier checking successfully completed acr
oss datasets.

```

## Create Test Set

```

In [22]: X = processed_df.drop(columns=CFG.TARGET_FEATURE)
y = processed_df[CFG.TARGET_FEATURE].map({"Non-Default": 0, "Default": 1})

split = StratifiedShuffleSplit(n_splits=CFG.SPLIT["n_splits"], test_size=CFG
                                random_state=CFG.RANDOM_STATE)

for train_idx, test_idx in split.split(X, y):
    X_train = X.iloc[train_idx]
    X_test = X.iloc[test_idx]

```

```
y_train = y.iloc[train_idx]
y_test = y.iloc[test_idx]
```

## Scale and Encode

```
In [23]: list_feature_num_stand = ["PT_person_age", "PT_loan_percent_income", "PT_loan_term",
                                   "credit_utilization_ratio"]
list_feature_num_robust = ["PT_person_income", "PT_person_emp_length", "PT_loan_term",
                           "PT_debt_to_income_ratio"]

num_robust_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", RobustScaler())
])

num_stand_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

cat_onehot_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(transformers=[
    ("num_robust", num_robust_transformer, list_feature_num_robust),
    ("num_standard", num_stand_transformer, list_feature_num_stand),
    ("cat_onehot", cat_onehot_transformer, cat_features)
])

preprocessor.fit(X_train)
X_train_prepared = preprocessor.transform(X_train)
list_feature_prepared = preprocessor.get_feature_names_out().tolist()
clean_features = [col.replace("num_standard__", "").replace("num_robust__", "") for col in list_feature_prepared]
clean_features
```

```
Out[23]: ['person_income',
          'person_emp_length',
          'loan_amnt',
          'loan_int_rate',
          'other_debt',
          'debt_to_income_ratio',
          'person_age',
          'loan_percent_income',
          'loan_to_income_ratio',
          'cb_person_cred_hist_length',
          'open_accounts',
          'credit_utilization_ratio',
          'person_home_ownership_MORTGAGE',
          'person_home_ownership_OTHER',
          'person_home_ownership_OWN',
          'person_home_ownership_RENT',
          'loan_intent_DEBTCONSOLIDATION',
          'loan_intent_EDUCATION',
          'loan_intent_HOMEIMPROVEMENT',
          'loan_intent_MEDICAL',
          'loan_intent_PERSONAL',
          'loan_intent_VENTURE',
          'loan_grade_A',
          'loan_grade_B',
          'loan_grade_C',
          'loan_grade_D',
          'loan_grade_E',
          'loan_grade_F',
          'loan_grade_G',
          'cb_person_default_on_file_N',
          'cb_person_default_on_file_Y',
          'gender_Female',
          'gender_Male',
          'marital_status_Divorced',
          'marital_status_Married',
          'marital_status_Single',
          'marital_status_Widowed',
          'education_level_Bachelor',
          'education_level_High School',
          'education_level_Master',
          'education_level_PhD',
          'employment_type_Full-time',
          'employment_type_Part-time',
          'employment_type_Self-employed',
          'employment_type_Unemployed',
          'country_Canada',
          'country_UK',
          'country_USA',
          'state_BC',
          'state_California',
          'state_England',
          'state_New York',
          'state_Ontario',
          'state_Quebec',
          'state_Scotland',
          'state_Texas',
```

```
'state_Wales',  
'city_Buffalo',  
'city_Cardiff',  
'city_Dallas',  
'city_Edinburgh',  
'city_Glasgow',  
'city_Houston',  
'city_London',  
'city_Los Angeles',  
'city_Manchester',  
'city_Montreal',  
'city_New York City',  
'city_Ottawa',  
'city_Quebec City',  
'city_San Francisco',  
'city_Swansea',  
'city_Toronto',  
'city_Vancouver',  
'city_Victoria',  
'loan_term_months_12',  
'loan_term_months_24',  
'loan_term_months_36',  
'loan_term_months_60',  
'past_delinquencies_0',  
'past_delinquencies_1',  
'past_delinquencies_2',  
'past_delinquencies_3',  
'past_delinquencies_4',  
'past_delinquencies_5']
```

## Metric

### AUC-ROC Overview

**AUC-ROC** evaluates a model's ability to **distinguish between two classes**.

- **ROC curve** plots **TPR vs FPR**
- **AUC** = Area under ROC curve → **ranking quality of predictions**

### Why AUC-ROC?

- Robust to **class imbalance**
- Measures **ranking ability**, not fixed threshold (0.5)
- Common metric in **credit risk & fraud prediction**

Higher AUC → better separation between *Paid* vs *Default*

## Interpretation

| AUC Score | Meaning                  |
|-----------|--------------------------|
| 1.0       | Perfect model            |
| 0.9+      | Excellent discrimination |
| 0.7 - 0.9 | Good                     |
| 0.5       | Random guessing          |
| < 0.5     | Worse than random        |

(Source: [ROC AUC](#))

# Modeling

## CatBoostClassifier

```
In [24]: study_cat = optuna.create_study(direction="maximize")
study_cat.optimize(
    lambda trial: objective_catboost(trial, X_train_prepared, y_train, logger,
    n_trials=10, show_progress_bar=True
)
```

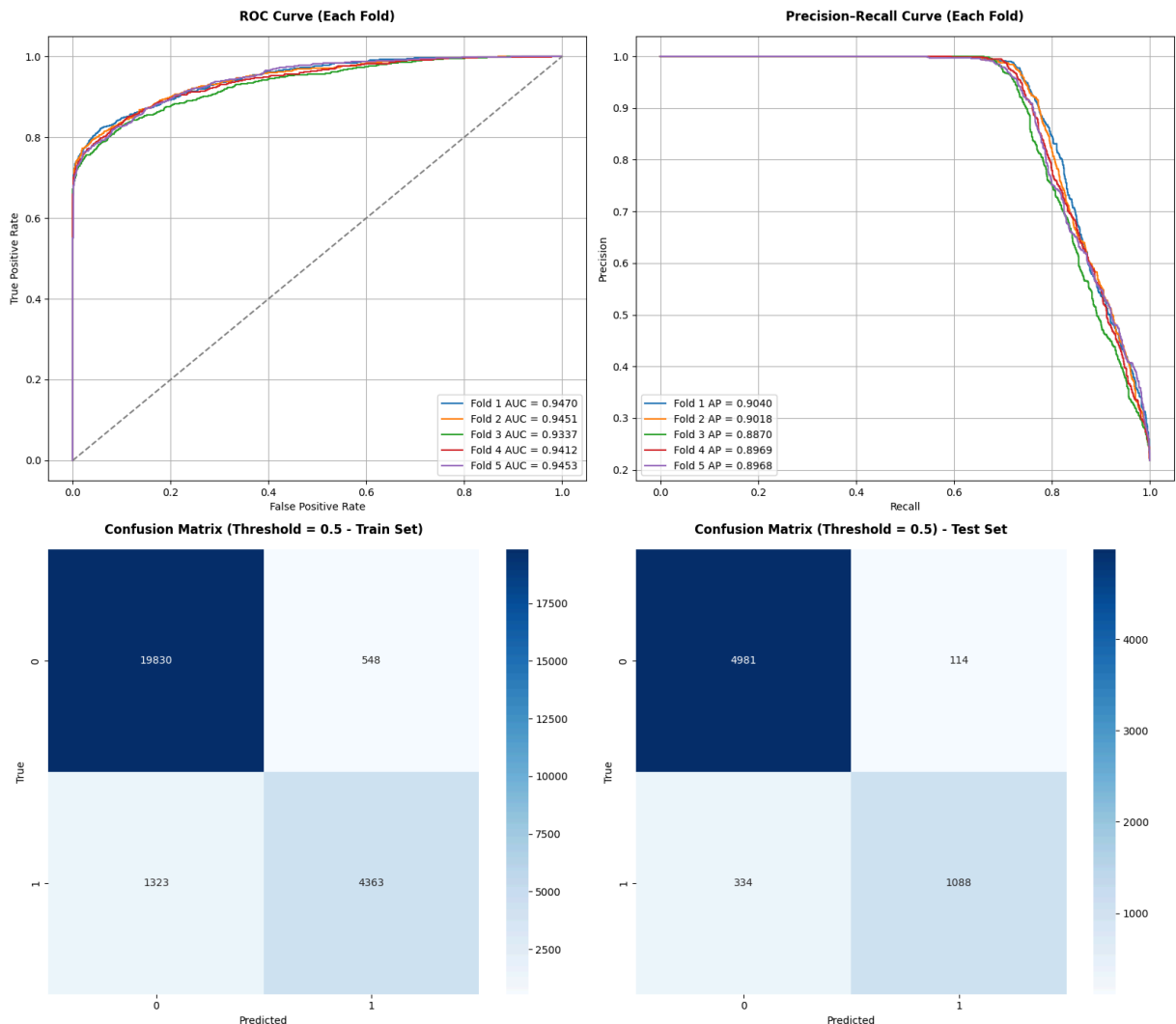
```
0%|          | 0/10 [00:00<?, ?it/s]
2026-04-18 19:02:17 | INFO      | Trial 0 finished | Mean AUC: 0.94123
2026-04-18 19:03:50 | INFO      | Trial 1 finished | Mean AUC: 0.94255
2026-04-18 19:04:39 | INFO      | Trial 2 finished | Mean AUC: 0.93542
2026-04-18 19:06:37 | INFO      | Trial 3 finished | Mean AUC: 0.93232
2026-04-18 19:07:22 | INFO      | Trial 4 finished | Mean AUC: 0.92501
2026-04-18 19:09:07 | INFO      | Trial 5 finished | Mean AUC: 0.94011
2026-04-18 19:10:13 | INFO      | Trial 6 finished | Mean AUC: 0.93276
2026-04-18 19:11:12 | INFO      | Trial 7 finished | Mean AUC: 0.92533
2026-04-18 19:12:37 | INFO      | Trial 8 finished | Mean AUC: 0.93866
2026-04-18 19:13:28 | INFO      | Trial 9 finished | Mean AUC: 0.93947
```

```
In [26]: model_cb = CatBoostClassifier(**study_cat.best_params, loss_function="Logloss",
    verbose = 0, thread_count=-1, eval_metric="AUC",
    allow_writing_files = False)
X_test_prepared = preprocessor.transform(X_test)
oof_pred_cb, test_pred_cb, fold_auc_cb, fold_models_cb = run_oof(model = model_cb, X_train = X_train_prepared,
    X_test = X_test_prepared)
```

```

2026-04-18 19:13:28 | INFO | Starting OOF training with CatBoostClassifier
2026-04-18 19:13:28 | INFO | Fold 1/5
2026-04-18 19:14:00 | INFO | Fold 1 AUC: 0.94701
2026-04-18 19:14:00 | INFO | Fold 2/5
2026-04-18 19:14:31 | INFO | Fold 2 AUC: 0.94510
2026-04-18 19:14:31 | INFO | Fold 3/5
2026-04-18 19:15:02 | INFO | Fold 3 AUC: 0.93367
2026-04-18 19:15:02 | INFO | Fold 4/5
2026-04-18 19:15:33 | INFO | Fold 4 AUC: 0.94118
2026-04-18 19:15:33 | INFO | Fold 5/5
2026-04-18 19:16:04 | INFO | Fold 5 AUC: 0.94527
2026-04-18 19:16:04 | INFO | OOF training completed
2026-04-18 19:16:04 | INFO | OOF AUC: 0.94237
2026-04-18 19:16:04 | INFO | Fold AUCs: [0.9470057905943623, 0.9450987525494065, 0.9336658026951423, 0.9411823563026298, 0.9452650662868056]

```



## LGBMClassifier

```

In [27]: study_lgbm = optuna.create_study(direction="maximize")
study_lgbm.optimize(lambda trial: objective_lgbm(trial, X_train_prepared, y,
n_trials=10, show_progress_bar=True
)

```

```

0%|          | 0/10 [00:00<?, ?it/s]
2026-04-18 19:29:54 | INFO      | Trial 0 finished | Mean AUC: 0.93586
2026-04-18 19:30:28 | INFO      | Trial 1 finished | Mean AUC: 0.94115
2026-04-18 19:31:47 | INFO      | Trial 2 finished | Mean AUC: 0.94117
2026-04-18 19:34:35 | INFO      | Trial 3 finished | Mean AUC: 0.94231
2026-04-18 19:35:49 | INFO      | Trial 4 finished | Mean AUC: 0.94454
2026-04-18 19:37:17 | INFO      | Trial 5 finished | Mean AUC: 0.94131
2026-04-18 19:37:53 | INFO      | Trial 6 finished | Mean AUC: 0.94361
2026-04-18 19:39:06 | INFO      | Trial 7 finished | Mean AUC: 0.94451
2026-04-18 19:40:24 | INFO      | Trial 8 finished | Mean AUC: 0.94322
2026-04-18 19:43:00 | INFO      | Trial 9 finished | Mean AUC: 0.94423

```

```

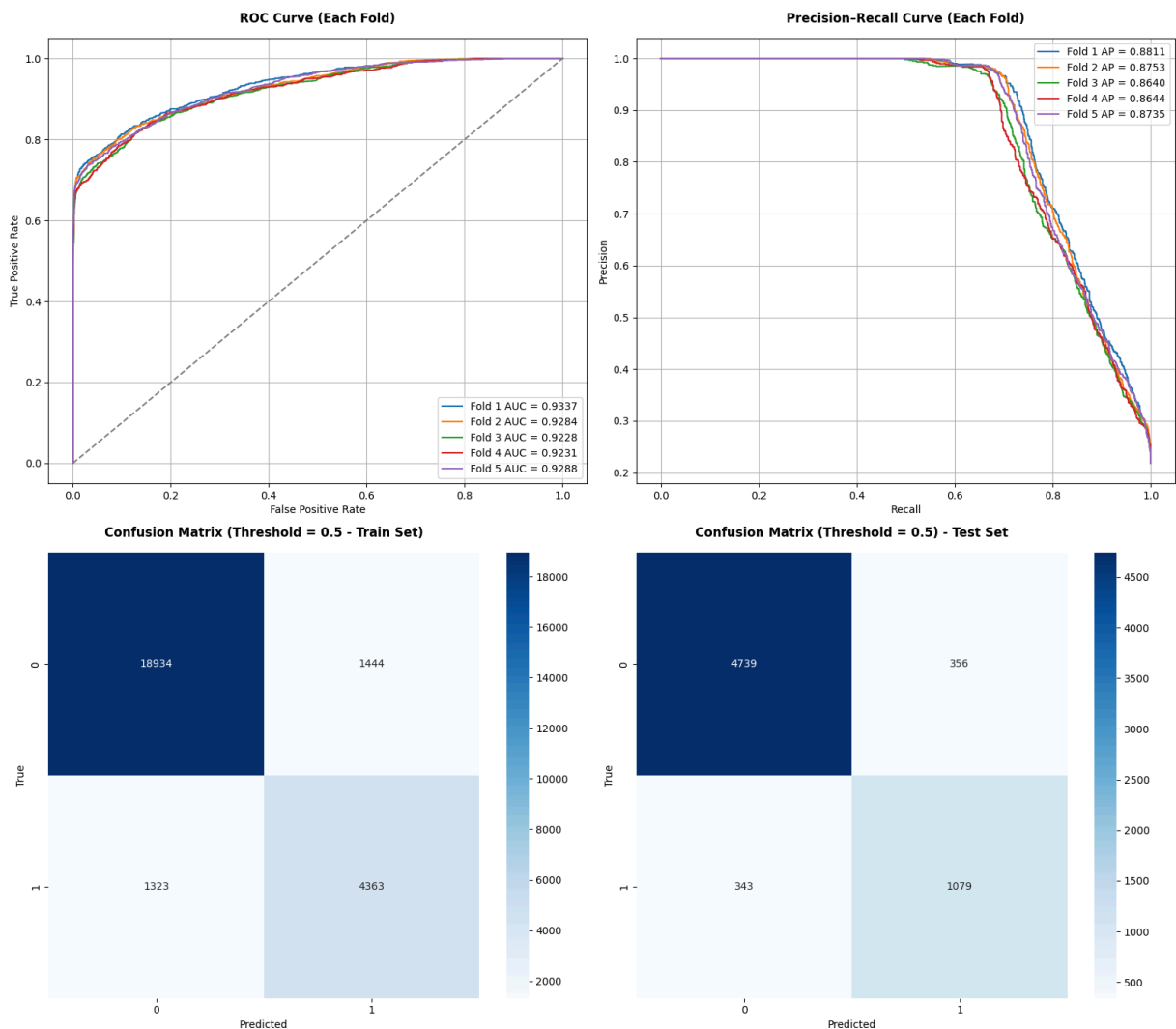
In [28]: model_lgbm = LGBMClassifier(**study_lgbm.best_params, objective="binary", me
        verbosity=-1, n_jobs=-1, boosting_type="gbdt",
        class_weight="balanced", random_state=CFG.RANDOM
        oof_pred_lgbm, test_pred_lgbm, fold_auc_lgbm, fold_models_lgbm = run_oof(moc
        X_t

```

```

2026-04-18 19:43:54 | INFO      | Starting OOF training with LGBMClassifier
2026-04-18 19:43:54 | INFO      | Fold 1/5
2026-04-18 19:43:55 | INFO      | Fold 1 AUC: 0.93369
2026-04-18 19:43:55 | INFO      | Fold 2/5
2026-04-18 19:43:55 | INFO      | Fold 2 AUC: 0.92841
2026-04-18 19:43:55 | INFO      | Fold 3/5
2026-04-18 19:43:55 | INFO      | Fold 3 AUC: 0.92279
2026-04-18 19:43:55 | INFO      | Fold 4/5
2026-04-18 19:43:55 | INFO      | Fold 4 AUC: 0.92307
2026-04-18 19:43:55 | INFO      | Fold 5/5
2026-04-18 19:43:56 | INFO      | Fold 5 AUC: 0.92877
2026-04-18 19:43:56 | INFO      | OOF training completed
2026-04-18 19:43:56 | INFO      | OOF AUC: 0.92721
2026-04-18 19:43:56 | INFO      | Fold AUCs: [0.9336923432789317, 0.928413140
6530105, 0.922789881434797, 0.9230737382341208, 0.9287739234127048]

```



## HistGradientBoostingClassifier

```
In [30]: X_train_dense = X_train_prepared.toarray()

study_hgbc = optuna.create_study(direction="maximize")
study_hgbc.optimize(lambda trial: objective_histgb(trial, X_train_dense, y_train_dense,
n_trials=10, show_progress_bar=True
)
```

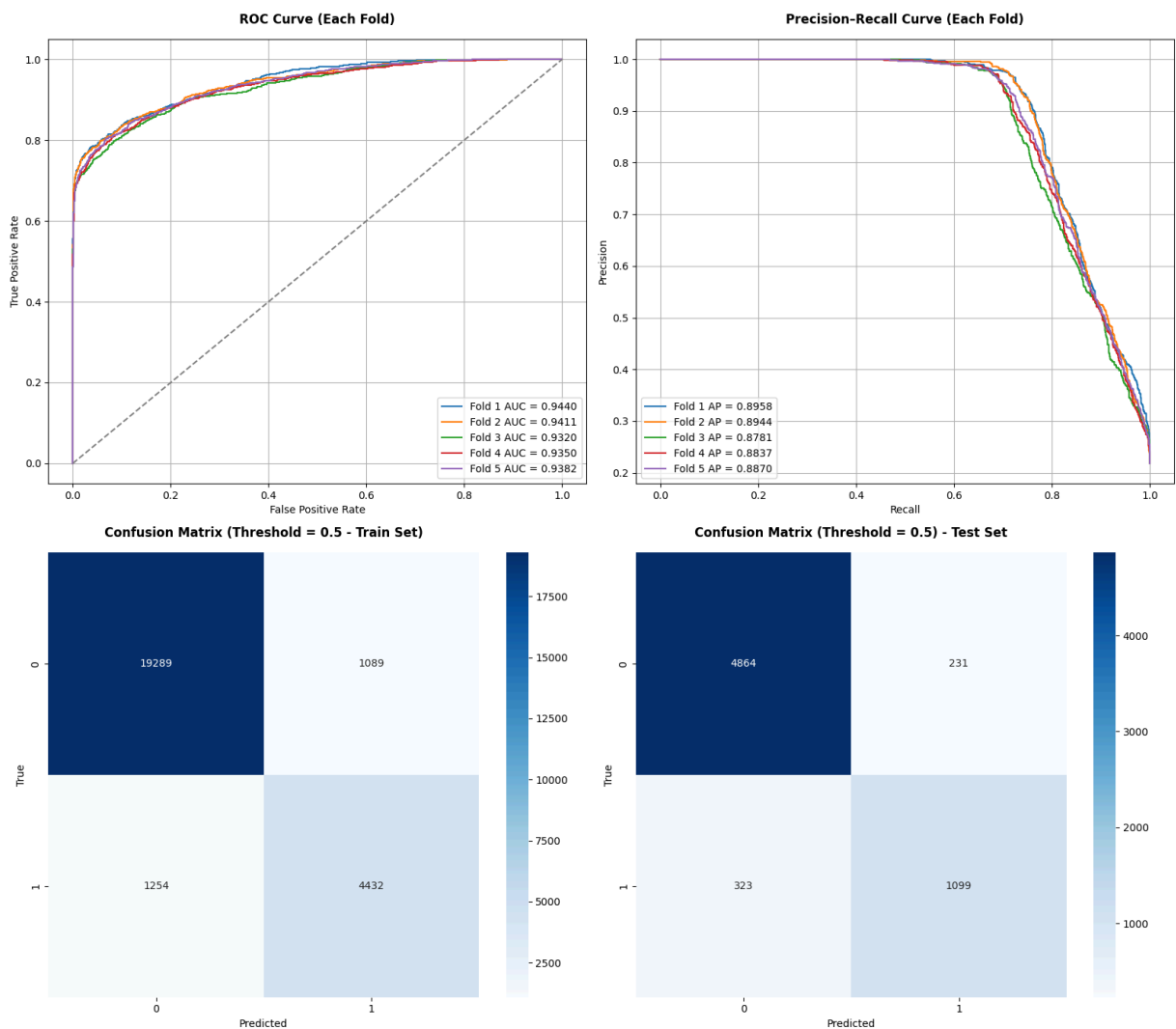
```
0%|          | 0/10 [00:00<?, ?it/s]
2026-04-18 19:48:25 | INFO | Trial 0 finished | Mean AUC: 0.93805
2026-04-18 19:48:31 | INFO | Trial 1 finished | Mean AUC: 0.93516
2026-04-18 19:48:47 | INFO | Trial 2 finished | Mean AUC: 0.93405
2026-04-18 19:48:56 | INFO | Trial 3 finished | Mean AUC: 0.93633
2026-04-18 19:49:05 | INFO | Trial 4 finished | Mean AUC: 0.93593
2026-04-18 19:49:14 | INFO | Trial 5 finished | Mean AUC: 0.93075
2026-04-18 19:49:18 | INFO | Trial 6 finished | Mean AUC: 0.93168
2026-04-18 19:49:26 | INFO | Trial 7 finished | Mean AUC: 0.93724
2026-04-18 19:49:30 | INFO | Trial 8 finished | Mean AUC: 0.93049
2026-04-18 19:49:37 | INFO | Trial 9 finished | Mean AUC: 0.93559
```



```
In [31]: model_hgbc = HistGradientBoostingClassifier(**study_hgbc.best_params, n_iter
                                                    class_weight="balanced", random_

X_test_dense = X_test_prepared.toarray()
oof_pred_hgbc, test_pred_hgbc, fold_auc_hgbc, fold_models_hgbc = run_oof(model_hgbc, X_train_dense, X_test_dense)
```

```
2026-04-18 19:52:19 | INFO | Starting OOF training with HistGradientBoos
tingClassifier
2026-04-18 19:52:19 | INFO | Fold 1/5
2026-04-18 19:52:22 | INFO | Fold 1 AUC: 0.94398
2026-04-18 19:52:22 | INFO | Fold 2/5
2026-04-18 19:52:25 | INFO | Fold 2 AUC: 0.94109
2026-04-18 19:52:25 | INFO | Fold 3/5
2026-04-18 19:52:27 | INFO | Fold 3 AUC: 0.93203
2026-04-18 19:52:27 | INFO | Fold 4/5
2026-04-18 19:52:29 | INFO | Fold 4 AUC: 0.93497
2026-04-18 19:52:29 | INFO | Fold 5/5
2026-04-18 19:52:31 | INFO | Fold 5 AUC: 0.93819
2026-04-18 19:52:31 | INFO | OOF training completed
2026-04-18 19:52:31 | INFO | OOF AUC: 0.93798
2026-04-18 19:52:31 | INFO | Fold AUCs: [0.9439801640423854, 0.941089182
403291, 0.9320256809278071, 0.9349693251533743, 0.9381944736714312]
```



# Evaluation

```
In [32]: max_index_cat = fold_auc_cb.index(max(fold_auc_cb))
max_index_lgbm = fold_auc_lgbm.index(max(fold_auc_lgbm))
max_index_hgbc = fold_auc_hgbc.index(max(fold_auc_hgbc))

list_model = [fold_models_cb[max_index_cat], fold_models_lgbm[max_index_lgbm]
shap_compare_models(models=list_model, X_test=X_test_prepared, logger=logger
```

```
2026-04-18 19:53:29 | INFO      | Starting SHAP comparison across models
2026-04-18 19:53:29 | DEBUG    | Converting sparse matrix to dense
2026-04-18 19:53:29 | INFO      | Sampling 1000 rows for SHAP computation
2026-04-18 19:53:29 | INFO      | Computing SHAP values for CatBoostClassifier
```

```
PermutationExplainer explainer: 100lit [00:56, 16.32it/s]
```

```
2026-04-18 19:54:26 | INFO      |
feature importance
8      loan_to_income_ratio 0.221547
7      loan_percent_income 0.176555
0      person_income 0.056426
25     loan_grade_D 0.051555
15     person_home_ownership_RENT 0.048084
3      loan_int_rate 0.039354
14     person_home_ownership_OWN 0.028232
21     loan_intent_VENTURE 0.021948
18     loan_intent_HOMEIMPROVEMENT 0.019900
4      other_debt 0.019029
22     loan_grade_A 0.018784
5      debt_to_income_ratio 0.017723
1      person_emp_length 0.015921
16     loan_intent_DEBTCONSOLIDATION 0.014913
2      loan_amnt 0.013168
6      person_age 0.010306
11     credit_utilization_ratio 0.009997
17     loan_intent_EDUCATION 0.009494
23     loan_grade_B 0.008815
26     loan_grade_E 0.008112
2026-04-18 19:54:26 | INFO      | *****
*****
```

```
2026-04-18 19:54:26 | INFO      | Computing SHAP values for LGBMClassifier
```

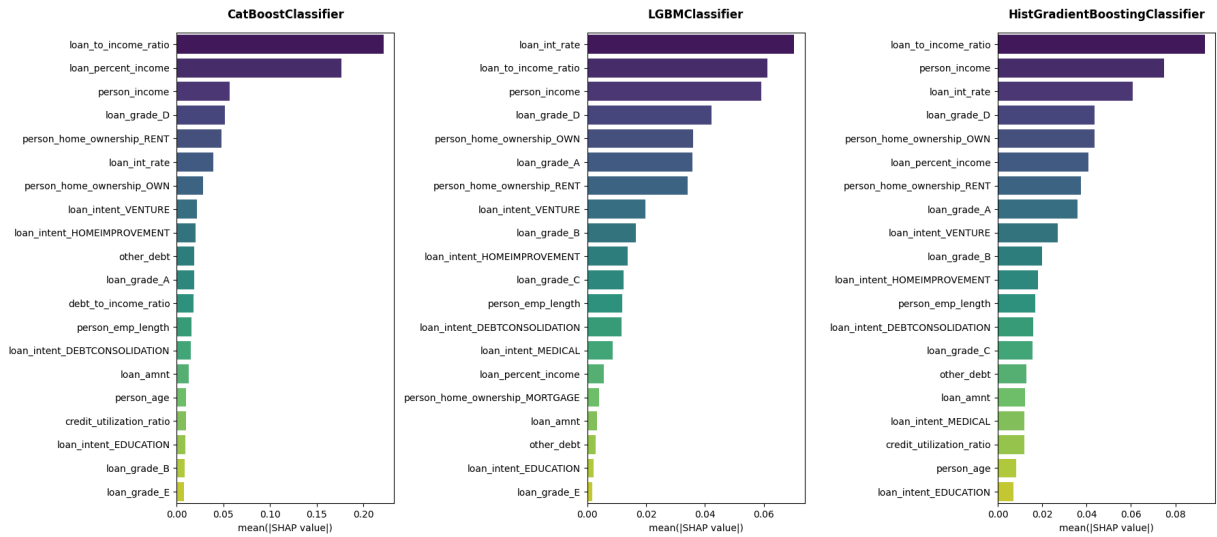
```
PermutationExplainer explainer: 100lit [00:33, 20.76it/s]
```

```
2026-04-18 19:54:59 | INFO |
                                feature importance
3                loan_int_rate    0.070301
8            loan_to_income_ratio  0.061172
0                person_income    0.059029
25               loan_grade_D     0.042168
14    person_home_ownership_OWN  0.036003
22               loan_grade_A     0.035737
15    person_home_ownership_RENT  0.034165
21               loan_intent_VENTURE 0.019733
23               loan_grade_B     0.016412
18    loan_intent_HOMEIMPROVEMENT 0.013639
24               loan_grade_C     0.012334
1                person_emp_length 0.011994
16    loan_intent_DEBTCONSOLIDATION 0.011554
19               loan_intent_MEDICAL 0.008621
7                loan_percent_income 0.005674
12    person_home_ownership_MORTGAGE 0.004015
2                loan_amnt        0.003412
4                other_debt       0.002774
17               loan_intent_EDUCATION 0.002205
26               loan_grade_E     0.001685
2026-04-18 19:54:59 | INFO | *****
*****
```

```
2026-04-18 19:54:59 | INFO | Computing SHAP values for HistGradientBoost
ingClassifier
```

```
PermutationExplainer explainer: 100lit [03:42, 4.30it/s]
```

```
2026-04-18 19:58:42 | INFO |
                                feature importance
8            loan_to_income_ratio  0.093277
0                person_income    0.075033
3                loan_int_rate    0.060877
25               loan_grade_D     0.043473
14    person_home_ownership_OWN  0.043447
7                loan_percent_income 0.040683
15    person_home_ownership_RENT  0.037344
22               loan_grade_A     0.035909
21               loan_intent_VENTURE 0.027003
23               loan_grade_B     0.019904
18    loan_intent_HOMEIMPROVEMENT 0.018163
1                person_emp_length 0.016851
16    loan_intent_DEBTCONSOLIDATION 0.015930
24               loan_grade_C     0.015702
4                other_debt       0.013002
2                loan_amnt        0.012279
19               loan_intent_MEDICAL 0.012023
11    credit_utilization_ratio    0.011811
6                person_age       0.008396
17               loan_intent_EDUCATION 0.007148
2026-04-18 19:58:42 | INFO | *****
*****
```



2026-04-18 19:58:42 | INFO | SHAP comparison completed

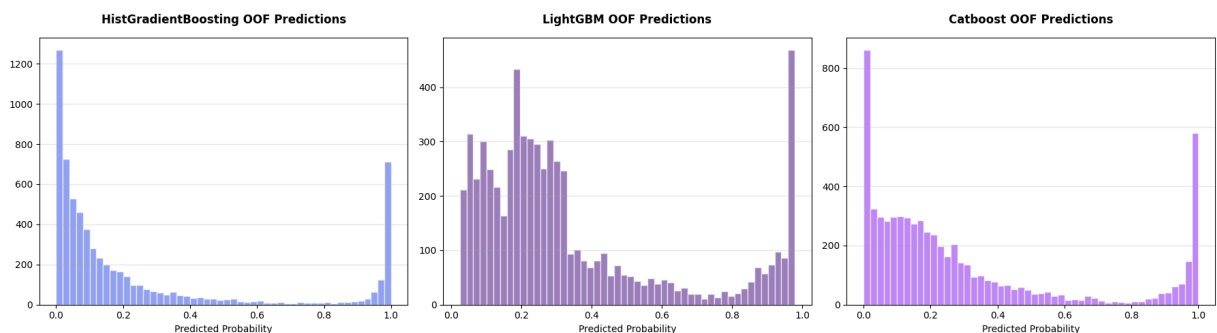
```
In [33]: fig, ax = plt.subplots(1, 3, figsize=(18, 5))
COLORS = ["#667eea", "#764ba2", "#a855f7", "#f093fb", "#fbc2eb"]

ax[0].hist(test_pred_cb, bins=50, color=COLORS[0], alpha=0.7, edgecolor="white")
ax[0].set_title("HistGradientBoosting OOF Predictions", fontsize=12, fontweight="bold")
ax[0].set_xlabel("Predicted Probability", fontsize=10)
ax[0].grid(axis="y", alpha=0.3)

ax[1].hist(test_pred_lgbm, bins=50, color=COLORS[1], alpha=0.7, edgecolor="white")
ax[1].set_title("LightGBM OOF Predictions", fontsize=12, fontweight="bold")
ax[1].set_xlabel("Predicted Probability", fontsize=10)
ax[1].grid(axis="y", alpha=0.3)

ax[2].hist(test_pred_hgbc, bins=50, color=COLORS[2], alpha=0.7, edgecolor="white")
ax[2].set_title("Catboost OOF Predictions", fontsize=12, fontweight="bold")
ax[2].set_xlabel("Predicted Probability", fontsize=10)
ax[2].grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.show()
```



```
In [34]: hgbc_binary = (test_pred_hgbc >= 0.5).astype(int)
lgb_binary = (test_pred_lgbm >= 0.5).astype(int)
cat_binary = (test_pred_cb >= 0.5).astype(int)

logger.info("HistGradientBoosting Classification Report:")
logger.info("\n" + classification_report(y_test, hgbc_binary, target_names=[
```

```

logger.info("*" * 80)

logger.info("LightGBM Classification Report:")
logger.info("\n" + classification_report(y_test, lgb_binary, target_names=['
logger.info("*" * 80)

logger.info("CatBoost Classification Report:")
logger.info("\n" + classification_report(y_test, cat_binary, target_names=['
logger.info("*" * 80)

```

2026-04-18 19:59:32 | INFO | HistGradientBoosting Classification Report:

```

2026-04-18 19:59:32 | INFO |
precision    recall  f1-score   support

      No      0.94      0.95      0.95      5095
      Yes      0.83      0.77      0.80      1422

 accuracy          0.91      6517
 macro avg          0.88      0.86      0.87      6517
weighted avg          0.91      0.91      0.91      6517

```

2026-04-18 19:59:32 | INFO | \*\*\*\*\*  
\*\*\*\*\*

2026-04-18 19:59:32 | INFO | LightGBM Classification Report:

```

2026-04-18 19:59:32 | INFO |
precision    recall  f1-score   support

      No      0.93      0.93      0.93      5095
      Yes      0.75      0.76      0.76      1422

 accuracy          0.89      6517
 macro avg          0.84      0.84      0.84      6517
weighted avg          0.89      0.89      0.89      6517

```

2026-04-18 19:59:32 | INFO | \*\*\*\*\*  
\*\*\*\*\*

2026-04-18 19:59:32 | INFO | CatBoost Classification Report:

```

2026-04-18 19:59:32 | INFO |
precision    recall  f1-score   support

      No      0.94      0.98      0.96      5095
      Yes      0.91      0.77      0.83      1422

 accuracy          0.93      6517
 macro avg          0.92      0.87      0.89      6517
weighted avg          0.93      0.93      0.93      6517

```

2026-04-18 19:59:32 | INFO | \*\*\*\*\*  
\*\*\*\*\*

# Recommendation

## □ Key Business Recommendations

*Key drivers of default are **financial capacity** and **credit behavior**.*

- **Focus on Repayment Capacity (Most Important)**
  - Use **DTI & income ratios** as core approval criteria
  - High ratio → **reject or increase interest rate**
- **Leverage Loan Grade for Risk Segmentation**
  - A-B: approve & low rate
  - C-D: controlled lending
  - E-G: **restrict / tighten conditions**
- **Flag High-Risk Customers**
  - Prior default (`cb_person_default_on_file = Y`) → **strong reject or strict policy**
  - Renters + low income → **higher monitoring**
- **Adjust by Loan Purpose**
  - Risky: **debt consolidation, medical** → stricter approval
  - Safer: **education, venture** → growth opportunity
- **Simplify Decision Factors**
  - Focus on: **income, DTI, loan grade, past default**
  - Ignore: **demographics & location (no impact)**

□ Thanks for Reading! □

Happy coding □😊